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# *From TCV to TRUST: Adaptation of the MEQ Framework for Equilibrium Reconstruction and Full-Pulse Simulation*

Nuclear Fusion

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# ABSTRACT

Equilibrium modelling is an essential part of tokamak scenario preparation and magnetic diagnostic interpretation. This thesis adapts the MEQ framework from TCV to the TRUST tokamak, integrating the machine geometry, active and passive circuits, diagnostic layout and local scenario data while retaining the generic solver core. The implementation supports inverse equilibrium generation with FBT, magnetic reconstruction with LIUQE and static forward calculations with FGS, together with interfaces to external equilibrium data.

The numerical assessment covers limiter, single-null and double-null configurations using synthetic magnetic signals and reference equilibria. It examines boundary geometry, global plasma quantities and active-current demand, distinguishing changes introduced by model transfer from those arising in subsequent calculations. Several LIUQE reconstructions recover the reference geometry closely while showing larger differences in internal quantities, including the on-axis safety factor, poloidal beta and plasma kinetic energy. A comparison of MEQ and EFIT reconstructions against six MAXFEA reference equilibria, using perturbed synthetic measurements, gives millimetre-scale RMS discrepancies in the sampled plasma-boundary-to-limiter gaps.

A coupling between the METIS system code and FBT extends the study to a reference scenario. Plasma parameters and associated boundary geometry define a sequence of quasi-static inverse equilibrium calculations that determine the active-circuit currents. All 44 sampled states converge and closely satisfy the prescribed global plasma targets. Although every state respects the FBT optimisation bounds, 42 exceed the separate common TRUST scenario current envelope in at least one circuit, identifying substantial current demand in the poloidal-field coils.

The resulting framework supports numerical preparation of TRUST scenarios and provides a basis for interpreting future experimental magnetic measurements. Improving the recovery of internal plasma quantities is the main priority for further reconstruction studies, alongside assessment of the scenario's actuator requirements.

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# 1 INTRODUCTION

## 1.1 CONTEXT AND MOTIVATION

Research on magnetic-confinement fusion brings together plasma physics, electromagnetics, diagnostics, control, and structural engineering. Among the available concepts, the tokamak is one of the most extensively studied. The magnetic configuration of a tokamak results from the combination of toroidal and poloidal magnetic fields. The toroidal field is generated by the toroidal-field coils, while the poloidal field results from the plasma current and the poloidal-field coils. Their combined effect must confine the plasma while remaining compatible with the geometry and electrical limits of the device.

A tokamak equilibrium describes this force balance and provides quantities needed for engineering analysis: magnetic-axis position, last closed flux surface shape and location, wall clearance, and, in diverted configurations, magnetic nulls and X-points. These quantities affect plasma-wall interaction, vertical stability, diagnostic interpretation, and plasma-control scenario design. Equilibrium calculations are therefore needed from preliminary machine design and pulse preparation to real-time monitoring and post-discharge analysis.

For a device still under development, equilibrium modelling begins before experimental discharges are available. Expected plasma states and magnetic signals must first be generated numerically. These models support the assessment of feasible equilibria, coil-current requirements, and the proposed diagnostic layout, while also providing test data for the reconstruction software. At this stage, numerical assessment relies on synthetic diagnostics and comparisons between independent numerical tools.

For the Tuscia Research University Small Tokamak (TRUST), this problem is addressed by adapting the MEQ equilibrium framework to the device geometry, electromagnetic system, data structures, and planned diagnostics. MEQ was developed at the Swiss Plasma Center (SPC) and has been used extensively for the Tokamak à Configuration Variable (TCV) [1]. The adapted framework brings equilibrium design, reconstruction, cross-code comparison, and time-dependent pulse simulation into the same computational environment.

## 1.2 TOKAMAK EQUILIBRIUM DESIGN AND RECONSTRUCTION

Under the usual axisymmetric approximation, a tokamak equilibrium represents the balance between plasma pressure and magnetic forces. Its mathematical description is based on the Grad–Shafranov equation, introduced in Chapter 2. Within this framework, three related computational problems are relevant to this work: inverse equilibrium design, forward equilibrium modelling, and equilibrium reconstruction.

Inverse equilibrium design starts from a prescribed plasma state and determines the active-coil currents required to reproduce it. In MEQ, this task is performed by FBT, which is used here to generate target limiter, single-null, and double-null equilibria [7]. Forward calculations instead determine the equilibrium associated with a specified electromagnetic state. MEQ provides FGS for static calculations and FGE for time-dependent plasma–circuit evolution [10].

Equilibrium reconstruction infers the plasma state from available diagnostic and global measurements. In MEQ, this task is performed by LIUQE, which determines an equilibrium consistent with magnetic measurements and a parameterised representation of the plasma sources [2]. Since experimental TRUST measurements are not yet available, the present work uses synthetic signals generated from FBT or MAXFEA reference equilibria. The reconstructed equilibria are then compared with their references using solver convergence, magnetic signals, active-coil currents, magnetic-axis and boundary positions, and X-point locations.

## 1.3 THE TRUST TOKAMAK AND ITS MODELLING REQUIREMENTS

TRUST is a compact university-class tokamak being developed at the University of Tuscia for research, education, and studies of enabling technologies for magnetic-confinement devices [3]. The reference scenario reported in the current project literature has a major radius of approximately 0.30 m, a toroidal magnetic field of approximately 0.8 T, and a nominal plasma current of 110 kA [4].

The poloidal-field system is designed to provide flexibility in plasma shaping. The present TRUST-MEQ configuration supports limiter, single-null, and double-null equilibria. A limiter plasma is bounded through contact with a material surface, while diverted configurations introduce one or two X-points and a separatrix. The double-null case is especially demanding for both equilibrium design and reconstruction because two magnetic nulls must be represented simultaneously and small variations in the field or in the diagnostic constraints may lead to a topologically different solution.

The TRUST active system comprises six poloidal-field coils and six central-solenoid modules, giving twelve active circuits [4]. In the present TRUST-MEQ model, the conducting vessel is discretised into 93 axisymmetric toroidal elements. The geometry file also includes the proposed magnetic probes and flux loops. Active circuits provide the fields for plasma formation and shaping, vessel currents modify the transient magnetic response, and the diagnostics supply the signals used for reconstruction.

Since TRUST has not yet produced experimental discharges against which these diagnostics can be validated, synthetic measurements are required. These measurements must be calculated at the same coordinates, with the same orientations and sign conventions, that will be assumed by the reconstruction software. This is also relevant to the physical design of the diagnostic system: if different equilibrium states produce signals that cannot be distinguished by the proposed sensor set, the reconstruction problem may remain poorly constrained even if the numerical solver itself is operating correctly.

Earlier work on TRUST examined stabilising plates, vertical dynamics, disruptions, and electromagnetic loads using MAXFEA [5]. The focus now shifts from passive-structure optimisation to an equilibrium workflow for scenario design, reconstruction, and pulse evolution. The two studies remain connected because the passive structures included in the electromagnetic model affect both vertical behaviour and the time-dependent plasma–circuit response.

## 1.4 FROM TCV-MEQ TO TRUST-MEQ: FRAMEWORK ADAPTATION

TCV is designed to explore a broad range of plasma shapes and magnetic configurations [6]. Supporting this flexibility requires an extensive infrastructure for modelling, reconstruction, and control. MEQ forms part of that infrastructure and provides MATLAB and C-based tools for inverse equilibrium design, equilibrium reconstruction, and static or evolutive forward calculations [1]. The numerical methods are reusable, but applying them to another tokamak requires machine-specific geometry, interfaces, and data.

This work retains the generic MEQ solvers rather than reimplementing the Grad–Shafranov equation. The main task was to identify the interfaces through which MEQ receives TCV geometry, shot information, magnetic measurements, equilibrium constraints, and circuit data, and then supply the corresponding information for TRUST. The adaptation therefore focuses on replacing the machine-specific interfaces required by MEQ while retaining the generic equilibrium solvers.

Several differences prevent direct use of the TCV workflow. TCV has an established experimental database and machine-specific services from which shot-dependent quantities can be retrieved. In the absence of experimental TRUST discharges, the TCV shot-based workflow cannot be reproduced directly. TRUST-MEQ therefore operates using design data, synthetic shots, externally supplied measurements, and equilibria generated by other numerical models. The TRUST coil arrangement, circuit ordering, number of turns, passive structures, limiter geometry, and diagnostic positions also differ from those of TCV.

TRUST-MEQ was therefore organised as a standalone machine layer, with the upstream MEQ repository retained as an external, unchanged dependency. TRUST geometry and default parameters are stored separately, and each run can be configured through a MATLAB structure or JSON file. Synthetic shot identifiers provide reproducible machine and scenario definitions without treating them as experimental discharges. Limiter, single-null, double-null, and custom cases are described in configuration data instead of separate branches in the calculation modules. This keeps the boundary between generic solvers and machine-specific data clear.

The adaptation supplies the geometry needed by FBT, LIUQE, FGS, and FGE, together with active-circuit and vessel models, magnetic probes, flux loops, and converters for external measurements. Current conventions require particular care. Physical conductor currents, coil turns, circuit currents in ampere-turns, virtual control currents, and code-specific channel ordering must be mapped consistently. Consistent mapping of these quantities is therefore essential to preserve the electromagnetic state across the different model interfaces.

The MAXFEA interface provides a second part of the workflow. It reads the magnetic-flux grid and equilibrium report, recovers plasma and circuit quantities, identifies the relevant separatrix, evaluates flux-loop signals, and derives magnetic-field components at the probe positions. MAXFEA control contributions are combined with the corresponding physical TRUST circuits before conversion to MEQ. The resulting data can define FBT boundary and plasma targets, LIUQE synthetic measurements, or initial equilibria for FGS and FGE.

Each conversion path tests a different part of the implementation. Direct MAXFEA-to-LIUQE reconstruction exercises the full data-conversion and reconstruction chain. Inserting FBT between MAXFEA and LIUQE separates the error introduced when reproducing the reference boundary from the subsequent reconstruction error. FGS provides a static comparison at fixed active currents. The numerical examples are presented in Chapter 4.

# 2 TOKAMAK EQUILIBRIUM MODELLING AND THE MEQ FRAMEWORK

Axisymmetric magnetohydrodynamic equilibrium provides the basis for the calculations discussed below. The following sections introduce the physical problem, the measurements that constrain it, and the numerical formulations implemented in MEQ. This background establishes the conventions needed to interpret the TRUST workflows presented in the following chapters.

A single equilibrium can be approached in several ways. A forward calculation predicts the plasma state from conductor currents and prescribed profile information; an inverse design calculation searches for the currents that realise a target boundary; and a reconstruction estimates the equilibrium from diagnostic measurements. These problems share the same force-balance equation, but differ in their known quantities, unknowns, and numerical constraints. That distinction is central to the roles of FBT, LIUQE, FGS, and FGE within MEQ.

## 2.1 AXISYMMETRIC MAGNETIC EQUILIBRIUM

On the time scale of an equilibrium calculation, the plasma is treated as stationary and the inertial term in the momentum equation is neglected. The pressure gradient is then balanced by the Lorentz force:

$$\nabla p = \mathbf{J} \times \mathbf{B}$$

Here $p$ is the plasma pressure, $J$ is the current density, and $B$ is the magnetic field. The equilibrium is described in cylindrical coordinates ($R$, $\varphi$, $Z$), assuming toroidal symmetry so that all equilibrium quantities are independent of $\varphi$. This approximation reduces the three-dimensional magnetic configuration to a two-dimensional problem in the poloidal plane while retaining the toroidal geometry of the device [10].

MEQ follows the COCOS 17 convention [10], [11]. In this convention, $\psi$ denotes the total poloidal magnetic flux in webers; its sign must remain consistent with the selected current and field directions. To write the standard Grad–Shafranov equation without an additional $2\pi$ factor, the equations below use the reduced flux

$$\Psi = \frac{\psi}{2\pi}$$

The axisymmetric magnetic field can then be expressed as:

$$\mathbf{B} = \frac{1}{R}\nabla\Psi \times \mathbf{e}_\varphi + \frac{F(\Psi)}{R}\mathbf{e}_\varphi$$

Here $e_\varphi$ is the toroidal unit vector, while the toroidal-field function is defined as

$$F(\Psi) = RB_\varphi$$

The poloidal field is tangent to surfaces of constant $\Psi$, so contours of poloidal flux provide a direct description of the magnetic topology. The radial, vertical, and toroidal components follow from

$$B_R = -\frac{1}{R}\frac{\partial\Psi}{\partial Z},\ B_Z = \frac{1}{R}\frac{\partial\Psi}{\partial R},\ B_\varphi = \frac{F(\Psi)}{R}$$

## 2.2 The Grad–Shafranov Equation

Axisymmetry and force balance imply that both the pressure *p* and the toroidal-field function *F* are constant on magnetic flux surfaces. They can therefore be written as functions of $\Psi$. Substituting the field representation into the force-balance equation gives the Grad–Shafranov equation:

$$\Delta^{*}\Psi \equiv R\frac{\partial}{\partial R}\left(\frac{1}{R}\frac{\partial \Psi}{\partial R}\right)+\frac{\partial^{2}\Psi}{\partial Z^{2}}=-\mu_{0}R^{2}\frac{dp}{d\Psi}-F\frac{dF}{d\Psi}$$

The two source terms describe the pressure gradient and the variation of the toroidal-field function; together they determine the toroidal plasma-current density. Although the differential operator is linear in $\Psi$, the equation is generally nonlinear because the source functions depend on the unknown flux. A numerical solution therefore requires a model for the plasma profiles together with boundary conditions and the magnetic contribution of the external conductors [10].

Inside the plasma, the source functions are commonly represented by a small set of basis functions whose coefficients are adjusted by the solver. In the vacuum region outside the plasma and conductors, the flux satisfies the homogeneous equation. Active coils and induced currents in conducting structures supply additional magnetic contributions through their Green-function coupling. Because the plasma boundary is itself a flux contour and is not known before the solution is obtained, tokamak equilibrium calculations are normally free-boundary problems.

A fixed-boundary formulation prescribes the last closed flux surface and solves only within that domain. A free-boundary formulation instead determines the plasma domain together with the flux map, accounting for the surrounding coils, vessel, and limiter. The latter is required when the quantities of interest include coil currents, wall gaps, X-point positions, or the response of passive structures.

## 2.3 Plasma Boundary and Magnetic Topology

The magnetic axis is an extremum of the poloidal flux around which the closed flux surfaces are nested. The last closed flux surface defines the plasma boundary. In a limiter configuration, that boundary is selected by contact with the limiter; in a diverted configuration, it is a separatrix that passes through a magnetic null where the poloidal field vanishes. The null is an X-point because the local flux contours form the characteristic hyperbolic pattern [9], [10].

For comparison between equilibria, the poloidal flux is often normalised between its value at the magnetic axis and its value at the boundary:

$$\psi_{N}=\frac{\psi-\psi_{A}}{\psi_{B}-\psi_{A}}$$

With this definition, the axis is at zero and the boundary is at one. The convention is useful for plasma profiles and for comparing equilibria with different absolute flux levels. It does not remove the need to track the sign and normalisation of the physical flux used by the solver.

A single-null equilibrium contains one dominant X-point. An ideal double-null equilibrium contains two X-points on the same separatrix, so their boundary fluxes must agree. In practice, small changes in coil currents, profile parameters, or diagnostic weights can make one null dominant. Accordingly, double-null design and reconstruction are assessed through the recovered topology and the individual X-point positions, rather than through a boundary-average error alone.

## 2.4 Forward, Inverse, and Reconstruction Problems

In a static forward problem, the active-coil and passive-conductor currents are given, together with either the plasma source profiles or a set of integral plasma constraints. The solver determines the flux map and derived quantities such as plasma current, magnetic-axis position, boundary, shape parameters, and magnetic signals. This formulation tests what equilibrium follows from a specified electromagnetic state [10].

An inverse design problem starts from a desired equilibrium description. Boundary points, flux values, field conditions, X-points, plasma current, or profile constraints are supplied as targets, and the active-coil currents are adjusted to minimise the associated errors while respecting circuit relationships and engineering limits [7], [12]. The resulting current demand can then be assessed against the available actuators and imposed engineering limits.

Equilibrium reconstruction has a different purpose. It uses measured magnetic quantities and known actuator currents to infer the plasma state that most consistently explains the observations. The solution is not a direct image of the plasma: it depends on the diagnostic geometry, uncertainties, profile parameterisation, regularisation, and physical constraints [2], [10]. This distinction is important when synthetic measurements are used, because agreement can otherwise be dominated by shared modelling assumptions.

## 2.5 Magnetic Measurements and Reconstruction

Magnetic probes measure one component of the local poloidal magnetic field [2]. For a probe whose sensitive axis makes an angle $\alpha$ with the radial direction, the ideal signal is

$$B_m = B_R \cos\alpha + B_Z \sin\alpha$$

Flux loops constrain the poloidal flux at their locations, while Rogowski-type measurements can provide the total plasma current. Active-coil currents and, where available, passive-structure currents are also required to separate the plasma contribution from the external field [2]. The geometric position, orientation, polarity, and calibration of every channel must use the same convention as the equilibrium model.

A reconstruction can be viewed as a weighted fit between measured and modelled signals [2], [10]. In a simplified form, the data contribution to the objective function is

$$\chi^2 = \sum_{i=1}^{N} \left( \frac{y_{i,\mathrm{meas}} - y_{i,\mathrm{model}}}{\sigma_i} \right)^2$$

where $\sigma_i$ represents the uncertainty or assigned error of channel i. The reconstruction also depends on the selected source-function basis and on any imposed physical constraints. If the model contains more degrees of freedom than the available measurements can constrain, the fit becomes increasingly sensitive to noise and non-uniqueness [2], [10]. In the present work, the assigned errors act primarily as reconstruction weights and do not represent experimentally calibrated TRUST sensor uncertainties. Reconstruction quality is assessed through convergence, residuals, recovered measurements, plasma current, axis and boundary position, and magnetic topology.

## 2.6 THE MEQ FRAMEWORK

MEQ is a MATLAB and C-based toolbox for axisymmetric tokamak equilibrium calculation, manipulation, and visualisation [1]. Its architecture separates generic numerical routines from machine-specific parameters, geometry, and data access. This separation makes it possible to retain the tested equilibrium solvers while replacing the interfaces that describe a particular tokamak.

The same naming pattern is used across the suite. A top-level solver assembles parameter functions, geometry functions, solver-specific input functions, consolidation routines, and the numerical solver. Device-specific functions carry the tokamak suffix, while the generic routines remain common.

MEQ uses a small set of standard data structures. L contains the assembled model; L.P stores the user and device parameters; and L.G stores the geometry, labels, grids, Green matrices, mutual inductances, resistances, and circuit transformations. LX contains shot and time-dependent input data, such as currents, voltages, measurements, and equilibrium constraints. LY contains the calculated equilibrium and derived quantities, including the flux map, current distribution, magnetic axis, plasma boundary, X-points, integral parameters, and solver diagnostics.

Many electromagnetic relationships are represented as linear Green matrices. For example, active-coil currents can be mapped to flux on the computational grid, flux-loop signals, or probe fields. The nonlinear part of the problem remains in the plasma response and the moving boundary. This organisation lets the same machine geometry support design, reconstruction, forward calculation, plotting, and cross-code comparison.

## 2.7 EQUILIBRIUM SOLVERS IN THE TRUST-MEQ WORKFLOW

### 2.7.1 FBT: Inverse Equilibrium Design

FBT solves an inverse free-boundary equilibrium problem. The desired equilibrium is expressed through control points and global constraints, and the code determines the active-coil currents and plasma-profile coefficients that best satisfy them [7]. Flux targets define boundary or internal surfaces; field targets can impose magnetic nulls; current constraints control the plasma and conductor currents; and relative weights determine how incompatible requirements are balanced.

Within TRUST-MEQ, FBT is used to generate limiter, single-null and double-null equilibria and to construct target sequences for the plasma-current ramp. The resulting boundaries and active-current demands are assessed in Chapters 4 and 5.

### 2.7.2 LIUQE: Equilibrium Reconstruction

LIUQE reconstructs the free-boundary equilibrium from magnetic and global measurements. Its input structure can include probe signals, flux-loop values, active- and passive-conductor currents, and total plasma current [2]. It adjusts a parameterised representation of the plasma sources and iterates between fitting the measurements and solving the equilibrium. This formulation belongs to a broader class of reconstruction methods in which magnetic and auxiliary measurements constrain a Grad–Shafranov solution through weighted fitting [8].

With the data available for this work, TRUST reconstruction currently relies on synthetic diagnostics. The reconstruction input is kept separate from the reference equilibrium so that signal generation, channel mapping, signs, units, assigned errors, and solver performance can be checked independently.

### 2.7.3 FGS: Static Forward Equilibrium

FGS solves the static forward free-boundary problem [10]. Given conductor currents and plasma-profile information, it computes the equilibrium that follows from that state. FGS provides a consistency check for selected operating points and a bridge between externally generated equilibria and the MEQ representation. Because the inputs and outputs use the same MEQ structures as the other solvers, FGS results can be examined with the same plotting and diagnostic tools.

### 2.7.4 FGE: Evolutive Free-Boundary Simulation

FGE extends the forward formulation to time-dependent evolution. It couples the free-boundary equilibrium to the electrical dynamics of active and passive conductors and to a model of plasma-current diffusion. In compact form, the circuit evolution is governed by the flux-balance relation

$$\frac{d\boldsymbol{\lambda}}{dt} = \mathbf{V} - \mathbf{RI}$$

where $\lambda$ is the vector of circuit flux linkages, which depends on the mutual inductances and on the evolving plasma state. FGE is therefore suited to voltage-driven simulations in which active-coil currents, vessel currents, and the equilibrium influence one another [10]. Its use requires a consistent circuit model and an initial equilibrium compatible with the selected conductor state.

FGE remains part of the implemented TRUST-MEQ framework, although evolutive FGE calculations are not included in the numerical assessment presented in Chapters 4 and 5.

## 2.8 Connection to the TRUST-MEQ Implementation

The TRUST adaptation provides the machine-specific function chain required by the MEQ modules to operate independently of the TCV database infrastructure. It handles local geometry loading, fictitious shot identifiers, TRUST control-point definitions, diagnostic coordinates, and synthetic data interfaces.

The current implementation includes centralised JSON defaults, scenario templates, wrappers for reproducible solver calls, interfaces to MAXFEA outputs, static and evolutive forward workflows, and additional consistency checks. Together, these components form the present TRUST-MEQ toolchain. Chapter 3 describes its architecture and the main implementation choices [12].

The framework establishes a controlled and reproducible environment for TRUST-MEQ calculations. Chapters 4 and 5 assess the resulting equilibria using solver diagnostics, comparisons with reference states, magnetic and geometric quantities, and active-current demand, following established verification and validation principles for scientific computing [13].

# 3 ADAPTATION OF MEQ FROM TCV TO TRUST

The adaptation of MEQ to TRUST centres on the information that defines the machine: conductor geometry, circuit coordinates, diagnostic channels and equilibrium targets. The numerical solvers introduced in Chapter 2 receive this information through a set of device interfaces. This chapter follows those interfaces from the local input files to the assembled equilibrium model, describing the machine-specific implementation choices that affect the subsequent reconstruction and scenario calculations [12].

## 3.1 SEPARATION OF THE NUMERICAL CORE AND THE MACHINE INTERFACES

In the TCV implementation, machine information is closely connected to the experimental data infrastructure. For example, `meqgtcv` retrieves winding positions, vessel elements, diagnostic geometry and selected electromagnetic matrices from the MDSplus STATIC tree [12]. The parameter and measurement functions also provide access to shot-dependent settings and signals. Reusing this machinery directly would require TRUST to reproduce both the TCV data organisation and its machine assumptions.

TRUST-MEQ instead supplies the corresponding information locally. The upstream `meq` directory contains the common equilibrium algorithms and numerical libraries, while `trust-meq` contains the TRUST functions, configuration files, machine data and examples. The numerical core remains an external dependency. This arrangement makes changes to a TRUST scenario independent of changes to the Grad–Shafranov solver and allows the origin of a calculation to be identified through the versions of both components.

The device interfaces follow the MEQ naming convention. Functions with the suffix `p` prepare machine-specific parameters, those with `g` provide the machine geometry, and those with `x` assemble the solver-specific time-dependent inputs. Appending `trust` selects the device implementation.

Thus, a call to fbt with the machine identifier trust resolves fbtptrust, fbtgtrust and fbtxtrust. The generic functions then complete the parameters and geometry, assemble the pre-computed model quantities through fbtc, complete and validate the time-dependent input structure through fbtx, and solve the problem through fbtt. LIUQE follows the same pattern through liu, its TRUST interfaces, liuc and liut.

Table 3.1 lists the device functions used by the four solvers.

*Table 3.1. Device-specific entry points for the TRUST machine [12].*

| Solver | Parameters | Geometry | Input data |
|---|---|---|---|
| FBT | `fbtptrust` | `fbtgtrust` | `fbtxtrust` |
| LIUQE | `liuptrust` | `liugtrust` | `liuxtrust` |
| FGS | `fgsptrust` | `fgsgtrust` | `fgsxtrust` |
| FGE | `fgeptrust` | `fgegtrust` | `fgextrust` |

The common starting points are `meqptrust` and `meqgtrust`. The first supplies the machine parameters from which the solver-specific parameter structures are built. The second defines the conductors, limiter and diagnostics used to construct the electromagnetic model. Solver-specific geometry functions add the quantities needed for their own calculation, such as current constraints for FBT or circuit data for FGE.

Initialisation is handled by `setup_trust_meq`, which configures the MATLAB environment and locates the external MEQ dependency. The high-level runner `run_trust_meq` coordinates the FBT and LIUQE workflows, while FGS and FGE are accessed through their MEQ entry points and dedicated TRUST workflows.

The assembled calculation retains the standard MEQ structures. `L` holds the model, with parameters in `L.P` and geometry and electromagnetic matrices in `L.G`. `LX` contains the supplied targets or measurements, normally with time samples arranged in columns. `LY` contains the solution and its diagnostics. Keeping these structures together is essential: a flux array or current vector only has a physical meaning when its grid, conductor ordering and normalisation are known. Conversion between two models therefore uses the associated `L` structures, rather than copying arrays on the assumption that their dimensions imply compatibility [12].

## 3.2 Local Configuration and Fictitious Shots

Machine data and run settings are stored separately. `data/trust.json` provides the conductor and diagnostic geometry. The defaults file, `config/trust_meq_defaults.json`, provides machine normalisation values, numerical settings, material parameters and solver defaults. Files under `config/shots` identify reference cases, while the templates under `config/templates` collect the settings for editable limiter, single-null and double-null calculations. The circuit scenario is stored in a separate MAT file because it includes matrices and waveform data.

The fictitious-shot mechanism preserves MEQ's shot-based interface without implying the existence of an experimental discharge. The configured range is 99000–99999. Within that range, `load_shot_config` reads the corresponding local JSON file. The main templates use 99002 for a limiter case, 99003 for a lower single-null case and 99004 for a double-null case. The identifier selects the input metadata, while the magnetic topology is determined by the resulting equilibrium calculation. Experimental TRUST shot retrieval is outside the scope of the current implementation [12].

A run configuration is supplied through `Config`, either as a MATLAB structure or as a JSON filename. Its main fields describe the plasma shape, global plasma targets, sensor selection and solver options. The `Config` shape fields `R0`, `Z0`, `a`, `kappa` and `delta` specify the geometric centre $(R_0, Z_0)$, minor radius $a$, elongation $\kappa$ and triangularity $\delta$; its target fields `Ip`, `bp`, `qA` and `rBt` specify $I_p$, $\beta_p$, $q_A$ and $RB_t$. Here $q_A$ is the safety factor at the magnetic axis. It must not be confused with an edge safety factor. Likewise, $RB_t$ is the vacuum radius–toroidal-field product, in T·m, rather than a field in teslas.

Reference values stored in the machine defaults are used for numerical configuration and should not be interpreted as scenario targets or equilibrium outputs; in particular, the magnetic-axis position is determined by the equilibrium solution.

Configuration precedence is implemented in the wrappers. For FBT, default run options are merged with `Config.fbt.options`; shape fields fill options that have not already been specified. Explicit `FbtOptions`, sensor-count arguments and general overrides are then passed to model construction. The wrapper subsequently applies the configuration when preparing the boundary and plasma targets. Repeating the same setting in several places can therefore make the intended case difficult to reconstruct, particularly when a shape value also appears among solver options. The resolved parameters and final `LX` provide a more reliable record than the template filename alone.

LIUQE has its own option path. Its configuration can contain direct settings, a `parameters` structure and an `options` structure, with explicit `LiuqeOptions` appended last. A general override accepted by the wrapper is not necessarily forwarded as a solver option. Changes to the source-function basis, reconstruction errors or

iteration settings should therefore be made through the LIUQE option fields and checked in the resulting `L.P` [12].

Local caching reduces repeated file reads but also affects reproducibility during interactive work. The geometry loader compares the filename, modification time and file size before reusing its stored data. Defaults and shot metadata are cached separately and do not use the same file-change check. A fresh MATLAB session, or deliberate clearing of the relevant loader functions, is consequently needed after editing these inputs. For reproducibility, the configuration associated with each calculation should therefore correspond to the inputs actually used during that run [12].

## 3.3 Geometry and the Electromagnetic Model

### 3.3.1 Active Coils and Circuit Coordinates

The TRUST active system comprises six central-solenoid modules and six poloidal-field coils [4]. In the default model they are ordered as `CS_001` to `CS_006`, followed by `PF_001` to `PF_006`. This order is recorded in `G.dima` and used to interpret the columns of the electromagnetic matrices. External current data are matched to these labels before they enter the solver.

Each coil is described by its centroid and rectangular poloidal cross-section. The winding generator divides that section into smaller current elements using nominal radial and vertical subdivision lengths of 0.02 m. Each element receives an equal fraction of the circuit's total ampere-turns through `G.Twa`. The column sum of this transformation is one for each coil. Refining the winding representation changes the spatial approximation of the conductor, without increasing the number of independently controlled circuits [12].

The active-current variable $I_a$ is expressed in ampere-turn coordinates. The unit turn counts stored in the geometry JSON are part of this normalisation; they are not the physical winding counts. The latter are loaded from `Scenario_TRUST_SN.mat` by `trust_load_current_scenario`. If *N* is the diagonal matrix of turn counts and $i$ is the vector of physical conductor currents, the coordinate transformation is

$$I_{\mathrm{a}} = N\, i, \quad V_{\mathrm{a}} = N^{-1}\, v$$

where $v$ denotes the terminal-voltage vector and $V_{\mathrm{a}}$ is the voltage vector used with the ampere-turn currents. The corresponding circuit matrices are

$$M_{\mathrm{a}} = N^{-1} M_{\mathrm{phys}} N^{-1}, \quad R_{\mathrm{a}} = N^{-1} R_{\mathrm{phys}} N^{-1}$$

Here $M_{\mathrm{phys}}$ and $R_{\mathrm{phys}}$ are the physical inductance and resistance matrices, and $M_{\mathrm{a}}$ and $R_{\mathrm{a}}$ are their representations in the MEQ active-current coordinates. Both sides of the transformation are required. Multiplying the current by the turn count while retaining the physical resistance or terminal voltage would change the circuit equation being solved [12].

In the current scenario, PF2–PF5 use 36 turns, while PF1 and PF6 use 100 turns [4]. The corresponding conductor currents and circuit quantities are transformed consistently using the same winding counts, preserving the ampere-turn representation adopted by MEQ. The scenario loader also checks the coil ordering and the consistency between conductor-current and ampere-turn coordinates before the circuit data are passed to the FBT and FGE interfaces. Couplings to the plasma grid and passive conductors are then constructed by the generic MEQ geometry routines [12].

The TRUST interfaces use current and voltage envelopes derived from the scenario data as numerical operating constraints. fbtgtrust loads the current envelopes and derives the voltage envelope from the stored circuit voltages, while FBT and FGE apply their respective solver-specific limits. Thermal limits and additional protection constraints are not included in the present model and are treated separately from these numerical operating bounds.

### 3.3.2 Passive Conductors

The present vessel model contains 93 axisymmetric toroidal elements. Their centroid coordinates and rectangular cross-sections come from the geometry JSON. Each element is treated as a closed conducting ring with its own current. This is the discretisation adopted by TRUST-MEQ, rather than a count of physical vessel components or a detailed representation of ports and joints.

The resistance assigned to element k is

$$R_{\mathrm{v,k}} = \frac{2\pi R_{\mathrm{k}}}{\sigma w_{\mathrm{k}} h_{\mathrm{k}}}$$

where $R_{\mathrm{k}}$ is its centroid radius, $w_{\mathrm{k}}$ and $h_{\mathrm{k}}$ define its poloidal cross-sectional area, and $\sigma$ is the assumed conductivity. The configured steel conductivity is $1.35 \times 10^6 \frac{S}{m}$. This model supplies a finite resistive response for the induced currents, while leaving temperature dependence and three-dimensional current paths outside the present description [12].

In the present implementation, $G.T_{vu}$ is the identity and all 93 vessel-element currents are retained; no reduced passive-current representation is used.

Before assembling the FGE circuit model, `fgegtrust` symmetrises the passive inductance matrix. If required, a diagonal correction is applied to the passive inductance matrix and its magnitude is recorded together with the assembled model [12].

### 3.3.3 Limiter and Computational Grids

The limiter contour and the conducting vessel have different roles. The contour bounds the accessible plasma region, while vessel elements describe induced electrical currents. When a limiter is not supplied in the geometry data, `meqgtrust` uses the polygon in the defaults file and interpolates along its segments. The current geometry JSON uses this fallback. The polygon is therefore part of the machine definition even though it is stored separately from the vessel coordinates.

The machine parameter function leaves the computational limits unspecified by assigning NaN to the radial and vertical bounds. The generic grid constructor then derives these bounds from the limiter with a small grid-dependent extension. The configured values nr = 65 and nz = 32 define the default grid resolution, although the resulting grid dimensions follow the conventions of the MEQ grid constructor rather than corresponding directly to these values. Explicit bounds or solver options can modify the grid, so comparisons should use the actual stored coordinates [12].

The optional izgrid extension provides a wider fixed grid for diagnostics and plotting. Grid resolution, winding subdivision and the vessel discretisation are separate modelling choices, each with a different effect on the magnetic calculation [12].

## 3.4 Translating Plasma Targets into FBT Constraints

The FBT wrapper converts a compact description of the desired plasma into control points. In `meqconfig_trust`, a parameterised contour is generated from the target centre, minor radius, elongation and a shaping parameter. With $\theta$ denoting the contour parameter, the implemented expressions are

$$R(\theta) = R_0 + a\cos(\theta + \delta_g \sin\theta),$$
$$Z(\theta) = Z_0 + \kappa\, a \sin\theta.$$

The code field `delta` corresponds here to $\delta_g$, the parameter inside the angular distortion. The default construction samples 39 points and excludes the duplicate endpoint. These points are targets for the inverse calculation; the shape and triangularity of the converged LCFS are subsequently obtained from the equilibrium. An explicit set of boundary coordinates can replace the parameterised contour through `trust_apply_boundary_config`, which checks that the radial and vertical arrays have the same length [12].

The wrapper adds the boundary points through the generic `fbtgp` interface. Their flux is constrained relative to a common boundary value. An X-point entry adds a point with the boundary flux together with zero radial and vertical field targets. In the single-null template one lower null is specified; the double-null template specifies upper and lower nulls. Nearby ordinary contour points can be removed before adding these exact constraints, reducing competition between the smooth parameterised boundary and the requested null. This preprocessing expresses the intended geometry without modifying the underlying optimisation algorithm.

The error fields determine how the constraints enter the calculation. A zero error denotes an exact condition, a finite positive error gives a weighted contribution, and an infinite error disables the corresponding penalty. Requested values marked `NaN` are left unspecified. For flux constraints, the global error scale and the relative point error are combined. Their numerical values therefore describe the adopted optimisation weights, rather than a direct tolerance on the distance between two plasma boundaries [12].

Global targets constrain the plasma profiles alongside the geometric conditions. The default FBT combination is plasma current, safety factor at the axis and poloidal beta. The profile basis is configured separately. Changing the set of integral constraints requires a compatible number of free profile coefficients; adding more targets does not automatically make an equilibrium better determined.

Shot currents are also kept distinct from enforced current targets. The wrapper retains explicit `gpia` constraints when they have been requested through FBT options, rather than automatically forcing the currents stored in a shot file. This leaves the inverse problem free to find currents that support the chosen shape. With `dohzero` enabled, the FBT geometry adapter adds difference equations between successive CS entries. In the ampere-turn representation these impose equal CS ampere-turns; interpreting them as equal conductor currents additionally depends on the turn counts [12].

There are two related input paths. A direct MEQ call can obtain `LX` from `fbtxtrust`, which checks for a saved autogen file and fills missing fields from the parameter structure. `trust_run_fbt` instead constructs a fresh set of constraints from `Config`, consolidates them with `fbtx`, and calls `fbtt`. An existing autogen file can consequently influence a direct call even when the shot identifier is unchanged. The final constraints must be archived whenever this route is used [12].

For time sequences, the distinction between static targets and circuit evolution remains explicit. With circuit coupling disabled, an FBT sequence represents a set of quasi-static operating points. Their dynamic accessibility is assessed separately with the circuit-coupled or evolutive formulation.

## 3.5 Diagnostic Geometry and Reconstruction Inputs

The default JSON contains three flux-loop entries and twelve magnetic-field channels. The latter occupy six distinct positions, repeated for two intended orientations at each position. Counting channels and counting sensor locations therefore gives different numbers. `meqgtrust` maps their coordinates and orientations into `G.rf`, `G.zf`, `G.rm`, `G.zm` and `G.am`; the generic geometry routines construct the corresponding responses to coil, plasma and vessel currents.

Sensor coordinates can also be supplied explicitly. If a non-default count is requested without custom coordinates, `meqconfig_trust` distributes sensors along the contour formed by the vessel centroids. It assigns probe orientations from the local contour normal and recomputes the relevant Green matrices. This provides a convenient way to construct alternative numerical layouts. Their information content and observability are assessed through the reconstruction response rather than from the channel count alone.

Probe orientations are converted at the interface to MEQ. The geometry JSON stores angles in degrees; meqgtrust converts them to radians when constructing G.am. The default angles of 0 and 90 degrees therefore select the radial and vertical field components, respectively. Explicit arrays supplied through P.sensor_probes or Config.sensors.probes already use radians, including arrays in a configuration JSON, and are passed through unchanged. Layouts generated from the vessel contour also use radians. The geometry plot reads G.am directly, so the displayed directions and the response matrices share the same convention [12].

Because synthetic signal generation and reconstruction can use the same geometry, agreement within that path does not independently verify the probe orientations. Reconstruction cases must therefore retain the angle convention with their inputs and check the assembled diagnostic response against the intended $B_r$ and $B_z$ components.

External measurements are organised by `trust_measurements_to_lx`. The converter accepts channel matrices and, where labels are available, named current or diagnostic fields. It checks array sizes and can replicate a single column over the requested time samples. The main quantities retain the MEQ units: magnetic-field channels in teslas, poloidal flux in webers, active currents in ampere-turns, and plasma and full vessel-element currents in amperes. Missing channels can receive fallback values, including zeros, so dimensional consistency alone does not confirm that a complete measurement set was supplied [12].

The high-level LIUQE workflow uses external measurements when present. Otherwise, it obtains an FBT equilibrium, or reuses one supplied by the caller, and converts it into the reconstruction model through `meqxconvert`. The reference equilibrium and its model remain available in the returned result. This makes it possible to inspect the measurements supplied to LIUQE separately from the equilibrium it reconstructs.

The low-level fallback in `liuxtrust` serves a narrower purpose. It forms flux and field signals from the active currents using `Mfa` and `Bma`, then adds Gaussian perturbations and supplies a plasma-current target separately. It does not include the magnetic contribution of a self-consistent plasma in those generated signals. The fallback can exercise the measurement interface, but it does not replace conversion from a complete FBT or external equilibrium for a reconstruction study [12].

Assigned errors and relative weights are set in liuptrust and can be overridden for each case. Their role in the fit was discussed in Chapter 2. The current values are numerical settings for the reconstruction and should not be presented as experimental uncertainty estimates. Similarly, fixed random seeds are used to make synthetic perturbations reproducible. The TRUST interface uses the ideal two-dimensional flux-loop response; a three-dimensional flux-loop model is not implemented [12].

## 3.6 INTERFACES TO MAXFEA AND EQDSK

The MAXFEA interface separates data reading from the preparation of solver inputs. The MATLAB function `maxfea_read_ans` reads the flux-grid data, while `maxfea_read_rep` extracts plasma quantities, circuit data and first-wall coordinates from the MAXFEA output. Prepared MAT files can provide the same two data structures.

The available input set includes double-null equilibria at the start and end of flat-top, stored in `examples/input/trust_dn_sof.mat` and `examples/input/trust_dn_eof.mat`. Each file contains `ans_data` and `rep_data`. The function build_maxfea_liuque_input combines these data into a common descriptor containing the flux map, conductor currents, diagnostic signals and plasma targets. A related builder derives the corresponding FBT configuration from the same descriptor [12].

Flux normalisation is applied when the MAXFEA data are read. The ANS reader retains the original grid and records the scale and offset applied during conversion; its default multiplicative scale is $2\pi$. After this conversion, `psi` is expressed as total poloidal flux in webers throughout the interface [12]. Magnetic-field components are obtained by differentiating the flux map using the corresponding factor $\frac{1}{2\pi R}$, consistent with the flux convention introduced in Chapter 2 [11]. Prepared inputs are expected to follow the same convention.

The MAXFEA input includes the virtual control-current contributions ctrlr and ctrlv. The interface maps these contributions onto the physical PF-coil currents before passing the circuit data to MEQ: ctrlr is added to PF3 and PF4, while ctrlv is added to PF2 and subtracted from PF5. Both the original and effective currents are retained in the descriptor for traceability [12].

For the available input set, diagnostic sampling uses `examples/input/trust_sensors.mat` in preference to the standard geometry JSON. This file defines 36 flux-loop locations and 36 magnetic-probe locations. At each probe position, both the radial and vertical magnetic-field components are sampled, resulting in 72 magnetic-field channels. The corresponding probe angles are assigned explicitly as zero and $\frac{\pi}{2}$. If the MAT file is unavailable, the interface falls back to the geometry JSON, where probe angles are converted from degrees to radians. Probe arrays supplied directly through configurations or example scripts are instead interpreted in radians. The MAXFEA reconstruction and the standard TRUST diagnostic configuration therefore use different diagnostic information, which must be taken into account when comparing their results.

The descriptor records the diagnostic source and the number of selected channels, while the LIUQE runner verifies that the expected channels are present [12].

The direct MAXFEA reconstruction reuses the CREATE-compatible data interface distributed with MEQ. The adapter generates the geometry and measurement MAT files required by this interface, which are then accessed through the MAXFEA-specific device functions. Along this path, the geometry includes the first-wall contour obtained from the MAXFEA data and applies a minimum coil cross-section width or height of 0.025 m in the serialised geometry. As a result, the geometry assembled through the MAXFEA path is not necessarily identical to that used by the standard TRUST-MEQ interface, despite both representing the same machine [12].

To prepare the FBT targets, the builder extracts the flux contour corresponding to the boundary-flux value provided by MAXFEA. The appropriate contour is selected using its relation to the magnetic axis and the reported X-point and is subsequently resampled into boundary-control points. For a requested double-null configuration, the procedure also searches the flux grid for a second saddle-point candidate close to the boundary flux. The identification of this second point is therefore part of the numerical topology assessment

rather than a guarantee that a double-null equilibrium has been obtained. A warning is generated if no suitable second candidate is identified [12].

The direct LIUQE reconstruction also uses the MAXFEA flux map to initialise the nonlinear equilibrium state and the initial profile coefficients. The final reconstruction is determined by the supplied diagnostic measurements, but its initial state therefore contains information from the reference equilibrium. Results obtained through this path consequently represent reconstructions with reference-based initialisation. Sensitivity to this choice can be assessed separately by repeating the reconstruction from alternative initial states [12].

The EQDSK interface is used for equilibrium exchange and for preparing the low-current seed equilibrium. The function `maxfea_read_eqdsk` reads the flux grid, plasma profiles, magnetic-axis position and boundary. The breakdown-input builder retains a resampled version of the boundary, assigns a target plasma current and scales the magnetic-axis safety-factor target inversely with current under the assumption of an approximately preserved profile. A poloidal-beta target is also assigned. These operations define an equilibrium target for FBT; they do not model gas ionisation or the physical breakdown process. Their implications for the pulse scenario are discussed in Chapter 5 [12].

## 3.7 Connection to Static and Evolutive Forward Calculations

The FGS and FGE adapters extend the same machine interface to forward calculations. `fgsxtrust` can initialise FGS from FBT, a MAXFEA descriptor or a prescribed vacuum-current state. An FBT equilibrium is transferred with `meqxconvert`; an external flux map is handled by `trust_external_equilibrium_to_lx`. The latter reorders the active currents, interpolates the flux onto the destination grid, identifies the plasma domain and fits the source-function coefficients. It rejects a grid that does not cover the required domain and initially sets the passive currents to zero [12].

Transferred equilibria provide initial states for the destination model, which evaluates them using its own grid, conductor representation and profile basis. The FGS result can then be compared with the transferred input.

FGE obtains its initial equilibrium through the same source adapter and adds the quantities required for circuit evolution. Its geometry interface supplies the active-circuit matrices, passive couplings, voltage limits and configured power-supply delays.

The available trust_run_fge_ramp workflow evolves the external circuits under a prescribed voltage trajectory while $I_p$, $\beta_p$ and $q_A$ are supplied along a reference trajectory. Plasma-current diffusion is not activated in that configuration. The full FGE formulation can incorporate current diffusion through an appropriate CDE model and compatible profile constraints [12].

The FBT ramp targets are obtained from the available METIS scenario in the associated TRUST-METIS dataset, which provides the reference trajectory used by this workflow [12].

Together, these interfaces provide a common TRUST representation for inverse equilibrium design, reconstruction, static forward calculations and evolutive simulations. The same machine geometry and circuit conventions can be transferred between FBT, LIUQE, FGS and FGE. The following chapter presents the selected FBT, LIUQE and FGS numerical examples.

# 4 EQUILIBRIUM RECONSTRUCTION AND CROSS-CODE VALIDATION

## 4.1 SCOPE AND ASSESSMENT CRITERIA

This chapter presents the TRUST equilibrium-generation, reconstruction, and model-transfer results obtained with FBT, LIUQE and FGS. The assessment includes six FBT cases, eight LIUQE reconstructions and two FGS cases, summarised in Table 4.1. Together, they cover limiter, single-null and double-null configurations, and low-current and flat-top equilibria. The reconstruction cases use synthetic magnetic signals or inputs derived from existing MAXFEA equilibria; experimental TRUST measurements are not yet available. Chapter 5 is reserved for the METIS-to-FBT scenario coupling [12].

The following sections assess these numerical results through solver status, agreement with available references and active-current demand [12,13].

The assessment is based on the quantities that are most relevant to each workflow. Solver behaviour is described through convergence residuals and iteration counts, using the criteria defined for each numerical formulation. When a reference equilibrium is available, the comparison includes plasma current, magnetic-axis and X-point positions, flux surfaces, and magnetic signals. Numerical differences are reported where they have been explicitly calculated, while comparisons based only on overlaid contours remain qualitative. For transfers between solvers, particular attention is given to changes in the equilibrium geometry and active-circuit currents.

Active-current demand is compared with the common TRUST scenario envelope introduced in Chapter 3. For each circuit, this reference value is obtained from the prescribed current-density limit and the corresponding coil-pack cross-sectional area, and is expressed in ampere-turns. Unless otherwise stated, the reported scenario-envelope utilisation is the largest absolute current-to-limit ratio among the circuits and states included in each example. The current figures instead use the assessment limits stored for the displayed solver state, which are not necessarily identical to the common scenario envelope. For the six FBT cases considered here, the two sets of assessment limits coincide. They are also distinct from the FBT optimisation bounds of ±800 kA-turn for the CS circuits and ±200 kA-turn for the PF circuits. The common scenario envelope is therefore used throughout the chapter as a consistent reference for comparing current demand, while a complete operational assessment would also require voltage, thermal, structural, and conductor-level constraints [12].

Seven of the 16 retained examples exceed the common scenario envelope in at least one state. Several of these cases originate from the same source equilibrium or reuse related synthetic inputs, so the exceedances are not independent operating points. The examples examine equilibrium generation, external-data import, reconstruction and static model transfer. The cross-code comparison then evaluates MEQ and EFIT reconstructions against MAXFEA reference geometries using synthetic magnetic measurements with added noise [12].

*Table 4.1. Examples included in the Chapter 4 assessment [12].*

| Family | Cases | Convergence | Purpose | Scenario envelope |
|---|---|---|---|---|
| FBT | 6 | All selected cases | Equilibrium design and external-boundary transfer | 2 exceeded |
| FGS | 2 | All selected cases | Static transfer at fixed active currents | All within |
| LIUQE | 8 | All selected cases | Synthetic reconstruction and MAXFEA-derived inputs | 5 exceeded |

## 4.2 FBT Equilibrium Generation

### 4.2.1 Direct FBT Shot Workflow

The first case considers a 15 kA TRUST equilibrium generated directly with FBT. The target boundary is defined by 20 control points, with geometric centre $R_0$ = 0.230 m and $Z_0$ = 0, minor radius $a$ = 0.045 m, elongation $\kappa$ = 1.15 and zero triangularity. These quantities define the requested geometry rather than the dimensions of the final equilibrium. The toroidal-field parameter is $RB_t$ = 0.32 T m. The 12 active-circuit currents are allowed to vary around the breakdown-scenario reference values, while circuit evolution is not included [12].

The equilibrium is constrained through the plasma current and plasma kinetic energy, together with a fixed profile coefficient. A reference value of $\beta_p$ = 0.06 corresponds to an energy target of 2.036 J. Although $q_A$ = 8 is included in the input data, it is not imposed in this calculation, and the on-axis safety factor is therefore determined by the resulting equilibrium. Error scales of 5 kA-turn for the current targets and $5 \times 10^{-5}$ Wb for the flux targets allow FBT to balance the current references against the requested boundary [12].

The calculation converged in 17 iterations, with a residual of $6.31 \times 10^{-7}$ below the prescribed tolerance of $10^{-6}$. The resulting plasma current is 15.000 kA. The magnetic axis is at $R$ = 0.231541 m and $Z$ = −0.0000218 m. The equilibrium is limited on the inboard side, with limiter contact at $R$ = 0.185 m and $Z$ = 0.01947 m. Table 4.2 gives the principal equilibrium quantities.

*Table 4.2. Principal quantities of the native 15 kA FBT equilibrium [12].*

| Quantity | Result |
|---|---|
| Poloidal beta, $\beta_p$ | 0.0600 |
| Normalised internal inductance, $l_i(3)$ | 0.9638 |
| Safety factor on axis, $q_A$ | 1.2790 |
| Plasma kinetic energy, $W_k$ | 2.0358 J |
| Plasma volume, $V_p$ | 0.011469 m³ |
| Toroidal beta, $\beta_t$ | 0.01541% |
| Maximum absolute active current | 425.45 kA-turn (CS2, CS5) |
| Maximum scenario-envelope utilisation | 40.4% |

The agreement with the requested boundary can be examined through the flux residuals at the 20 control points. Relative to the common fitted flux value, their RMS value is $2.91 \times 10^{-4}$ Wb and the maximum absolute residual is $8.72 \times 10^{-4}$ Wb. These residuals describe the consistency of the imposed flux constraints and are not geometrical boundary errors. The fitted control-point flux differs from the equilibrium boundary flux by $1.48 \times 10^{-4}$ Wb.

Figure 4.1 shows the resulting equilibrium, while the corresponding active-circuit currents are reported in Figure 4.2. The largest absolute current is 425.45 kA-turn in CS2 and CS5, corresponding to a maximum scenario-envelope utilisation of 40.4%. The remaining circuits have lower utilisation, and all 12 currents remain within both the common scenario envelope and the FBT optimisation bounds of ±800 kA-turn for the CS circuits and ±200 kA-turn for the PF circuits.

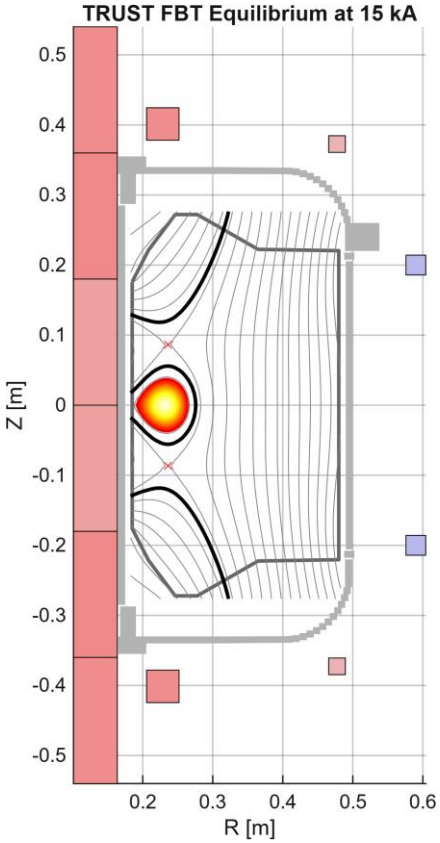



*Figure 4.1. Native 15 kA FBT equilibrium in the TRUST poloidal cross-section [12].*

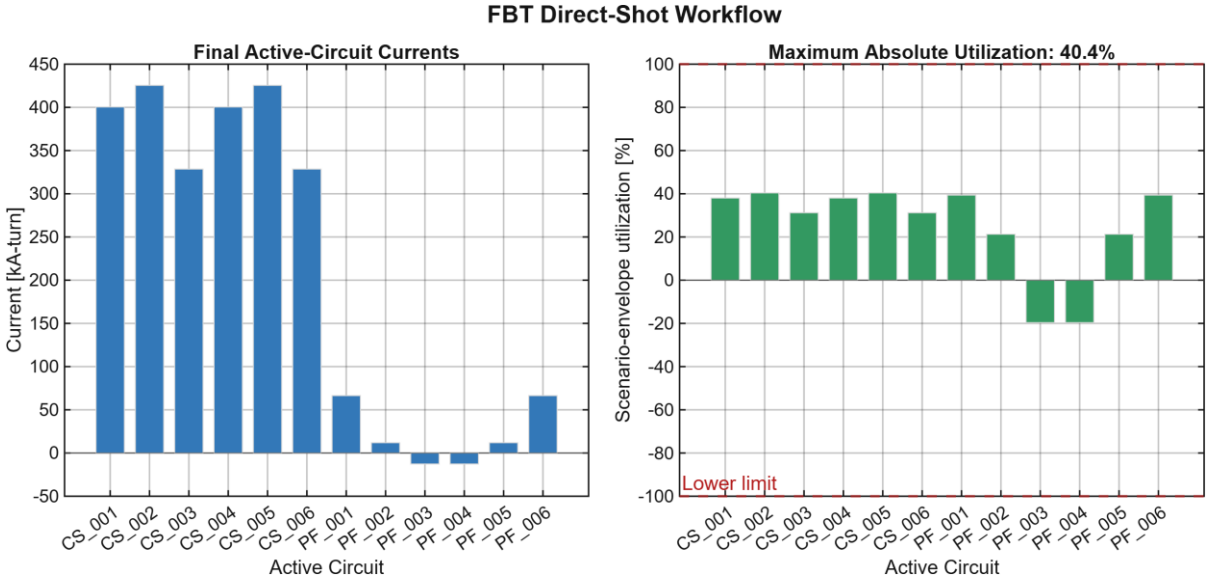



*Figure 4.2. Final active-circuit currents and utilisation of the common scenario envelope [12].*

### 4.2.2 FBT Limiter

The limiter case considers a static FBT inverse equilibrium at 110 kA. The target boundary has geometric centre $R_0$ = 0.290 m and $Z_0$ = 0, minor radius $a$ = 0.120 m, elongation $\kappa$ = 1.5 and triangularity $\delta$ = 0.1. It is represented by 39 control points. These dimensions describe the target contour. The toroidal-field parameter is $RB_t$ = 0.32 T m. The calculation uses a 66 × 65 grid and adjusts 12 active-circuit currents, without circuit evolution [12].

The active profile constraints are the plasma current, the axial safety factor $q_A$ = 3 and the poloidal beta $\beta_p$ = 0.1245. All active-current references are zero, with a finite error scale of 52.65 kA-turn. The flux-target error scale is $1.575 \times 10^{-3}$ Wb. These weights allow the inverse calculation to adjust the coil currents while fitting the prescribed boundary [12].

The calculation converged in 15 iterations, with a residual of $6.49 \times 10^{-7}$ below the prescribed tolerance of $10^{-6}$. The resulting plasma current is 110.000 kA. The magnetic axis is at $R$ = 0.301101 m and $Z$ = 0.0000254 m. The equilibrium is limited on the inboard side, with contact at $R$ = 0.184999 m and $Z$ = 0.0001698 m, and no boundary X-point. Table 4.3 gives the principal equilibrium quantities; $q_A$ and $\beta_p$ reproduce their prescribed values to the reported precision.

*Table 4.3. Principal quantities of the limiter FBT equilibrium [12].*

| Quantity | Result |
|---|---|
| Poloidal beta, $\beta_p$ | 0.1245 |
| Normalised internal inductance, $l_i(3)$ | 0.6961 |
| Safety factor on axis, $q_A$ | 3.0000 |
| Plasma kinetic energy, $W_k$ | 227.167 J |
| Plasma volume, $V_p$ | 0.095696 m³ |
| Toroidal beta, $\beta_t$ | 0.3179% |
| Maximum absolute active current | 51.502 kA-turn (PF4) |
| Maximum scenario-envelope utilisation | 79.9% |

The flux-target residuals, measured relative to the common flux value fitted at the control points, have an RMS of $2.29 \times 10^{-4}$ Wb and a maximum absolute value of $5.50 \times 10^{-4}$ Wb. These measure the fit of the flux constraints, rather than geometric distances. The fitted flux is lower than the equilibrium boundary flux by $4.18 \times 10^{-3}$ Wb. The requested control points extend to $R$ = 0.1703 m, whereas limiter contact occurs at $R \approx$ 0.185 m. The final boundary therefore differs from the requested contour on the inboard side.

Figure 4.3 shows the resulting limiter equilibrium, while Figure 4.4 reports the active-circuit currents and their utilisation of the common scenario envelope. PF4 carries the largest absolute current, 51.502 kA-turn, corresponding to a utilisation of 79.9%. All active-circuit currents remain within the common scenario envelope and also satisfy the FBT optimisation bounds of ±800 kA-turn for the CS circuits and ±200 kA-turn for the PF circuits.

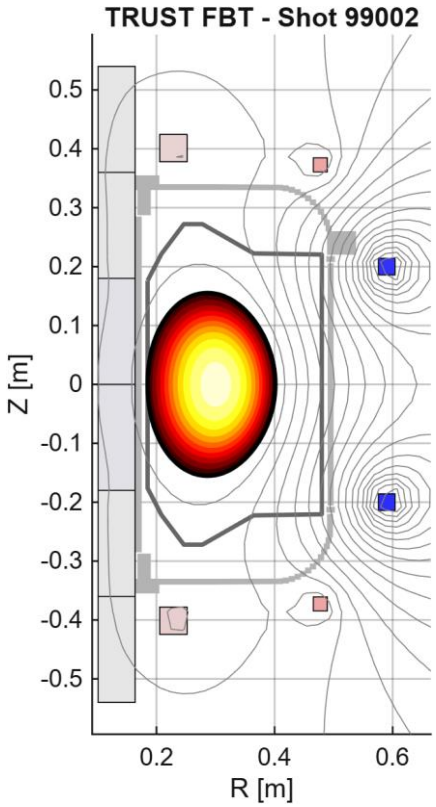



*Figure 4.3. Limiter FBT equilibrium obtained from the TRUST topology template [12].*

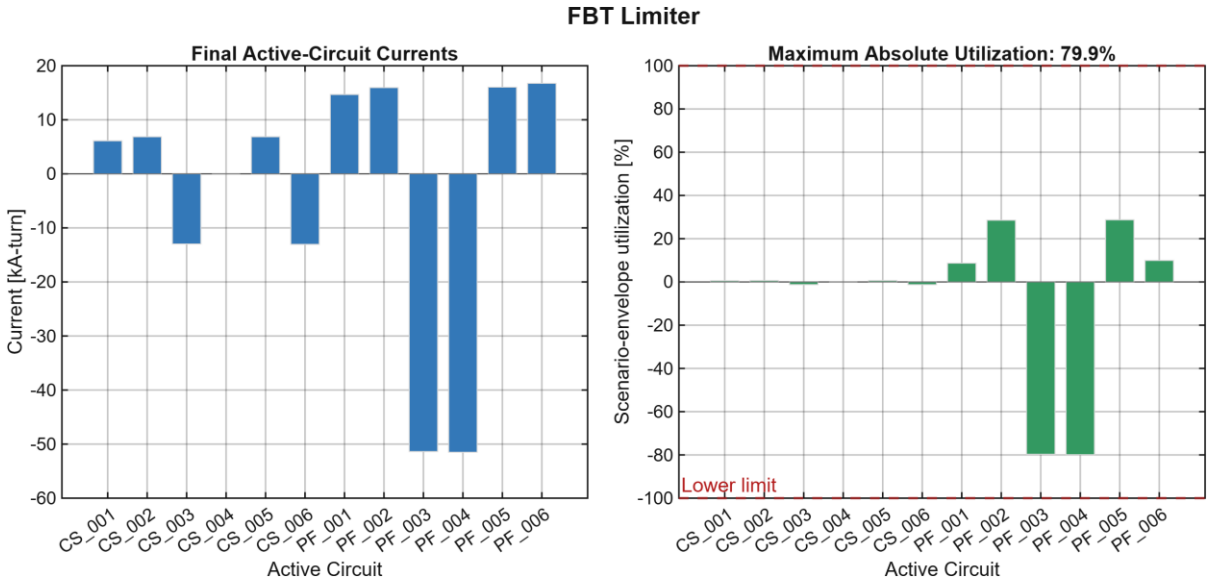



*Figure 4.4. Final active-circuit currents and utilisation of the common scenario envelope [12].*

### 4.2.3 FBT Single Null

The single-null template defines a static FBT inverse equilibrium calculation at 110 kA, with a prescribed lower X-point. The target contour has geometric centre $R_0$ = 0.318 m and $Z_0$ = −0.030 m, minor radius $a$ = 0.100 m, elongation $\kappa$ = 1.8 and triangularity $\delta$ = 0.35. The input contains 37 regular boundary control points and one X-point target at $R$ = 0.2426 m and $Z$ = −0.1987 m. The toroidal-field parameter is $RB_t$ = 0.32 T m. The calculation uses a 66 × 65 grid and 12 active circuits, without circuit evolution [12].

The active profile constraints are $I_p$ = 110 kA, $q_A$ = 3 and $\beta_p$ = 0.1245. At the prescribed X-point, the poloidal-field components are constrained to zero and the flux is constrained to the common fitted boundary value. The regular flux targets have an error scale of $1.575 \times 10^{-3}$ Wb. All active-current references are zero, with an error scale of 52.65 kA-turn [12].

The calculation converged in 19 iterations, with a residual of $6.10 \times 10^{-7}$ below the tolerance of $10^{-6}$. The resulting plasma current is 110.000 kA and the magnetic axis is at $R$ = 0.323889 m and $Z$ = −0.022435 m. The boundary passes through the lower X-point at $R$ = 0.242532 m and $Z$ = −0.198537 m. A second null above the midplane has a different flux and does not belong to the plasma boundary. Table 4.4 summarises the principal results; $q_A$ and $\beta_p$ reproduce their targets to the reported precision.

*Table 4.4. Principal quantities of the single-null FBT equilibrium [12].*

| Quantity | Result |
|---|---|
| Poloidal beta, $\beta_p$ | 0.1245 |
| Normalised internal inductance, $l_i(3)$ | 0.6838 |
| Safety factor on axis, $q_A$ | 3.0000 |
| Safety factor at 95% normalised poloidal flux, $q_{95}$ | 3.4810 |
| Plasma kinetic energy, $W_k$ | 227.167 J |
| Plasma volume, $V_p$ | 0.102957 m³ |
| Toroidal beta, $\beta_t$ | 0.3391% |
| Maximum absolute active current | 134.916 kA-turn (CS6) |
| Maximum scenario-envelope utilisation | 132.0% (PF3) |

For the 37 regular flux constraints, the RMS residual is $1.80 \times 10^{-3}$ Wb and the maximum absolute residual is $4.60 \times 10^{-3}$ Wb. These are flux differences relative to the fitted value, rather than geometric distances. The fitted flux differs from the equilibrium boundary flux by only $2.21 \times 10^{-6}$ Wb, but the larger residuals at the regular control points show that the requested contour is not reproduced exactly. The input shape parameters should therefore not be interpreted as measured dimensions of the final boundary.

Figure 4.5 shows the equilibrium and Figure 4.6 the current demand. The maximum absolute current is 134.916 kA-turn in CS6. PF3 and PF5 exceed the assessment limits: their currents are −85.071 and −71.698 kA-turn, corresponding to 132.0% and 128.4% utilisation. These limits coincide with the common scenario envelope. All currents satisfy the FBT optimisation bounds of ±800 kA-turn for CS and ±200 kA-turn for PF. The single-null equilibrium therefore satisfies the optimisation bounds used by FBT, while exceeding the adopted scenario envelope in PF3 and PF5.

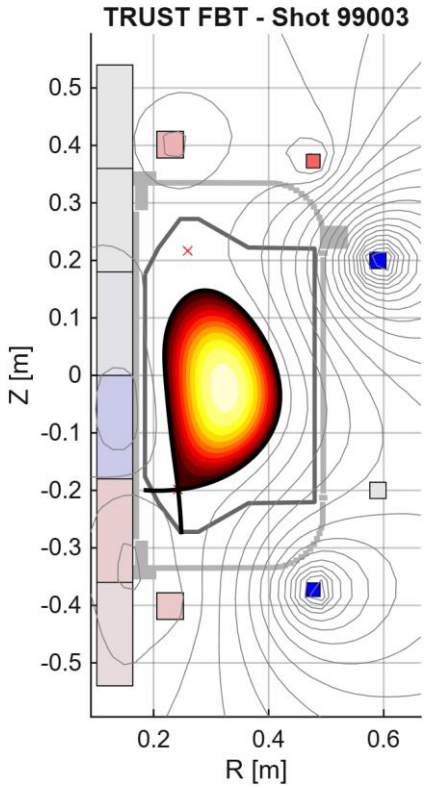



*Figure 4.5. Single-null FBT equilibrium with the lower X-point defining the plasma boundary [12].*

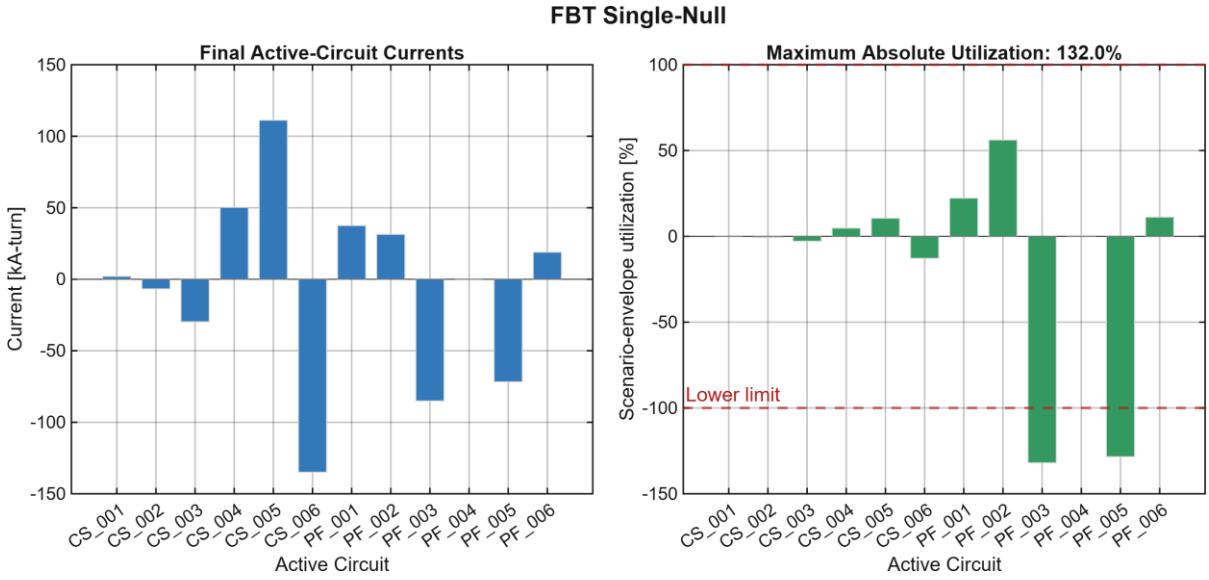



*Figure 4.6. Final active-circuit currents and utilisation of the common scenario envelope. PF3 and PF5 exceed their assessment limits [12].*

### 4.2.4 FBT Double Null

The double-null template defines a static FBT inverse equilibrium calculation at 110 kA, with prescribed upper and lower X-points. The target contour has geometric centre $R_0$ = 0.318 m and $Z_0$ = 0, minor radius $a$ = 0.100 m, elongation $\kappa$ = 1.8 and triangularity $\delta$ = 0.35. The input contains 35 regular boundary control points and two X-point targets at $R$ = 0.2426 m and $Z$ = ±0.1987 m. The toroidal-field parameter is $RB_t$ = 0.32 T m. The calculation uses a 66 × 65 grid and 12 active circuits, without circuit evolution [12].

The active profile constraints are $I_p$ = 110 kA, $q_A$ = 3 and $\beta_p$ = 0.1245. At both prescribed X-points, the poloidal-field components are constrained to zero and the flux is constrained to the same fitted boundary value. The regular flux targets have an error scale of $1.575 \times 10^{-3}$ Wb. All active-current references are zero, with an error scale of 52.65 kA-turn [12].

The calculation converged in 15 iterations, with a residual of $4.56 \times 10^{-7}$ below the tolerance of $10^{-6}$. The resulting plasma current is 110.000 kA and the magnetic axis is at $R$ = 0.321583 m and $Z$ = 0.0000668 m. The lower and upper X-points are at ($R$, $Z$) = (0.242502, −0.198534) m and (0.242487, 0.198532) m. Their fluxes differ by $1.96 \times 10^{-7}$ Wb, with the lower null defining the calculated boundary flux. The result is therefore close to a balanced double-null configuration, with a small residual flux imbalance. Table 4.5 summarises the principal results.

*Table 4.5. Principal quantities of the double-null FBT equilibrium [12].*

| Quantity | Result |
|---|---|
| Poloidal beta, $\beta_p$ | 0.1245 |
| Normalised internal inductance, $l_i(3)$ | 0.6829 |
| Safety factor on axis, $q_A$ | 3.0000 |
| Safety factor at 95% normalised poloidal flux, $q_{95}$ | 4.4856 |
| Plasma kinetic energy, $W_k$ | 227.167 J |
| Plasma volume, $V_p$ | 0.112322 m³ |
| Toroidal beta, $\beta_t$ | 0.3014% |
| Maximum absolute active current | 83.588 kA-turn (CS6) |
| Maximum scenario-envelope utilisation | 72.1% (PF2) |

For the 35 regular flux constraints, the RMS residual is $6.01 \times 10^{-4}$ Wb and the maximum absolute residual is $1.24 \times 10^{-3}$ Wb. These measure flux differences relative to the fitted value, rather than geometric distances. The fitted flux differs from the equilibrium boundary flux by $1.75 \times 10^{-6}$ Wb. Although the two null fluxes are closely matched, the finite residuals at the regular control points show that the requested contour is not reproduced exactly.

Figure 4.7 shows the resulting double-null equilibrium, while Figure 4.8 reports the active-current demand. The largest absolute current is 83.588 kA-turn in CS6, whereas PF2 reaches the highest scenario-envelope utilisation at 72.1%. All active-circuit currents remain within both the common scenario envelope and the FBT optimisation bounds of ±800 kA-turn for the CS circuits and ±200 kA-turn for the PF circuits. For the targets considered here, the maximum utilisation is substantially lower than in the single-null case, where PF3 reached 132.0%.

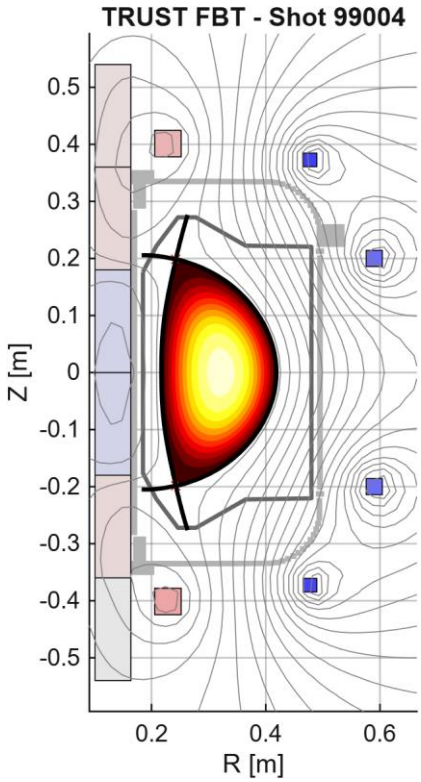



*Figure 4.7. Nearly balanced double-null FBT equilibrium with prescribed upper and lower X-points [12].*

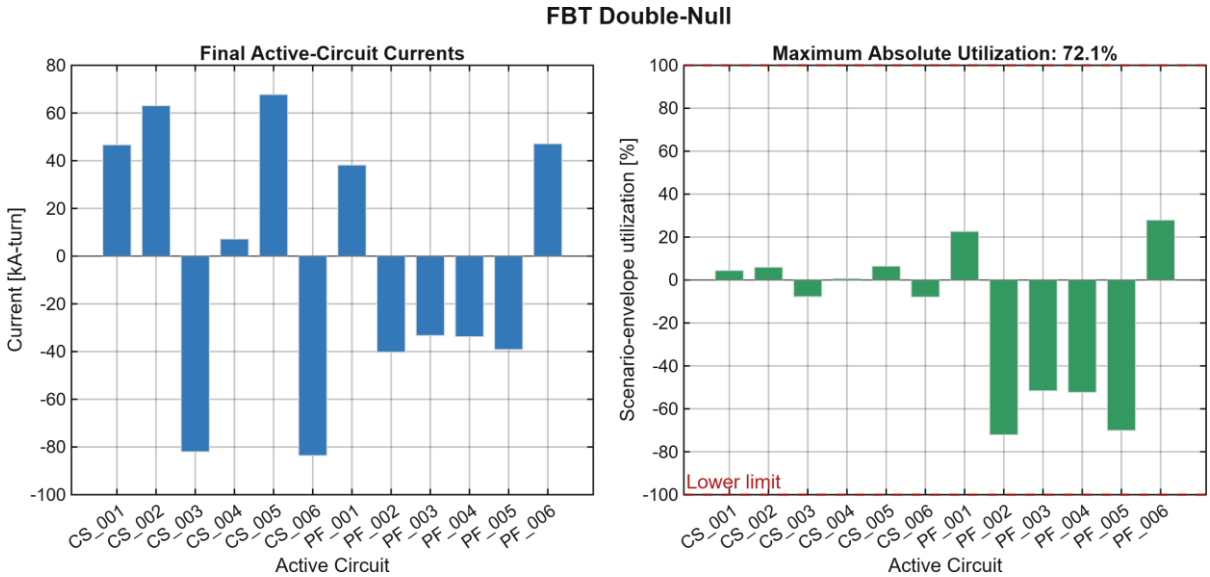



*Figure 4.8. Final active-circuit currents and utilisation of the common scenario envelope. All currents remain within the assessment limits [12].*

### 4.2.5 FBT from gEQDSK Breakdown

This example constructs a static 15 kA FBT equilibrium from the boundary of the 110 kA equilibrium in trust24bd.eqdsk. The imported plasma contour is resampled into 64 boundary control points. Active-current references are supplied separately from the TRUST current scenario. The calculation uses a 66 × 65 grid and 12 active circuits. The 93 passive-conductor currents are constrained to zero, and circuit evolution is disabled [12].

The active profile constraints are $I_p$ = 15 kA, $q_A$ = 6.8931 and $\beta_p$ = 0.063. The axial safety-factor target is obtained by scaling the imported value inversely with plasma current. The toroidal-field parameter is retained at $RB_t$ = 0.240 T m. Error scales of 0.500 kA-turn for the current references and $2.147 \times 10^{-4}$ Wb for the flux targets define the relative weighting of the supplied currents and the imported contour [12].

The calculation converged in 32 iterations, with a residual of $7.44 \times 10^{-7}$ below the tolerance of $10^{-6}$. The resulting plasma current is 15.000 kA and the magnetic axis is at $R$ = 0.282182 m and $Z$ = 0.0001200 m. Although the imported contour comes from a limited equilibrium, the reconstructed boundary is determined by an inboard X-point at $R$ = 0.196182 m and $Z$ = −0.0000531 m. Table 4.6 summarises the principal results.

*Table 4.6. Principal quantities of the gEQDSK-initialised 15 kA FBT equilibrium [12].*

| Quantity | Result |
|---|---|
| Poloidal beta, $\beta_p$ | 0.0630 |
| Normalised internal inductance, $l_i(3)$ | 0.6251 |
| Safety factor on axis, $q_A$ | 6.8931 |
| Safety factor at 95% normalised poloidal flux, $q_{95}$ | 4.6467 |
| Plasma kinetic energy, $W_k$ | 2.1375 J |
| Plasma volume, $V_p$ | 0.023316 m³ |
| Toroidal beta, $\beta_t$ | 0.02031% |
| Maximum absolute active current | 396.025 kA-turn (CS1–CS6) |
| Maximum scenario-envelope utilisation | 37.6% (CS1–CS6) |

The 64 flux-target residuals, measured relative to the common fitted flux, have an RMS of $7.10 \times 10^{-4}$ Wb and a maximum absolute value of $2.27 \times 10^{-3}$ Wb. The fitted flux is lower than the equilibrium boundary flux by $2.08 \times 10^{-3}$ Wb. Together with the appearance of the inboard X-point, these differences show that the final FBT equilibrium does not preserve the imported boundary exactly. Figure 4.9 compares the resulting boundary with the imported target and the final currents with their supplied references.

The current-reference mismatch has an RMS value of 2.029 kA-turn and reaches a maximum of 4.657 kA-turn in PF4. The six CS circuits carry approximately 396.025 kA-turn each. As shown in Figure 4.10, the maximum scenario-envelope utilisation is 37.6%, and all active-circuit currents remain within both the common scenario envelope and the FBT optimisation bounds of ±800 kA-turn for the CS circuits and ±200 kA-turn for the PF circuits. The case provides a low-current equilibrium target for subsequent calculations; gas ionisation and plasma-current formation are outside the model.

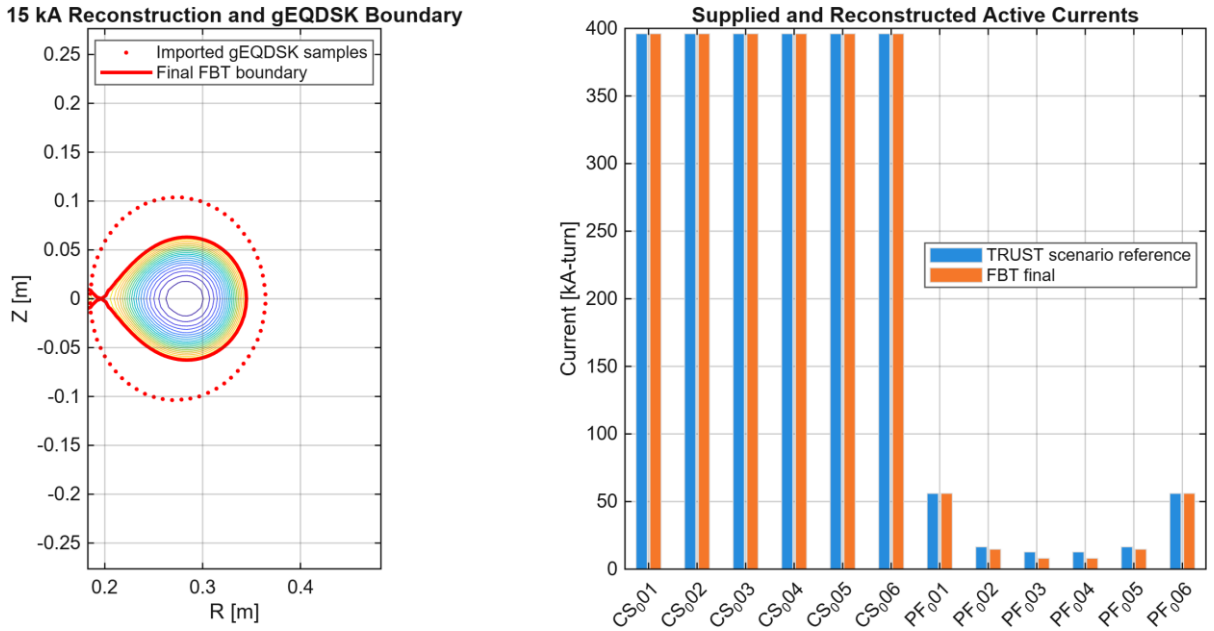



*Figure 4.9. Final 15 kA FBT boundary (red contour) and imported gEQDSK boundary samples (red points), left; final FBT currents and supplied TRUST scenario current references, right [12].*

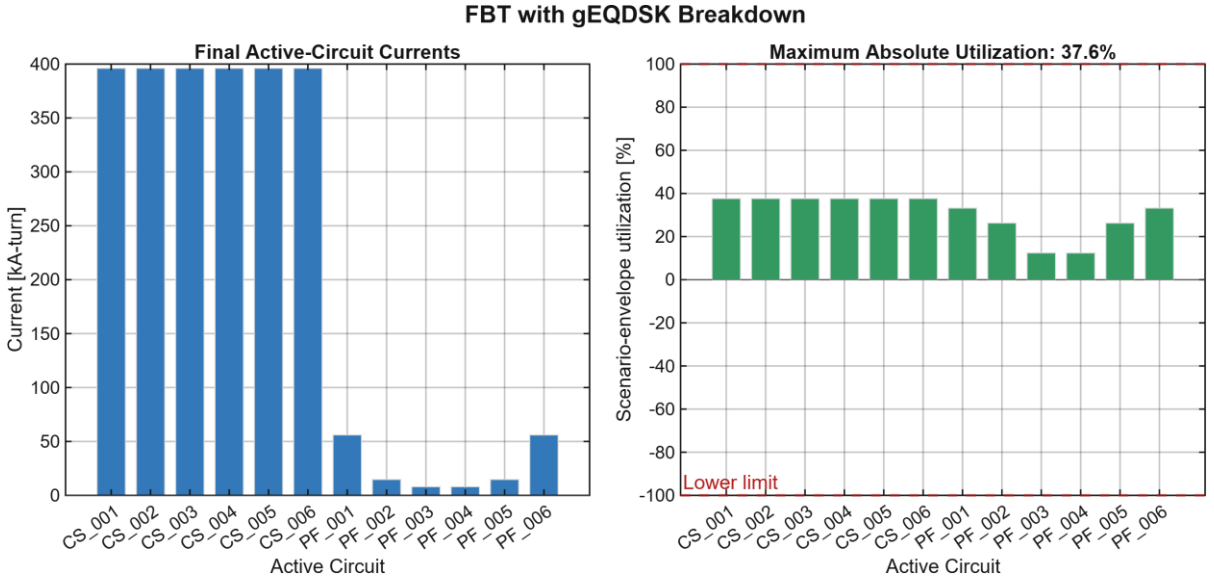



*Figure 4.10. Final active-circuit currents and utilisation of the common scenario envelope. All currents remain within the assessment limits [12].*

### 4.2.6 FBT with MAXFEA Boundary

This example transfers a precomputed MAXFEA separatrix and global plasma targets into a static FBT inverse problem at 110 kA. The imported contour is sampled at 96 positions; after removing samples near the prescribed lower X-point, the effective input contains 94 regular flux targets and one X-point target at $R$ = 0.27302051 m and $Z$ = −0.22406231 m. The calculation uses 12 active circuits and a 66 × 65 grid spanning $R$ = 0.10–0.65 m and $Z$ = −0.60–0.60 m, without circuit evolution [12].

The active profile constraints are $I_p$ = 110 kA, $q_A$ = 1.3810902 and $\beta_p$ = 0.11507934, with $RB_t$ = 0.240 T m. At the prescribed X-point, both poloidal-field components are constrained to zero and the flux is constrained to the common fitted boundary value. The regular flux-target error scale is $1.244 \times 10^{-3}$ Wb. The MAXFEA currents, including the virtual-control contributions mapped to physical PF circuits, provide an initial guess; deviations from these currents are not penalised in the fit [12].

The calculation converged in 20 iterations, with a residual of $6.46 \times 10^{-7}$ below the tolerance of $10^{-6}$. The resulting plasma current is 110.000 kA and the magnetic axis is at $R$ = 0.319147 m and $Z$ = −0.0009910 m. The boundary is determined by the lower X-point at $R$ = 0.271192 m and $Z$ = −0.224047 m. Table 4.7 summarises the principal results.

*Table 4.7. Principal quantities of the FBT equilibrium from the MAXFEA boundary [12].*

| Quantity | Result |
|---|---|
| Poloidal beta, $\beta_p$ | 0.1151 |
| Normalised internal inductance, $l_i(3)$ | 0.8674 |
| Safety factor on axis, $q_A$ | 1.3811 |
| Safety factor at 95% normalised poloidal flux, $q_{95}$ | 3.2285 |
| Plasma kinetic energy, $W_k$ | 209.978 J |
| Plasma volume, $V_p$ | 0.128300 m³ |
| Toroidal beta, $\beta_t$ | 0.4312% |
| Maximum absolute active current | 573.846 kA-turn (CS1) |
| Maximum scenario-envelope utilisation | 118.6% (PF1) |

The 94 regular flux-target residuals have an RMS of $9.82 \times 10^{-5}$ Wb and a maximum absolute value of $2.20 \times 10^{-4}$ Wb. These are measured relative to the fitted flux, which is $3.02 \times 10^{-4}$ Wb above the equilibrium boundary flux. Separately, evaluating the normalised flux on all 96 imported contour samples gives an RMS deviation from unity of $8.47 \times 10^{-3}$, with zero flux assigned to the axis and unity to the final boundary. This provides a dimensionless measure of the agreement between the imported contour and the final FBT boundary.

Figure 4.11 compares the processed and resampled MAXFEA contour with the final FBT boundary, and the mapped MAXFEA initial currents with the final FBT currents. The maximum current change is 589.023 kA-turn in CS5; the maximum absolute final current is 573.846 kA-turn in CS1. All currents satisfy the optimisation bounds of ±800 kA-turn for CS and ±200 kA-turn for PF. PF1 reaches its 200.000 kA-turn upper bound, corresponding to 118.6% utilisation of the common scenario limit. It is the only circuit exceeding the assessment limits shown in Figure 4.12. The final current distribution therefore differs considerably from the mapped MAXFEA initial values and exceeds the common scenario envelope in PF1.

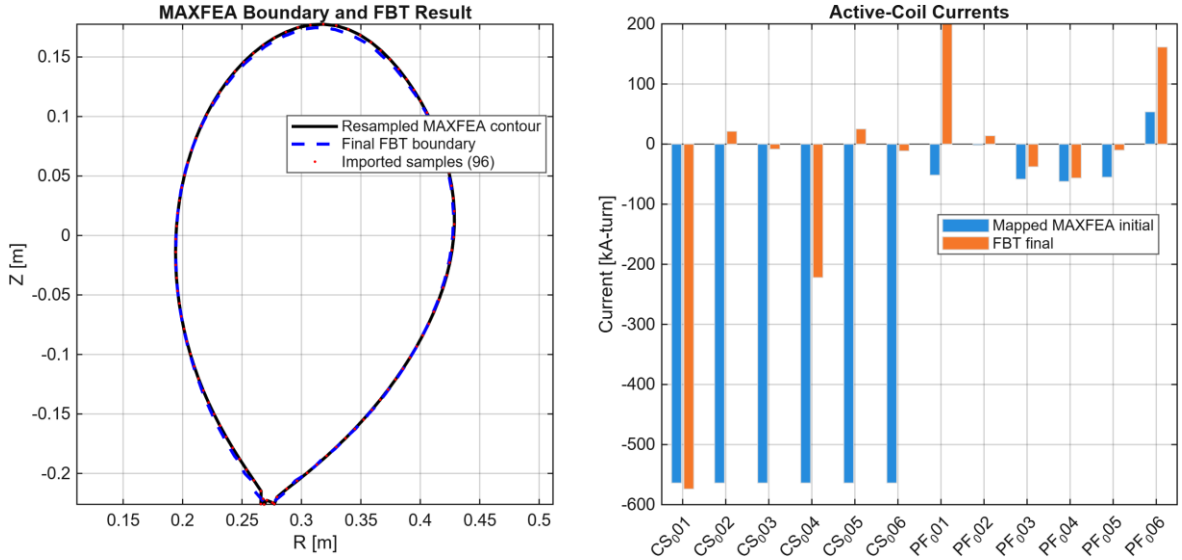



*Figure 4.11. Processed and resampled MAXFEA contour (black), its 96 imported samples (red points), and the final FBT boundary (blue dashed), left; mapped MAXFEA initial currents and final FBT currents in kA-turn, right. The displayed samples precede removal of two points near the prescribed X-point; the effective FBT input contains 94 regular flux targets and one X-point target [12].*

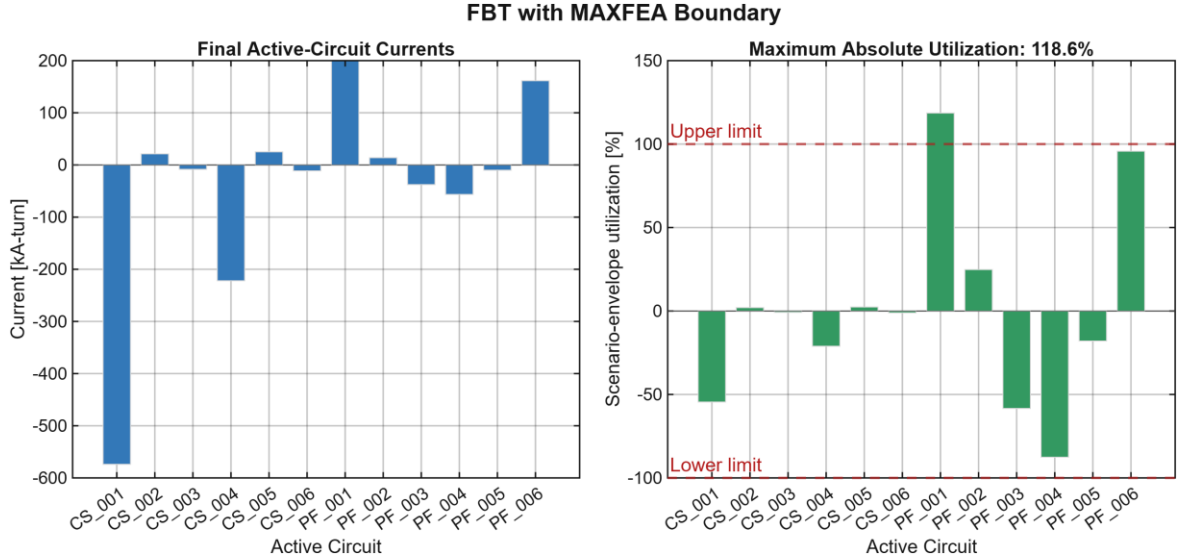



*Figure 4.12. Final active-circuit currents and utilisation of the common scenario envelope. PF1 exceeds its assessment limit [12].*

Across the six FBT cases, all convergence residuals remain below $10^{-6}$. The single-null template and the MAXFEA boundary-transfer case exceed the common scenario envelope, whereas the other four cases remain within it. The examples also show different levels of agreement with their prescribed or imported boundaries, particularly when external equilibrium data are transferred into FBT. In the two external-data cases, the imported boundary provides the target geometry while the active currents are free to adjust during the inverse calculation.

## 4.3 LIUQE RECONSTRUCTION

The LIUQE reconstruction method is described by Moret et al. [2] and summarised in Sections 2.5 and 2.7.2, while the TRUST-specific diagnostic geometry and reconstruction inputs are discussed in Section 3.5. The present section focuses on the reconstruction results obtained with the different synthetic input sets. The first three examples use FBT equilibria as reference states and generate the corresponding synthetic magnetic signals at the TRUST diagnostic locations. The remaining cases use the external-data interface to supply synthetic signals derived either from FBT or from MAXFEA equilibria. No artificial measurement noise is added in any of the eight reconstructions considered in this section. The reconstructed equilibria are compared with their source states in terms of magnetic signals, plasma geometry, global quantities, and, where relevant, internal profile parameters. Active-current demand is evaluated against the common TRUST scenario envelope and, separately, against the current limits stored in the LIUQE model; the latter are model-local values rather than TRUST hardware limits.

### 4.3.1 FBT-LIUQE Limiter

The limiter reconstruction uses the FBT equilibrium of Section 4.2.2 as its reference at $t$ = 0 s, with $I_p = 110kA$ and $RB_t = 0.32\,T\cdot m$. The synthetic diagnostic set consists of three flux-loop signals, twelve magnetic-probe signals, twelve active-circuit currents, the plasma current, and the diamagnetic flux. The LIUQE model also includes 93 passive circuits with zero reference currents and finite fitting weights. Both equilibria are calculated on a 66 × 65 grid. The reconstruction uses the LIH initialisation available in MEQ. In this procedure, the supplied measurements are used to construct an initial estimate of the plasma current density that seeds the LIUQE iterations, consistently with the measurement-based initialisation described by Moret et al. [2]. The converged FBT nonlinear state is therefore not passed directly to LIUQE. No artificial noise is added to the signals, while finite measurement weights and regularisation are retained in the fit. The profile coefficients are reconstructed without imposing the FBT values of $q_A\ or\ \beta_p$ [12].

LIUQE converges in 15 iterations, with a numerical residual of $3.38 \times 10^{-5}$ and a weighted fit indicator $\chi = 2.18 \times 10^{-2}$. The equilibrium remains limiter-bounded, with contact at $R$ = 0.185 m and $Z$ = 0.000131 m. Table 4.8 compares the principal results with the FBT reference.

*Table 4.8. Principal quantities of the FBT limiter reference and its LIUQE reconstruction [12].*

| Quantity | FBT reference | LIUQE reconstruction |
|---|---|---|
| Plasma current, $I_p$ [kA] | 110.000 | 109.562 |
| Magnetic-axis radius, $R$ [m] | 0.301101 | 0.302587 |
| Magnetic-axis height, $Z$ [m] | 0.0000254 | 0.0000290 |
| On-axis safety factor, $q_A$ | 3.0000 | 1.2527 |
| Poloidal beta, $\beta_p$ | 0.1245 | 0.0719 |
| Normalised internal inductance, $l_i(3)$ | 0.6961 | 0.8347 |
| Plasma kinetic energy, $W_k$ [J] | 227.167 | 130.234 |
| Plasma volume, $V_p$ [m³] | 0.095696 | 0.094818 |
| Toroidal beta, $\beta_t$ [%] | 0.3179 | 0.1853 |

Relative to reference: $I_p$ −0.438 kA (−0.40%); magnetic axis +1.486 mm outward; volume −0.92%; $qA$ −58.2%, $\beta p$ −42.2%, kinetic energy −42.7%, $l_i(3)$ +19.9%. RMS differences: loops $3.60 \times 10^{-4}$ Wb, probes $2.18 \times 10^{-4}$ T; diamagnetic-flux difference $5.02 \times 10^{-5}$ Wb. Current and geometry match better than internal quantities, sensitive to measurements, profile parameterisation and regularisation. Signal errors are distinct from numerical convergence residual and weighted fit indicator.

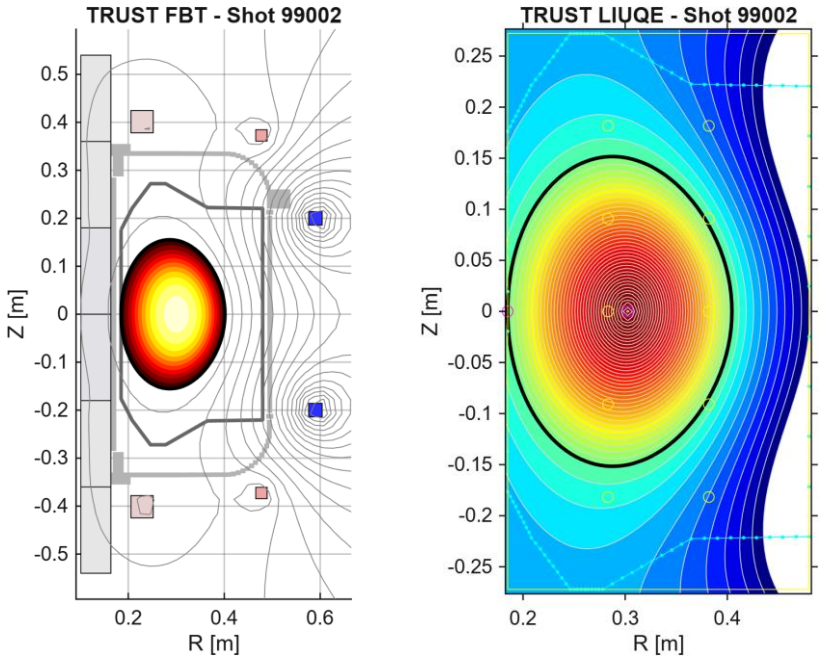



*Figure 4.13. FBT limiter reference used to generate the synthetic measurements (left) [12].*

*Figure 4.14. LIUQE reconstruction of the synthetic limiter equilibrium (right). The two views use different spatial windows. In the LIUQE views, magenta circles mark the magnetic axis and boundary-defining point, yellow circles mark finite-element current locations, and white crosses identify magnetic nulls where present [12].*

The active currents remain close to their reference values: the RMS change is 0.792 A-turn and the maximum change is 1.715 A-turn in PF4. The largest absolute current is 51.504 kA-turn, also in PF4. All active currents satisfy both the common scenario envelope and the stored LIUQE model-local current limits. Their maximum utilisation is 79.9% and 25.8%, respectively: these percentages refer to the same reconstructed currents assessed against different limits. The fitted passive currents remain small, with a maximum magnitude of 4.208 A.

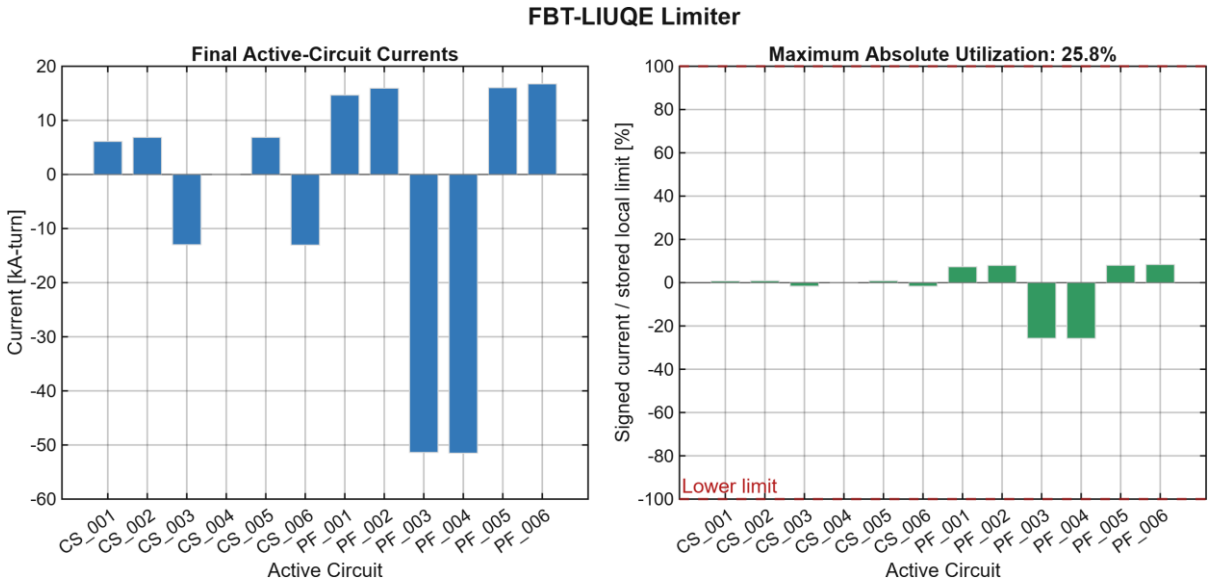



*Figure 4.15. Reconstructed active-circuit currents in kA-turn and signed current-to-limit ratios relative to the stored LIUQE model-local current limits. The common TRUST scenario envelope is assessed separately in the text [12].*

### 4.3.2 FBT-LIUQE Single Null

The single-null reconstruction uses the FBT equilibrium of Section 4.2.3 as its reference at $t$ = 0 s, with $I_p$ = 110 kA and $RB_t$ = 0.32 T m. The diagnostic inputs and weighting scheme are the same as in the limiter example: three flux loops, twelve magnetic probes, twelve active-circuit currents, the plasma current and the diamagnetic flux. LIUQE also fits 93 passive-circuit currents with zero reference values. Both equilibria use a 66 × 65 grid. The reconstruction uses the same LIH initialisation described in the previous example; the converged FBT nonlinear state is not used directly to initialise LIUQE. No noise is added, and the profile coefficients are fitted with regularisation, without imposing the FBT values of $q_A$ and $\beta_p$ [12].

LIUQE converges in 45 iterations, with a numerical residual of $4.02 \times 10^{-5}$ and a weighted fit indicator $\chi = 1.61 \times 10^{-3}$. The lower X-point defines the separatrix in both equilibria. A second, upper magnetic null lies at a different flux and does not make the configuration double-null. Table 4.9 compares the principal results.

*Table 4.9. Principal quantities of the FBT single-null reference and its LIUQE reconstruction [12].*

| Quantity | FBT reference | LIUQE reconstruction |
|---|---|---|
| Plasma current, $I_p$ [kA] | 110.000 | 110.011 |
| Magnetic-axis radius, $R$ [m] | 0.323889 | 0.324322 |
| Magnetic-axis height, $Z$ [m] | −0.022435 | −0.022165 |
| Lower X-point radius, $R$ [m] | 0.242532 | 0.241959 |
| Lower X-point height, $Z$ [m] | −0.198537 | −0.199109 |
| On-axis safety factor, $q_A$ | 3.0000 | 1.6850 |
| Poloidal beta, $\beta_p$ | 0.1245 | 0.1201 |
| Normalised internal inductance, $l_i(3)$ | 0.6838 | 0.6807 |
| Plasma kinetic energy, $W_k$ [J] | 227.167 | 219.234 |
| Plasma volume, $V_p$ [m³] | 0.102957 | 0.102041 |
| Toroidal beta, $\beta_t$ [%] | 0.3391 | 0.3304 |

The plasma current increases by only 11.402 A (+0.0104%). The magnetic axis shifts by +0.433 mm in $R$ and +0.271 mm in $Z$, while the lower X-point moves by 0.809 mm in the poloidal plane. The reconstructed $\beta_p$ and plasma kinetic energy are approximately 3.5% below the reference, and $l_i(3)$ and plasma volume differ by less than 1%. However, $q_A$ decreases by 43.8%. The close agreement in geometry and global quantities therefore does not extend to the on-axis safety factor.

The RMS differences between the reconstructed and reference signals are $9.62 \times 10^{-6}$ Wb for the flux loops and $4.71 \times 10^{-5}$ T for the magnetic probes. The diamagnetic-flux difference is $1.05 \times 10^{-6}$ Wb. These signal differences quantify the magnetic fit and remain distinct from the numerical convergence residual.

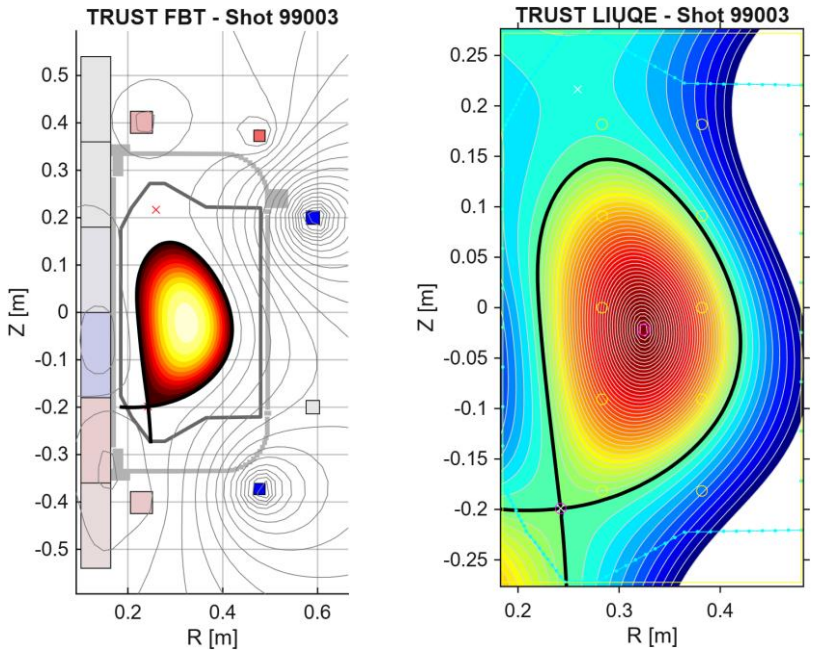



*Figure 4.16. FBT single-null reference used to generate the synthetic measurements (left) [12].*

*Figure 4.17. LIUQE reconstruction of the synthetic single-null equilibrium (right). The two views use different spatial windows [12].*

The RMS change in active currents is 0.033 A-turn, with a maximum of 0.091 A-turn in PF6. The largest absolute current is 134.916 kA-turn in CS6. All active currents satisfy the stored LIUQE model-local current limits, with a maximum local-limit utilisation of 42.5% in PF3. Against the common scenario envelope, however, PF3 and PF5 reach 132.0% and 128.4%, respectively. These exceedances belong to the reconstructed state itself: the close recovery of the FBT currents preserves the source case's current-limit violations. The fitted passive currents remain small, with a maximum magnitude of 0.612 A.

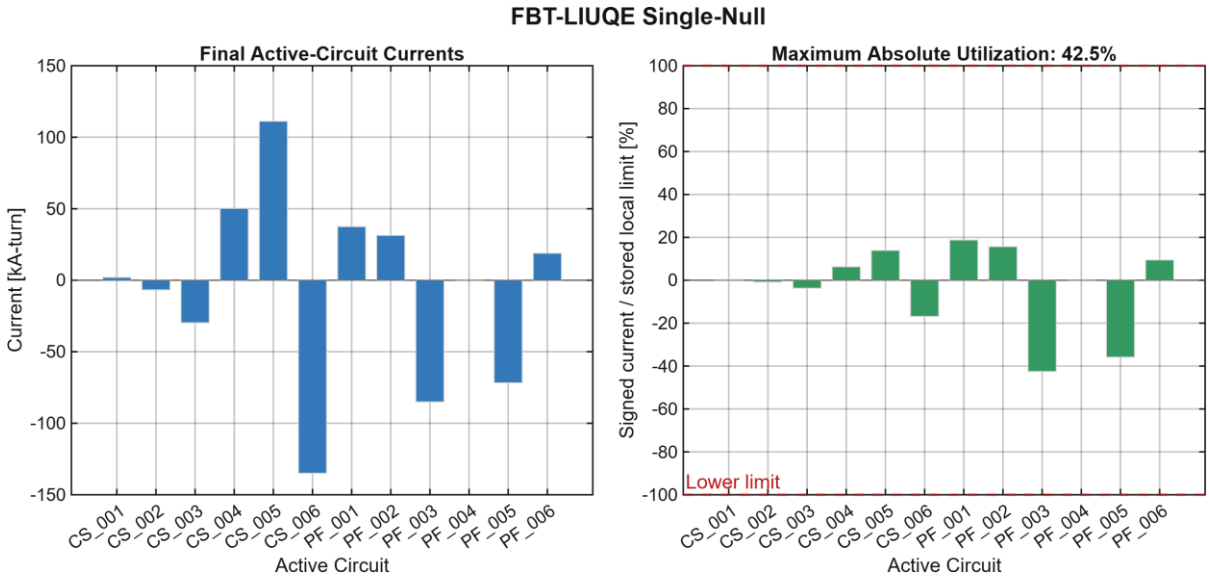



*Figure 4.18. Reconstructed active-circuit currents in kA-turn and signed current-to-limit ratios relative to the stored LIUQE model-local current limits. The common TRUST scenario envelope, exceeded in PF3 and PF5, is assessed separately in the text [12].*

### 4.3.3 FBT-LIUQE Double Null

The double-null reconstruction uses the FBT equilibrium of Section 4.2.4 as its reference at $t$ = 0 s, with $I_p$ = 110 kA and $RB_t$ = 0.32 T m. The diagnostic inputs and weighting scheme are unchanged from the preceding reconstructions: three flux loops, twelve magnetic probes, twelve active-circuit currents, the plasma current and the diamagnetic flux. LIUQE also fits 93 passive-circuit currents with zero reference values. Both equilibria use a 66 × 65 grid. The reconstruction uses the same LIH initialisation described in the previous example; the converged FBT nonlinear state is not used directly to initialise LIUQE. No noise is added, and the profile coefficients are fitted with regularisation, without imposing the FBT values of $q_A$ and $\beta_p$ [12].

LIUQE converges in 20 iterations, with a numerical residual of $3.33 \times 10^{-5}$ and a weighted fit indicator $\chi$ = $1.69 \times 10^{-3}$. Both equilibria contain a lower and an upper magnetic null. In each case, the lower null defines the reported boundary flux, while the upper-null flux differs from it by only $8.19 \times 10^{-6}$ of the plasma flux interval for FBT and $1.93 \times 10^{-5}$ for LIUQE. The reconstructed configuration therefore retains two magnetic nulls with the quantified small flux imbalance. Table 4.10 compares the principal results.

*Table 4.10. Principal quantities of the FBT double-null reference and its LIUQE reconstruction [12].*

| Quantity | FBT reference | LIUQE reconstruction |
|---|---|---|
| Plasma current, $I_p$ [kA] | 110.000 | 110.009 |
| Magnetic-axis radius, $R$ [m] | 0.321583 | 0.322089 |
| Magnetic-axis height, $Z$ [m] | 0.000067 | 0.000062 |
| Lower X-point radius, $R$ [m] | 0.242502 | 0.241975 |
| Lower X-point height, $Z$ [m] | −0.198534 | −0.199208 |
| Upper X-point radius, $R$ [m] | 0.242487 | 0.241952 |
| Upper X-point height, $Z$ [m] | 0.198532 | 0.199280 |
| On-axis safety factor, $q_A$ | 3.0000 | 1.6874 |
| Poloidal beta, $\beta_p$ | 0.1245 | 0.1183 |
| Normalised internal inductance, $l_i(3)$ | 0.6829 | 0.6875 |
| Plasma kinetic energy, $W_k$ [J] | 227.167 | 215.806 |
| Plasma volume, $V_p$ [m³] | 0.112322 | 0.111768 |
| Toroidal beta, $\beta_t$ [%] | 0.3014 | 0.2873 |

The plasma current increases by 9.393 A (+0.00854%). The magnetic axis shifts by +0.506 mm in $R$ and −0.005 mm in $Z$. The lower and upper nulls move by 0.855 mm and 0.920 mm, respectively, in the poloidal plane. The reconstructed $\beta_p$ and plasma kinetic energy are approximately 5.0% below the reference, while $l_i(3)$ increases by 0.68% and the plasma volume decreases by 0.49%. However, $q_A$ decreases by 43.8%. The close agreement in current, geometry and topology therefore does not imply equal internal profiles.

The RMS differences between reconstructed and reference signals are $1.17 \times 10^{-5}$ Wb for the flux loops and $4.94 \times 10^{-5}$ T for the magnetic probes. The stored diamagnetic-flux quantity differs by $8.44 \times 10^{-7}$ Wb. These values quantify the magnetic fit and remain distinct from the numerical convergence residual and the weighted fit indicator.

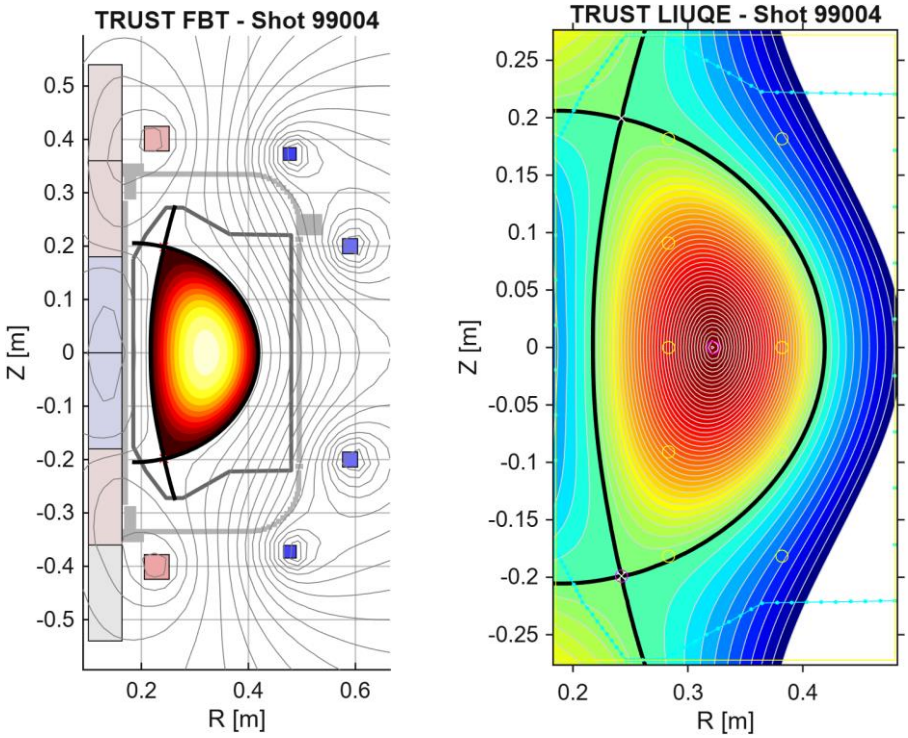



*Figure 4.19. FBT double-null reference used to generate the synthetic measurements (left) [12].*

*Figure 4.20. LIUQE reconstruction of the synthetic double-null equilibrium (right). The two views use different spatial windows [12].*

The RMS change in active currents is 0.030 A-turn, with a maximum of 0.058 A-turn. The largest absolute reconstructed current is 83.588 kA-turn in CS6. All active currents remain inside both sets of assessment limits: PF2 reaches the largest common-envelope utilisation, 72.1%, whereas PF6 reaches the largest stored model-local utilisation, 23.6%. The fitted passive currents remain below 0.406 A in magnitude. The difference between the two percentages reflects the different current limits used for the common scenario envelope and the stored LIUQE model.

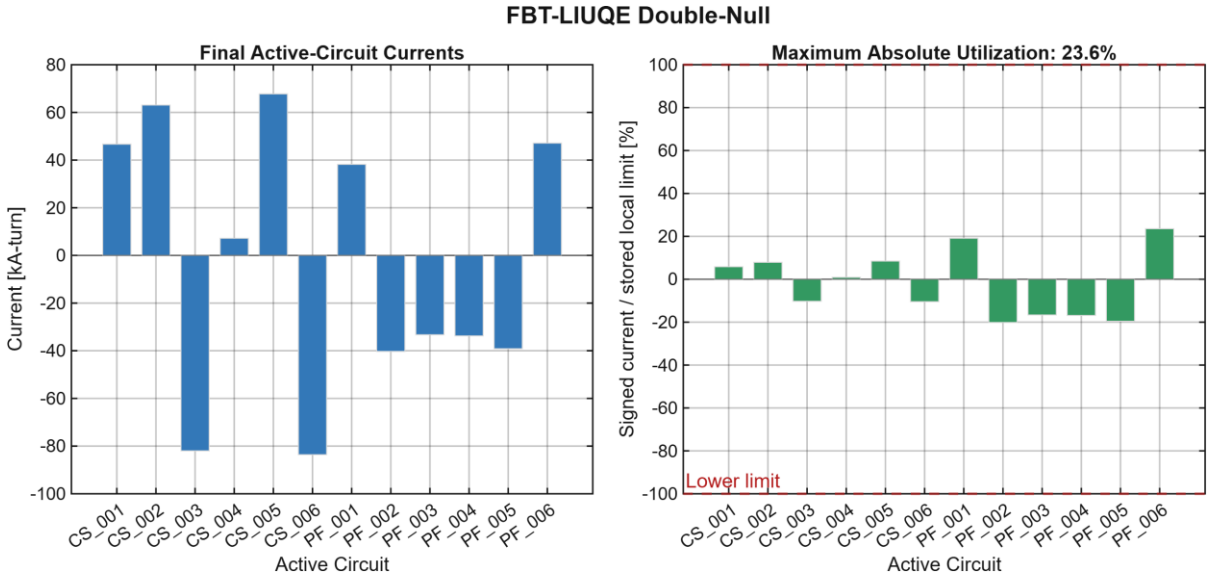



*Figure 4.21. Reconstructed active-circuit currents in kA-turn and signed current-to-limit ratios relative to the stored LIUQE model-local current limits. All currents also remain within the common TRUST scenario envelope, assessed separately in the text [12].*

### 4.3.4 LIUQE with External Measurements

This case examines the LIUQE reconstruction when the diagnostic data are supplied through the external-measurement interface. A*t* $t$ = 0 s, the input structure contains twelve active-circuit currents, ninety-three passive-circuit references, three flux-loop signals, twelve magnetic-probe signals, the plasma current, the diamagnetic signal, and the toroidal-field information. The data are generated from the FBT single-null equilibrium used in Section 4.3.2 and converted to the LIUQE input format before reconstruction. The supplied measurement structure is processed using the same LIH initialisation described in Section 4.3.1 [12].

LIUQE converges in 45 iterations, with a numerical residual of $4.02 \times 10^{-5}$ and a weighted fit indicator $\chi = 1.61 \times 10^{-3}$. The reconstructed plasma current is 110.011 kA; the magnetic axis is at $R$ = 0.324322 m and $Z$ = −0.022165 m. The main profile and integral quantities are $q_A$ = 1.6850, $\beta_p$ = 0.1201, $l_i(3)$ = 0.6807, $W_k$ = 219.23 J and $V_p$ = 0.102041 m³. These values, together with the convergence indicators and all twelve final active currents, agree with those obtained in the direct single-null reconstruction of Section 4.3.2 to the reported precision.

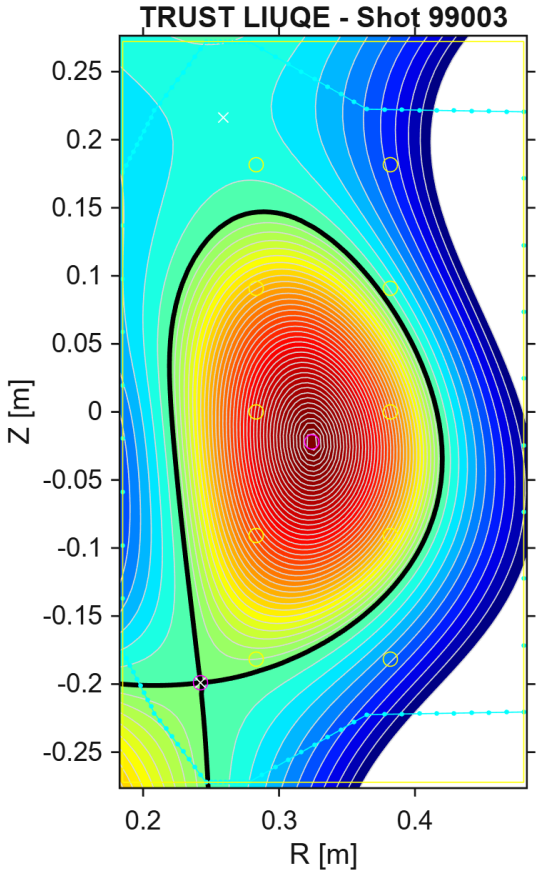



*Figure 4.22. LIUQE single-null equilibrium reconstructed from the synthetic external-measurement structure [12].*

The RMS fit differences are $9.62 \times 10^{-6}$ Wb for the flux loops and $4.71 \times 10^{-5}$ T for the magnetic probes. The maximum active-current difference from the supplied values is only 0.091 A-turn, and the fitted passive-current differences remain below 0.612 A. The largest reconstructed active current is 134.916 kA-turn in CS6. All currents satisfy the stored LIUQE model-local current limits; PF3 reaches 42.5% of its stored local current limit. As in the direct reconstruction of Section 4.3.2, PF3 and PF5 therefore exceed the common scenario envelope while remaining within the stored LIUQE model-local limits.

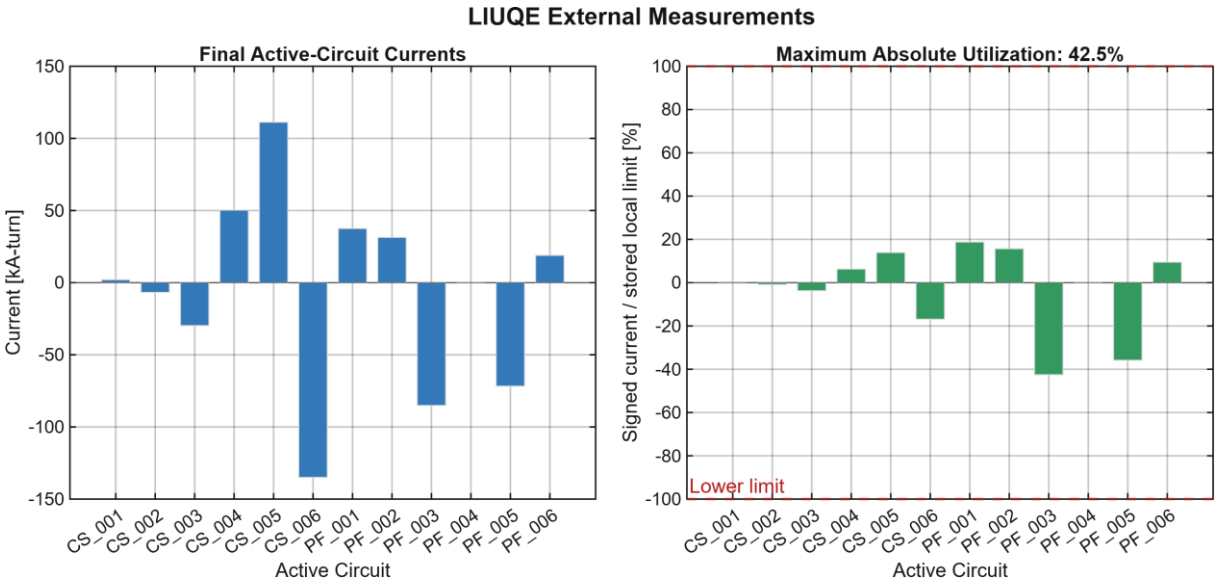



*Figure 4.23. Reconstructed active-circuit currents in kA-turn and signed current-to-limit ratios relative to the stored LIUQE model-local current limits. PF3 and PF5 exceed the distinct common TRUST scenario envelope discussed in the text [12].*

The agreement with the direct reconstruction of Section 4.3.2 confirms that supplying the same synthetic diagnostic data through the external-measurement interface leads to the same LIUQE equilibrium at the reported precision.

### 4.3.5 LIUQE with MAXFEA Data: Single Null

This case reconstructs a 110 kA MAXFEA single-null equilibrium using the CREATE-compatible LIUQE interface. The imported flux map provides 36 flux-loop signals and both magnetic-field components at 36 probe positions, giving 72 field signals. Twelve effective active-circuit currents include the MAXFEA virtual-control contributions. The imported MAXFEA flux map initialises the nonlinear state, rather than the LIH initialisation used in the FBT-based cases. This configuration uses a richer diagnostic set than the synthetic FBT cases; the diamagnetic term is disabled and the profile coefficients are fitted with regularisation, without imposing the source $q_A$ or $\beta_p$ [12].

LIUQE converges in 17 iterations with a numerical residual of $9.72 \times 10^{-5}$ and a weighted fit indicator $\chi$ = 1.414. Table 4.11 compares the principal quantities. The plasma-current difference is −0.645 A; the axis shifts by −0.733 mm in $R$ and +1.214 mm in $Z$, and the lower X-point moves by 1.377 mm in the poloidal plane. The reconstructed plasma kinetic energy is 159.563 J, the plasma volume is 0.127153 m³ and the toroidal beta is 0.3283%.

*Table 4.11. Principal quantities of the imported MAXFEA single-null equilibrium and its LIUQE reconstruction [12].*

| Quantity | MAXFEA reference | LIUQE reconstruction |
|---|---|---|
| Plasma current, $I_p$ [kA] | 110.000 | 109.999 |
| Magnetic-axis radius, $R$ [m] | 0.319690 | 0.318956 |
| Magnetic-axis height, $Z$ [m] | −0.001113 | 0.000101 |
| Lower X-point radius, $R$ [m] | 0.273021 | 0.271873 |
| Lower X-point height, $Z$ [m] | −0.224062 | −0.223302 |
| On-axis safety factor, $q_A$ | 1.3811 | 1.1027 |
| Poloidal beta, $\beta_p$ | 0.1151 | 0.0899 |
| Normalised internal inductance, $l_i(3)$ | 0.8038 | 0.8373 |

Figure 4.24 compares the equilibria over the same spatial window, with equal $R/Z$ aspect ratios and identical normalised-flux levels. The separatrix and internal contours are overlaid directly. Figure 4.25 compares the diagnostic signals: the loop RMS difference is $3.81 \times 10^{-4}$ Wb after removing the mean additive flux offset ($-3.83 \times 10^{-4}$ Wb), versus $5.40 \times 10^{-4}$ Wb without alignment. The $B_r$ and $B_z$ RMS differences are $5.91 \times 10^{-3}$ T and $5.61 \times 10^{-3}$ T. These RMS values describe the agreement of the reconstructed magnetic signals with the MAXFEA-derived inputs, whereas $\chi$ and the convergence residual refer to the LIUQE fitting and iterative solution.

The active currents reproduce the supplied values to numerical precision (RMS difference $2.71 \times 10^{-10}$ kA-turn). Their maximum magnitude is 563.885 kA-turn in the CS circuits. All lie inside the common scenario envelope, with a maximum utilisation of 98.6% in PF5. The stored LIUQE adapter-local current limits instead give 1253.1% in the CS circuits because they store ±45 kA-turn for CS and ±48 kA-turn for PF. The large difference between the two utilisation values reflects the different definitions of the two limit sets; the stored LIUQE values are not treated here as TRUST hardware limits.

Larger differences appear in the reconstructed profile quantities. The values of $q_A$ and $\beta_p$ are 20.2% and 21.9% below the MAXFEA reference, respectively. The reconstructed pressure reaches a minimum of −0.126 kPa, so the admissibility of the fitted pressure profile remains unresolved. The calculation also records 17 near-singular-matrix warnings.

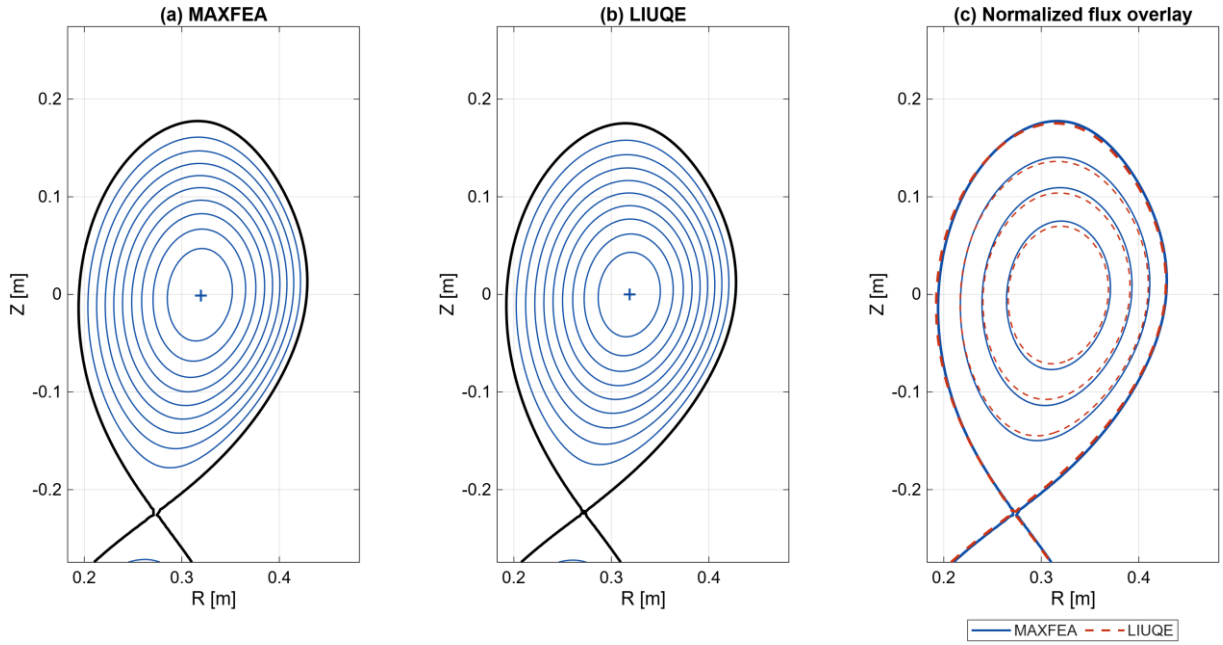



*Figure 4.24. MAXFEA and LIUQE single-null equilibria on identical R/Z axes. Panels (a–b): $\psi_N$ = 0.1 to 0.9, with $\psi_N$ = 1 in black and the magnetic axis marked by a cross. Panel (c): overlaid $\psi_N$ = 0.25, 0.50, 0.75 and 1 contours. Each map is normalised with its own axis and boundary flux; the view is restricted to the LIUQE grid [12].*

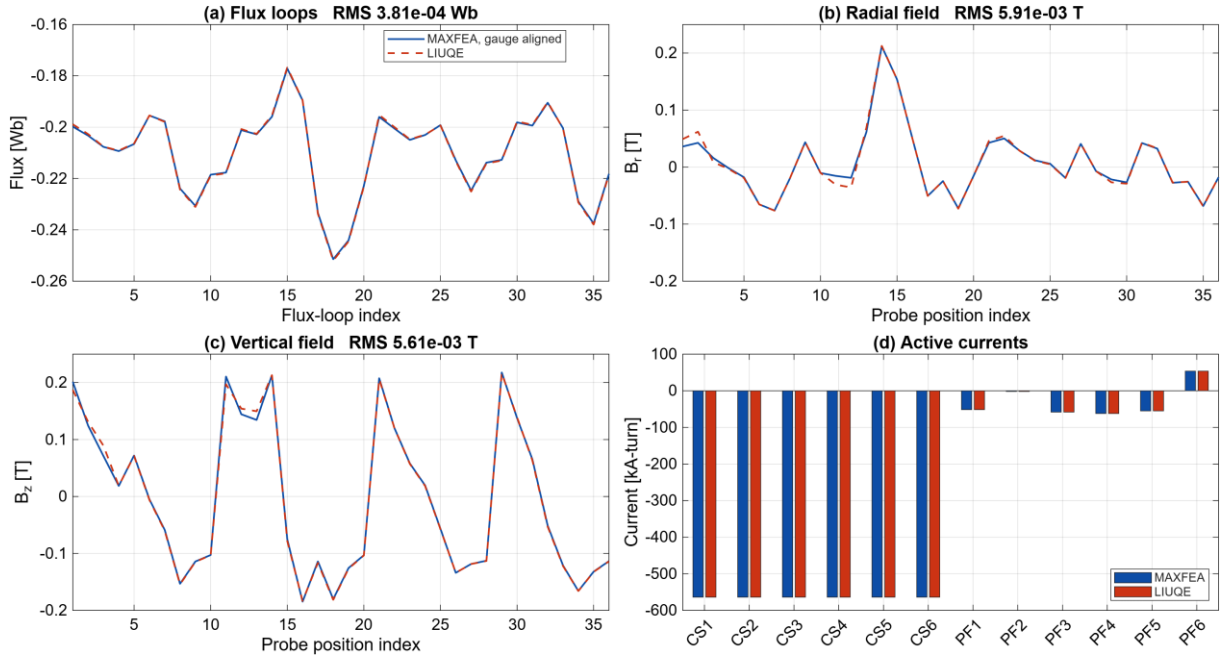



*Figure 4.25. MAXFEA-derived inputs and LIUQE fitted signals: (a) flux loops after mean-offset alignment, (b–c) $B_r$ and $B_z$ at the 36 probe positions, and (d) effective active-circuit currents in kA-turn. The two field panels share the same vertical scale; current limits are assessed separately in the text [12].*

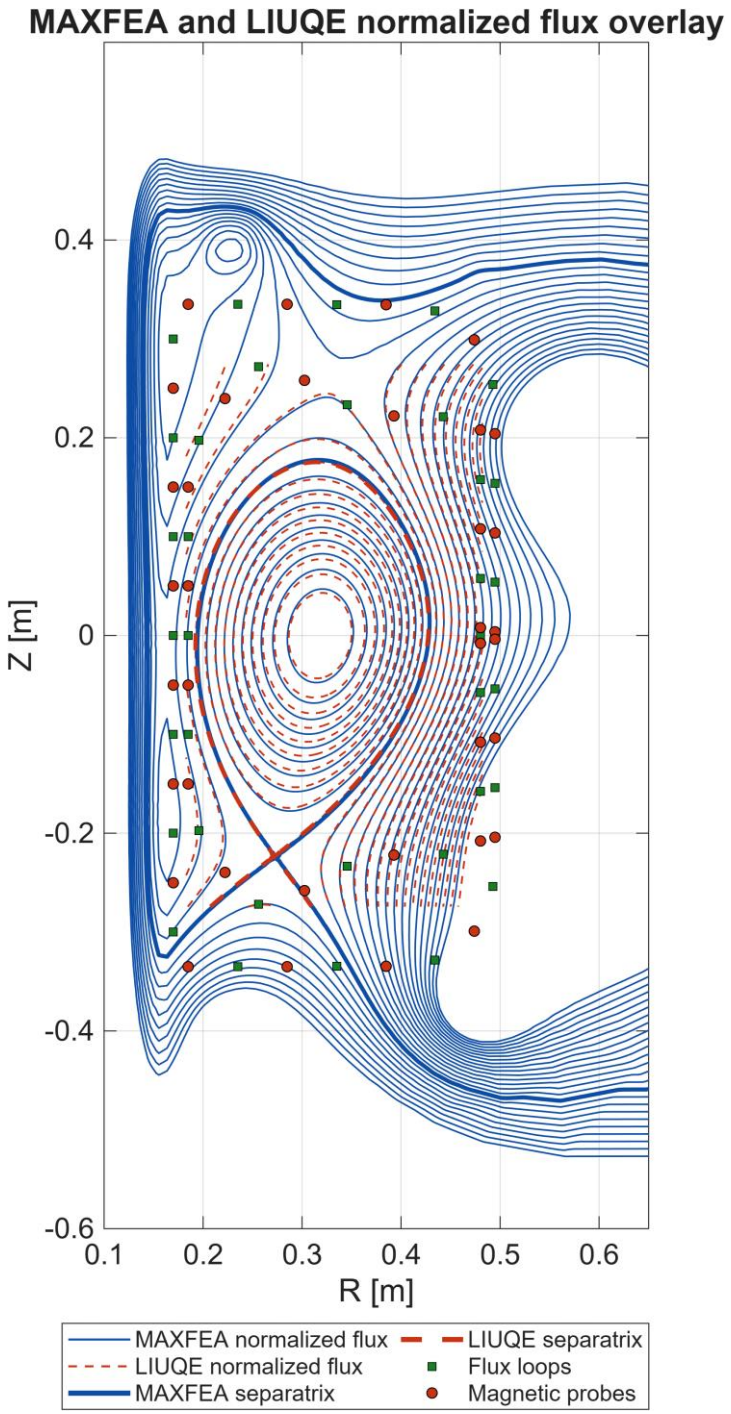



*Figure 4.26. Full-domain MAXFEA and LIUQE normalised-flux overlay, including the 36 flux-loop and 36 magnetic-probe positions. Both maps use the same contour levels ($\psi_N$ = 0.1 to 2.5, step 0.1); thick lines mark $\psi_N$ = 1. Each map is normalised with its own axis and boundary flux. MAXFEA extends beyond the LIUQE grid: the absence of red contours there reflects the available reconstruction domain [12].*

### 4.3.6 LIUQE with MAXFEA Data: Double Null

The direct double-null example reconstructs the nominal 110 kA MAXFEA start-of-flat-top equilibrium. As in the single-null case, the input contains 36 flux-loop signals, 72 field signals and 12 effective active-circuit currents, and the imported MAXFEA flux map initialises the nonlinear state rather than using LIH initialisation. The double-null configuration enables three computational domains; only the main domain carries plasma current. The diamagnetic term is disabled, and the profile fit does not impose the source $q_A$ or $\beta_p$ [12].

LIUQE converges in 31 iterations, with residual $9.74 \times 10^{-5}$ and weighted fit indicator $\chi = 0.636$, despite 30 near-singular-matrix warnings. Table 4.12 lists the principal results. The current differs by +0.753 A; the magnetic axis shifts by +2.652 mm in $R$ and −0.569 mm in $Z$, and the lower X-point moves by 1.539 mm in the poloidal plane.

*Table 4.12. Reported equilibrium quantities for the nominal MAXFEA double-null case and its LIUQE reconstruction [12].*

| Quantity | MAXFEA reference | LIUQE reconstruction |
|---|---|---|
| Plasma current, $I_p$ [kA] | 110.000 | 110.001 |
| Magnetic-axis radius, $R$ [m] | 0.316783 | 0.319435 |
| Magnetic-axis height, $Z$ [m] | 0.000225 | −0.000345 |
| Lower X-point radius, $R$ [m] | 0.277283 | 0.277510 |
| Lower X-point height, $Z$ [m] | −0.215238 | −0.213716 |
| On-axis safety factor, $q_A$ | 1.2534 | 1.0674 |
| Poloidal beta, $\beta_p$ | 0.1082 | 0.0651 |
| Normalised internal inductance, $l_i(3)$ | 0.8119 | 0.8033 |
| Plasma kinetic energy, $W_k$ [J] | 145.893 | 115.987 |
| Plasma volume, $V_p$ [m³] | 0.122526 | 0.121792 |
| Toroidal beta, $\beta_t$ [%] | 0.3682 | 0.2533 |

The two reconstructed nulls are not exactly flux-balanced. The upper null is at ($R$, $Z$) = (0.278165, 0.213750) m; their flux difference is $1.89 \times 10^{-4}$ Wb, or 0.642% of the axis-to-boundary flux interval. The upper null sets the LIUQE boundary flux, whereas the MAXFEA reference boundary passes through the lower null. Figure 4.27 therefore shows a change in the controlling null despite the close geometry. The MAXFEA scalar report supplies only the lower-null coordinates, so no upper-null displacement is inferred.

Figure 4.28 compares the fitted signals. The flux-loop RMS difference is $2.11 \times 10^{-4}$ Wb after removing the mean LIUQE-minus-input offset of $+6.59 \times 10^{-4}$ Wb, versus $6.92 \times 10^{-4}$ Wb without alignment. The $B_r$ and $B_z$ RMS differences are $3.10 \times 10^{-3}$ T and $2.46 \times 10^{-3}$ T. Figure 4.29 presents the normalised-flux overlay over the available MAXFEA domain, at the same spatial scale on the page as Figure 4.26.

Active currents reproduce the supplied inputs to numerical precision, with an RMS difference of $1.58 \times 10^{-10}$ kA-turn. The CS circuits reach 254.399 kA-turn, while PF1 and PF6 reach 188.519 kA-turn and 111.8% of the common scenario envelope. The stored LIUQE model-local CS current limits give a substantially higher utilisation of 565.3%, reflecting the different definition of these limits. The reconstructed $q_A$ and $\beta_p$ also differ from the MAXFEA reference values. In contrast to the single-null case, the sampled pressure in the main computational domain remains non-negative, ranging from 0 to 2.455 kPa. Together with the residual null-flux imbalance, these results show that the main remaining differences concern the double-null balance, the reconstructed profiles, and the current-limit assessment.

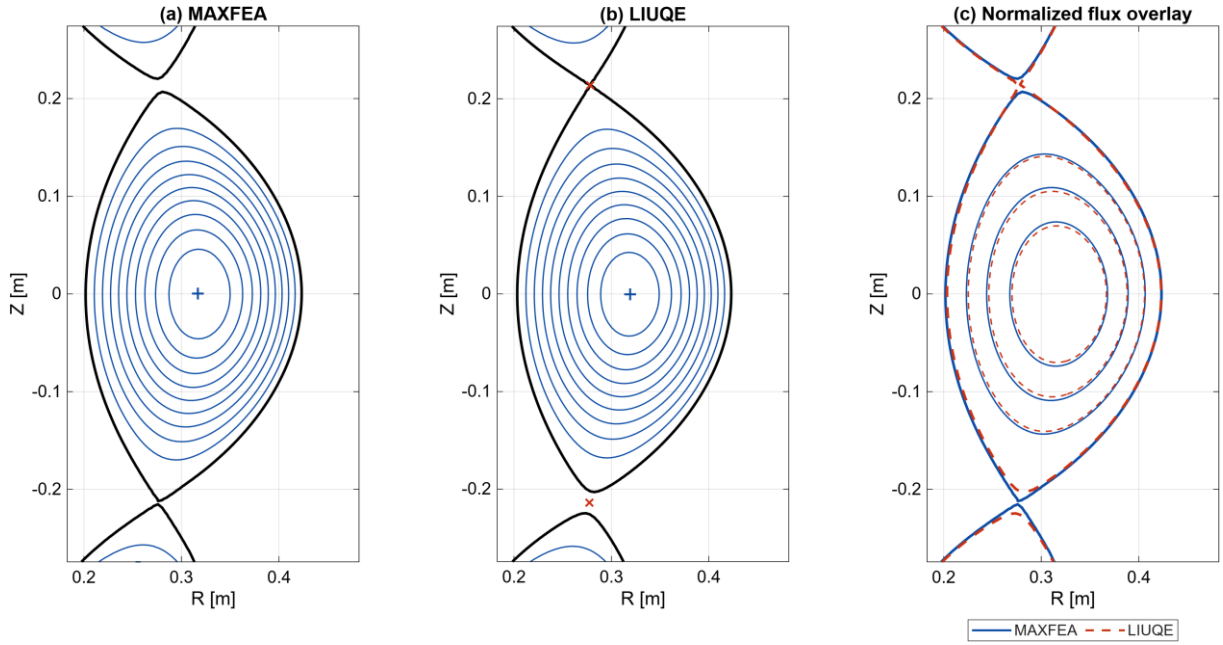



*Figure 4.27. Nominal double-null MAXFEA and LIUQE equilibria on identical R/Z axes. Panels (a–b): $\psi_N$ = 0.1 to 0.9 and boundary level $\psi_N$ = 1. Panel (c): overlaid $\psi_N$ = 0.25, 0.50, 0.75 and 1 contours. Each map uses its own axis and boundary flux. The two LIUQE nulls are marked; the boundary-flux contour connects to different nulls in the two solutions [12].*

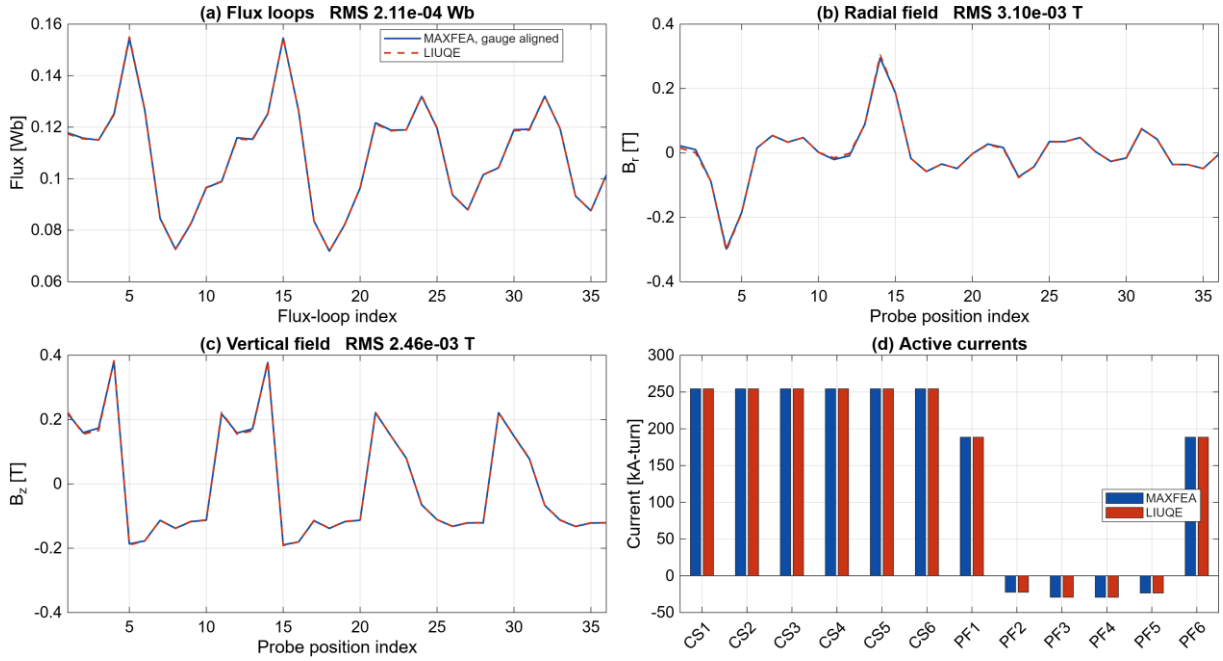



*Figure 4.28. MAXFEA-derived inputs and LIUQE fitted signals for the nominal double-null case: (a) flux loops after mean-offset alignment, (b–c) $B_r$ and $B_z$ at 36 probe positions, and (d) effective active-circuit currents. Both field panels share the same vertical scale. Current-limit exceedances are quantified in the text [12].*

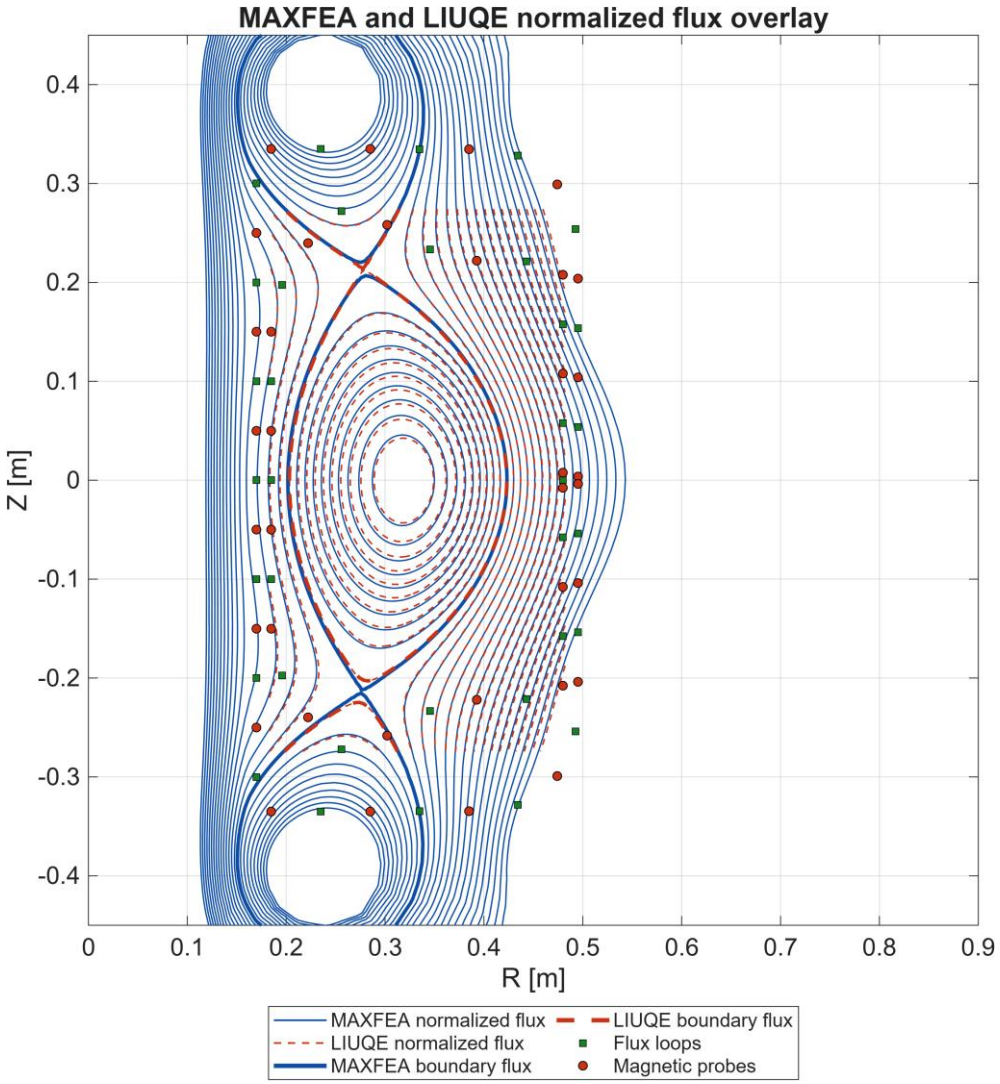



*Figure 4.29. MAXFEA and LIUQE normalised-flux overlay for the nominal double-null case, at the same spatial scale as Figure 4.26. The axes span the complete available MAXFEA grid (R = 0–0.90 m, Z = −0.45–0.45 m); contours reaching its edges are not extended beyond the data. LIUQE contours are restricted to its smaller reconstruction grid. Both maps use levels $\psi_N$ = 0.1 to 2.5, with $\psi_N$ = 1 thicker, and are normalised with their respective axis and boundary fluxes. Flux-loop and magnetic-probe positions are included [12].*

### 4.3.7 LIUQE with FBT Data from MAXFEA: Single Null

This example separates the MAXFEA-to-FBT transfer from the subsequent LIUQE reconstruction. The imported MAXFEA single-null contour contains 96 samples; after preprocessing, the effective saved FBT input contains 94 regular flux targets and one X-point constraint. FBT imposes $I_p$, $q_A$ and $\beta_p$. The effective MAXFEA coil currents, including virtual-control contributions, provide an initial guess and are free to change. LIUQE then receives 36 flux-loop signals, 72 field signals and 12 active currents from FBT and uses LIH initialisation. Its input retains default zero references for the diamagnetic signal and 93 passive currents; $q_A$ and $\beta_p$ are supplied but not imposed by active profile weights [12].

FBT converges in 20 iterations with residual $6.46 \times 10^{-7}$; LIUQE converges in 12 iterations with residual $3.33 \times 10^{-5}$ and $\chi = 0.2546$. The complete workflow records 14 near-singular-matrix warnings. Table 4.13 compares the principal quantities at all three stages.

*Table 4.13. Reported equilibrium quantities along the single-null MAXFEA → FBT → LIUQE workflow [12].*

| Quantity | MAXFEA | FBT | LIUQE |
|---|---|---|---|
| Plasma current, $I_p$ [kA] | 110.000 | 110.000 | 109.613 |
| Magnetic-axis radius, $R$ [m] | 0.319690 | 0.319147 | 0.322609 |
| Magnetic-axis height, $Z$ [m] | −0.001113 | −0.000991 | −0.000500 |
| Lower X-point radius, $R$ [m] | 0.273021 | 0.271192 | 0.273411 |
| Lower X-point height, $Z$ [m] | −0.224062 | −0.224047 | −0.224954 |
| On-axis safety factor, $q_A$ | 1.3811 | 1.3811 | 1.0335 |
| Poloidal beta, $\beta_p$ | 0.1151 | 0.1151 | 0.1470 |
| Normalised internal inductance, $l_i(3)$ | 0.8038 | 0.8674 | 0.8566 |
| Plasma kinetic energy, $W_k$ [J] | 163.385 | 209.978 | 266.383 |
| Plasma volume, $V_p$ [m³] | 0.127888 | 0.128300 | 0.128585 |
| Toroidal beta, $\beta_t$ [%] | 0.3916 | 0.4312 | 0.5552 |

Figures 4.30–4.32 compare geometry, signals and normalised-flux overlays. Both FBT and LIUQE retain a lower-null boundary; their upper nulls lie on different flux surfaces. Relative to FBT, LIUQE shifts the axis by +3.462 mm in $R$ and +0.491 mm in $Z$, and the lower null by 2.397 mm in the poloidal plane. Most of the changes in coil currents and external magnetic fields occur during the preceding MAXFEA-to-FBT transfer.

For MAXFEA → FBT, the aligned flux-loop RMS difference is $3.67 \times 10^{-3}$ Wb (22.82%); $B_r$ and $B_z$ differences are $4.71 \times 10^{-2}$ T (82.17%) and $1.01 \times 10^{-1}$ T (77.10%). For FBT → LIUQE they are $6.00 \times 10^{-4}$ Wb (3.76%), $5.71 \times 10^{-3}$ T (10.07%) and $4.51 \times 10^{-3}$ T (2.65%). These map-sampled comparisons remove mean flux offsets of +0.2138 Wb and $+8.27 \times 10^{-4}$ Wb, respectively. The percentages are normalised to the centred flux RMS for the loops and to the field RMS for the probes.

The active-current RMS change is 353.305 kA-turn from MAXFEA to FBT, but only 10.329 A-turn from FBT to LIUQE (0.0053%, calculated as RMS current difference / RMS FBT current × 100). Reconstructed PF1 reaches 199.970 kA-turn: 99.985% of its stored LIUQE model-local current limit, but 118.568% of the common scenario envelope. Relative to FBT, LIUQE changes $I_p$ by −386.590 A, $q_A$ by −25.2%, $\beta_p$ by +27.8% and plasma kinetic energy by +26.9%. Sampled main-domain pressures remain non-negative, reaching 5.417 kPa; passive currents reach 102.585 A despite their zero references. The comparison therefore shows two different effects along the chain: the largest changes in magnetic signals and active currents occur in the MAXFEA-to-FBT transfer, whereas the subsequent LIUQE reconstruction introduces smaller current changes but appreciable differences in the reconstructed profile quantities.

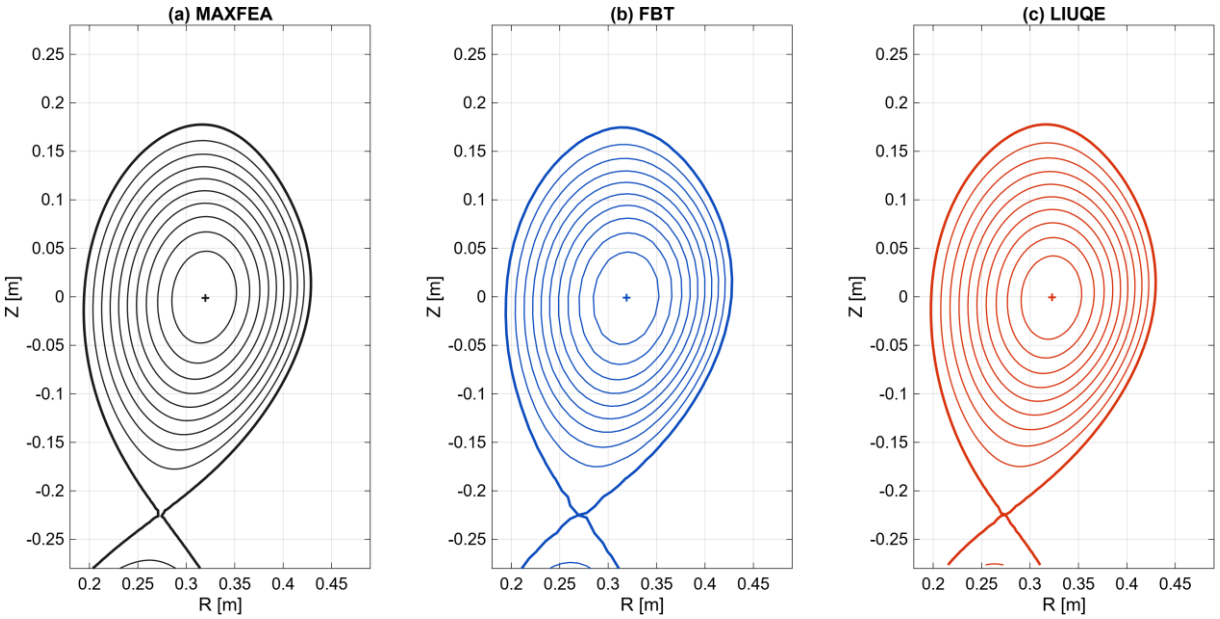



*Figure 4.30. MAXFEA, FBT and LIUQE single-null equilibria on identical R/Z axes and with equal spatial aspect ratios. Internal contours span $\psi_N$ = 0.1 to 0.9; the thicker curve marks $\psi_N$ = 1. Each map uses its own axis and boundary flux. Crosses mark the magnetic axes [12].*

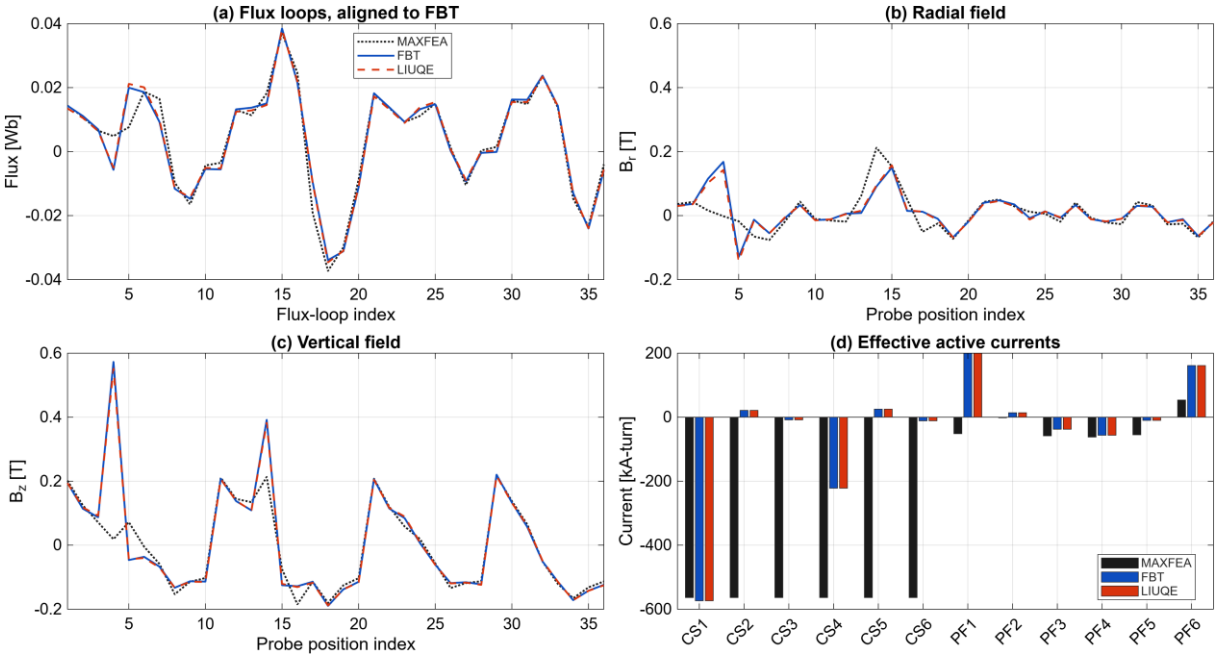



*Figure 4.31. Signals sampled from the three field maps and effective active-circuit currents. Flux-loop signals are aligned to the FBT mean by additive offsets only. The $B_r$ and $B_z$ panels share the same vertical scale. FBT and LIUQE curves nearly overlap in the current panel, while their differences from MAXFEA remain visible [12].*

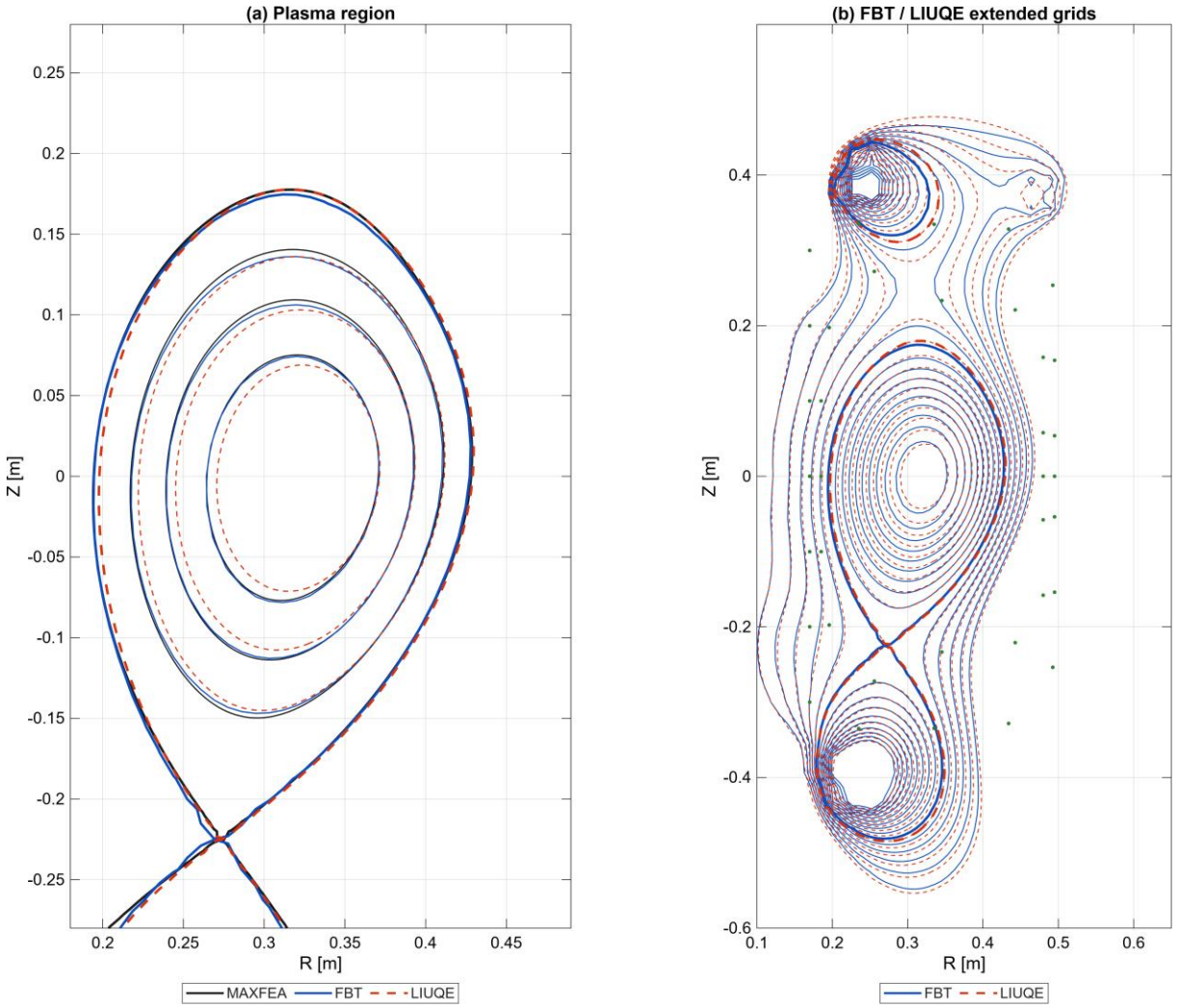



*Figure 4.32. Normalised-flux overlays with equal R/Z aspect ratios within each panel. (a) All three equilibria in the plasma region, at $\psi_N$ = 0.25, 0.50, 0.75 and 1. (b) FBT and LIUQE on their common extended domain, at $\psi_N$ = 0.1 to 1.5, with flux-loop positions in green. The boundary level $\psi_N$ = 1 is thicker. Each map is normalised separately; the two panels intentionally show different spatial windows [12].*

### 4.3.8 LIUQE with FBT Data from MAXFEA: Double Null

The double-null chain applies the same transfer-and-reconstruction workflow to the MAXFEA end-of-flat-top equilibrium at 110 kA. The imported contour contains 96 samples; after preprocessing, the effective FBT input contains 92 regular flux targets and two X-point constraints. FBT receives the $I_p$, $q_A$ and $\beta_p$ targets; the effective MAXFEA currents remain a free initial guess. LIUQE receives 36 flux-loop signals, 72 field signals and 12 active currents generated by FBT, using the same LIH initialisation as in the preceding chained case. As in the single-null chain, the diamagnetic and 93 passive-current references default to zero, while the supplied $q_A$ and $\beta_p$ are not imposed by active profile weights [12].

FBT converges in 17 iterations with residual $5.87 \times 10^{-7}$. LIUQE converges in 12 iterations with residual $3.22 \times 10^{-5}$ and $\chi = 0.2460$; the full workflow records 14 near-singular-matrix warnings. Table 4.14 lists the principal results. Relative to FBT, the reconstructed current increases by 1.533 kA (+1.394%), and the axis shifts by +2.243 mm in $R$ and −0.196 mm in $Z$.

*Table 4.14. Reported quantities along the nominal double-null MAXFEA → FBT → LIUQE workflow [12].*

| Quantity | MAXFEA | FBT | LIUQE |
|---|---|---|---|
| Plasma current, $I_p$ [kA] | 110.000 | 110.000 | 111.533 |
| Magnetic-axis radius, $R$ [m] | 0.316783 | 0.320009 | 0.322252 |
| Magnetic-axis height, $Z$ [m] | 0.000225 | 0.000301 | 0.000105 |
| Upper X-point radius, $R$ [m] | 0.279494 | 0.280145 | 0.280662 |
| Upper X-point height, $Z$ [m] | 0.218555 | 0.218966 | 0.220990 |
| On-axis safety factor, $q_A$ | 1.2662 | 1.2662 | 0.9955 |
| Poloidal beta, $\beta_p$ | 0.1088 | 0.1088 | 0.1380 |
| Normalised internal inductance, $l_i(3)$ | 0.8079 | 0.8642 | 0.8388 |
| Plasma kinetic energy, $W_k$ [J] | 145.301 | 198.501 | 258.901 |
| Plasma volume, $V_p$ [m³] | 0.124231 | 0.125852 | 0.127507 |
| Toroidal beta, $\beta_t$ [%] | 0.3620 | 0.4179 | 0.5446 |

The configuration is close to, but not exactly, a balanced double null. The null-flux separation, normalised to each axis-to-boundary interval, decreases from 0.226% in FBT to 0.0176% in LIUQE. The upper null determines the boundary flux in FBT, whereas the lower null determines it in LIUQE. The upper and lower nulls move by 2.088 and 1.979 mm, respectively; the reconstructed lower null is at (0.284797, −0.219469) m. Figures 4.33 and 4.35 compare the maps. The null-flux values provide a more reliable indication of the magnetic topology than the apparent contour connectivity on the plotting grids.

Figure 4.34 separates the two diagnostic comparisons. For MAXFEA → FBT, the aligned loop RMS difference is $2.86 \times 10^{-3}$ Wb (14.76%), with $B_r$ and $B_z$ differences of $3.90 \times 10^{-2}$ T (48.00%) and $4.71 \times 10^{-2}$ T (31.45%). For FBT → LIUQE, these become $3.23 \times 10^{-4}$ Wb (1.76%), $3.52 \times 10^{-3}$ T (6.07%) and $3.71 \times 10^{-3}$ T (2.17%). Mean flux offsets of +0.2252 Wb and $+5.42 \times 10^{-4}$ Wb are removed. The percentages are normalised to the centred flux RMS for the loops and to the corresponding field RMS for the probes.

The active-current RMS change is 380.034 kA-turn from MAXFEA to FBT, but only 7.028 A-turn from FBT to LIUQE (0.0060%, calculated as RMS current difference / RMS FBT current × 100). Reconstructed PF6 reaches 200.008 kA-turn, corresponding to 118.591% of the common scenario envelope. It is also marginally above the stored LIUQE model-local limit of 200 kA-turn, by 7.680 A-turn (0.00384%). Relative to FBT, $q_A$ decreases by 21.4%, while $\beta_p$ and plasma kinetic energy increase by 26.9% and 30.4%, respectively. Sampled main-

domain pressures remain non-negative, reaching 5.475 kPa, and the fitted passive currents reach 69.707 A despite their zero references. As in the single-null chain, the largest change in active currents occurs during the MAXFEA-to-FBT transfer, whereas the subsequent LIUQE reconstruction produces smaller current changes together with appreciable differences in the reconstructed profile quantities.

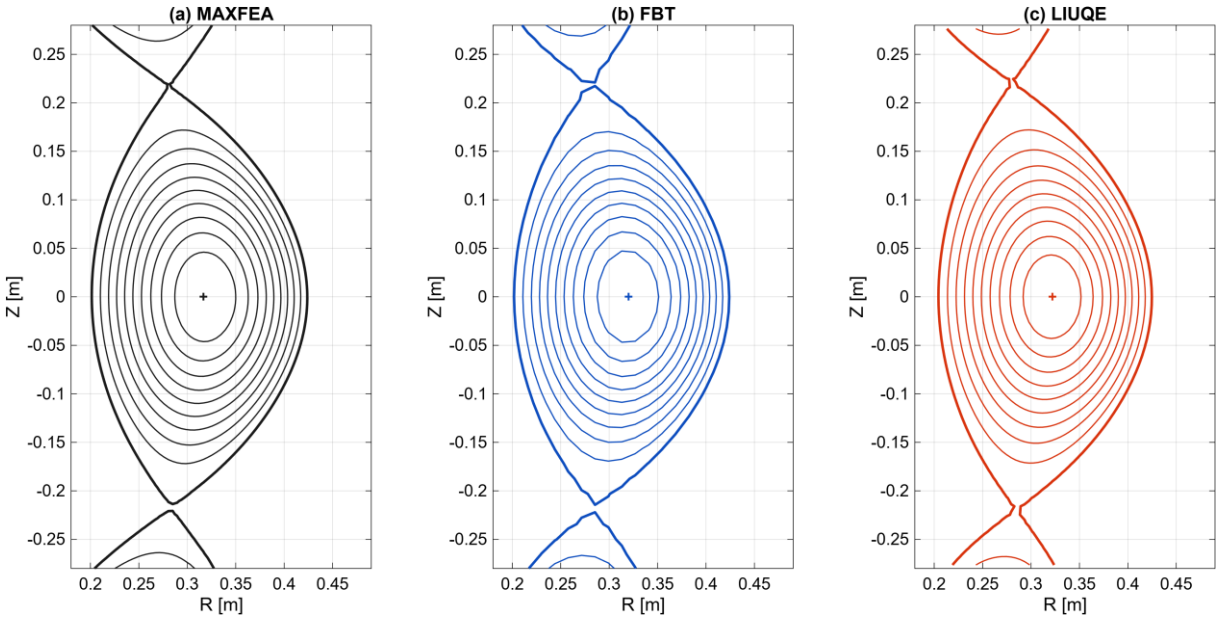



*Figure 4.33. Nominal double-null MAXFEA, FBT and LIUQE equilibria on identical R/Z axes, with equal spatial aspect ratios. Internal contours span $\psi_N$ = 0.1 to 0.9 and the thicker curve marks $\psi_N$ = 1. Each map uses its own axis and boundary flux. The MAXFEA scalar report provides the upper-null coordinates only [12].*

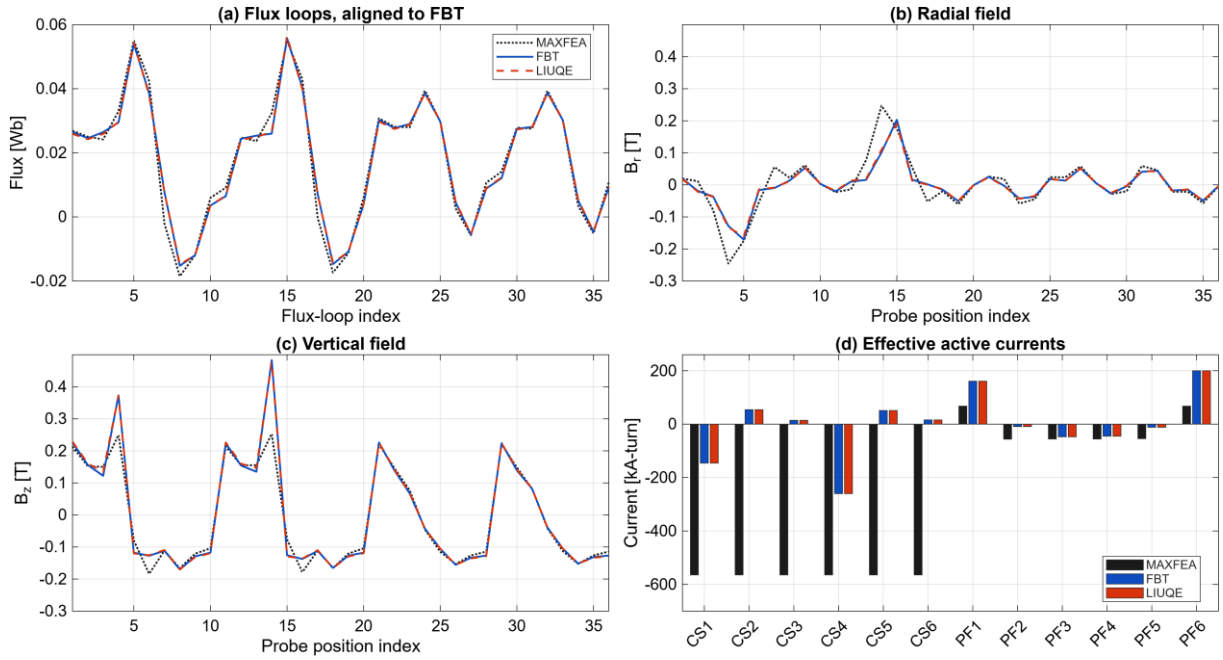



*Figure 4.34. Map-sampled diagnostic signals and effective active-circuit currents at the three stages. Flux-loop means are aligned to FBT by additive offsets only; $B_r$ and $B_z$ share the same vertical scale. The near overlap of FBT and LIUQE currents contrasts with the changes introduced during transfer from MAXFEA [12].*

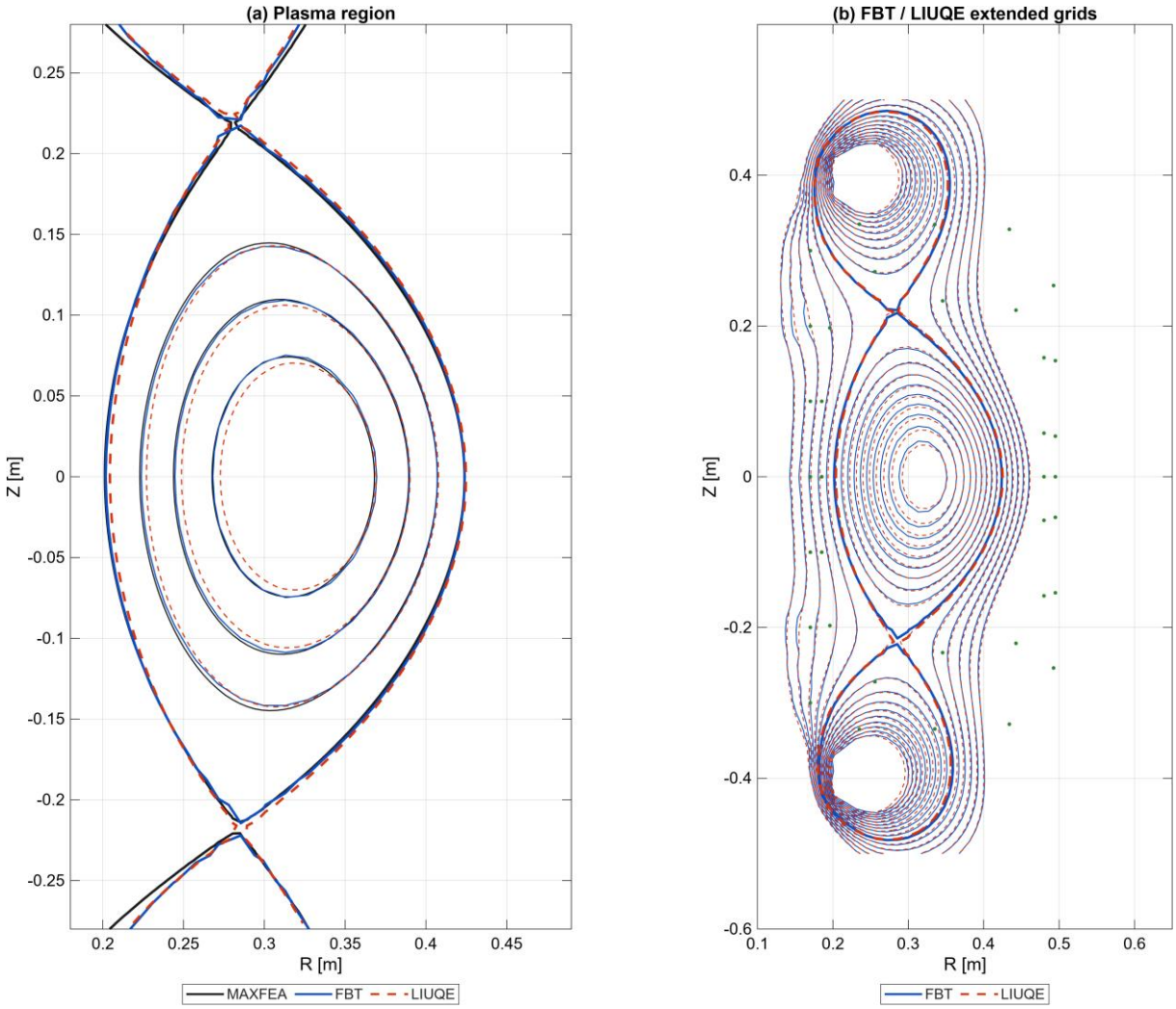



*Figure 4.35. Normalised-flux overlays with equal R/Z aspect ratios within each panel. (a) MAXFEA, FBT and LIUQE in the plasma region at $\psi_N$ = 0.25, 0.50, 0.75 and 1. (b) FBT and LIUQE on their common extended domain at $\psi_N$ = 0.1 to 1.5, with flux loops in green. The $\psi_N$ = 1 contour is thicker; normalisation is separate for each map. Panel windows differ intentionally [12].*

*Table 4.15. LIUQE reconstruction results. Current utilisation refers to the final reconstructed state and the common scenario envelope; warning counts cover the complete calculation chain [12].*

| Case | Iterations | $\chi$ | $I_p$ [kA] | Maximum common-envelope utilisation [%] | Workflow warnings |
|---|---|---|---|---|---|
| LIUQE limiter | 15 | 0.0218 | 109.562 | 79.89% | 0 |
| LIUQE single null | 45 | 0.0016 | 110.011 | 131.96% | 0 |
| LIUQE double null | 20 | 0.0017 | 110.009 | 72.10% | 0 |
| LIUQE external interface | 45 | 0.0016 | 110.011 | 131.96% | 0 |
| Direct MAXFEA single null | 17 | 1.4137 | 109.999 | 98.58% | 17 |
| Direct MAXFEA double null | 31 | 0.6363 | 110.001 | 111.78% | 30 |
| Indirect MAXFEA single null | 12 | 0.2546 | 109.613 | 118.57% | 14 |
| Indirect MAXFEA double null | 12 | 0.2460 | 111.533 | 118.59% | 14 |

Table 4.15 uses the final LIUQE state consistently, including for the current-envelope comparison. A small $\chi$ or convergence residual does not establish compliance with that envelope, and $\chi$ values from different diagnostic sets and weights should not be treated as a common accuracy ranking. Conditioning-warning counts include initialisation and preceding stages where present.

In both chained examples, LIUQE reproduces the FBT-generated magnetic measurements more closely than FBT reproduces the imported MAXFEA measurements. The largest signal and coil-current changes therefore occur during the boundary-and-target transfer to FBT. Remaining profile differences, default diagnostic references, null-flux imbalance and current-limit exceedances must still be assessed separately.

## 4.4 FGS Static Forward Equilibria

FGS solves the static free-boundary Grad–Shafranov problem with specified active-circuit currents and selected global plasma constraints [10]. In the following examples, each source equilibrium is converted to the FGS representation before the static calculation. Comparing the transferred input and FGS result shows which differences arise during transfer and which appear in the forward calculation.

### 4.4.1 FGS with FBT Data

The first case starts from an FBT equilibrium for the 99002 end-of-flat-top scenario. The adapter converts the flux map, current distribution, profiles and conductor currents, then reprocesses the transferred state on the 66 × 65 FGS grid. The comparison below uses this transferred input, rather than an independently retained pre-conversion FBT state. A non-conducting vessel and a single plasma domain are selected, with no evolving passive-current degrees of freedom. The nonlinear constraints preserve the input $I_p$, $\beta_p$ and $q_A$ while the active-circuit currents remain fixed [12].

FGS converges in five iterations and ten function evaluations, with residual $2.46 \times 10^{-13}$ against a tolerance of $10^{-7}$. No conditioning warnings are reported. Table 4.16 compares the input and final quantities. The input already carries 114.993 kA, which the final solution preserves to within $2.1 \times 10^{-9}$ A. The values $q_A$ = 2.51617 and $\beta_p$ = 0.113924 are also preserved to numerical precision. The 114.993 kA value therefore belongs to the transferred FBT state rather than being produced by the FGS calculation.

*Table 4.16. Transferred FBT input and converged FGS quantities on the destination grid [12].*

| Quantity | Transferred input | FGS result |
|---|---|---|
| Plasma current, $I_p$ [kA] | 114.993 | 114.993 |
| Magnetic-axis radius, $R$ [m] | 0.327216 | 0.331522 |
| Magnetic-axis height, $Z$ [m] | 0.000021 | 0.000054 |
| Limiting-point radius, $R$ [m] | 0.185000 | 0.345411 |
| Limiting-point height, $Z$ [m] | 0.019467 | −0.233444 |
| On-axis safety factor, $q_A$ | 2.51617 | 2.51617 |
| Poloidal beta, $\beta_p$ | 0.113924 | 0.113924 |
| Normalised internal inductance, $l_i(3)$ | 0.727113 | 0.702114 |
| Plasma kinetic energy, $W_k$ [J] | 227.167 | 227.167 |
| Plasma volume, $V_p$ [m³] | 0.186012 | 0.187477 |
| Toroidal beta, $\beta_t$ [%] | 0.1862 | 0.1906 |

Figure 4.36 shows the spatial adjustment on common axes. The magnetic axis shifts by +4.306 mm in $R$ and +0.033 mm in $Z$. Both states are classified as limiter equilibria, with no X-point, but their stored limiting-point coordinates differ. Internal inductance decreases by 3.44% and plasma volume increases by 0.79%, while the reported plasma kinetic energy remains 227.167 J. Although the selected global constraints are preserved, the forward solution produces a small shift of the magnetic axis and corresponding changes in some derived quantities, while the limiting point is identified at a different location on the limiter.

Figure 4.37 compares the profiles, fixed currents, and their limits. The sampled pressure remains non-negative, with its maximum increasing slightly from 2.836 to 2.866 kPa. The toroidal-field function is shown on the same normalised-flux coordinate. All 12 active currents remain unchanged through the FGS calculation. PF3 has the largest magnitude, 48.053 kA-turn, corresponding to 74.54% of the common scenario envelope and 24.03% of the stored FGS model-local current limit. Both current criteria are satisfied; the difference between the two utilisation values arises from the different limit sets applied to the same current.

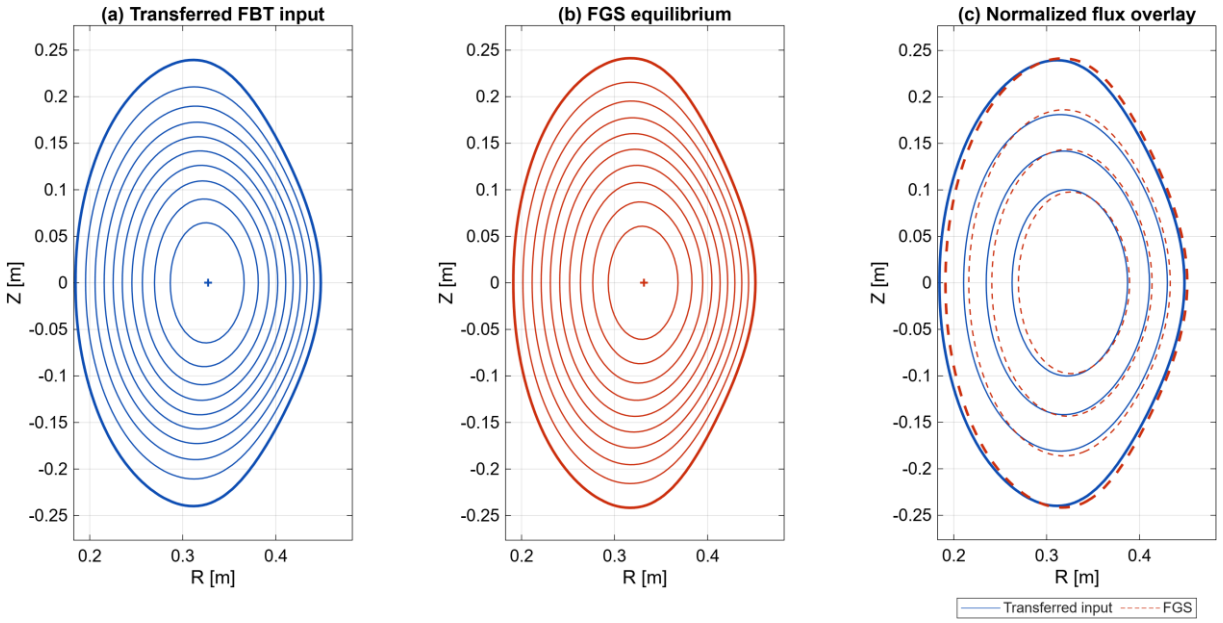



*Figure 4.36. Transferred FBT input and converged FGS equilibrium on identical R/Z axes and with equal spatial aspect ratios. Panels (a–b) show $\psi_N$ = 0.1 to 0.9 and the thicker boundary contour $\psi_N$ = 1; plus signs (+) mark the magnetic axes. Panel (c) overlays $\psi_N$ = 0.25, 0.50, 0.75 and 1. Each map uses its own axis and boundary flux [12].*

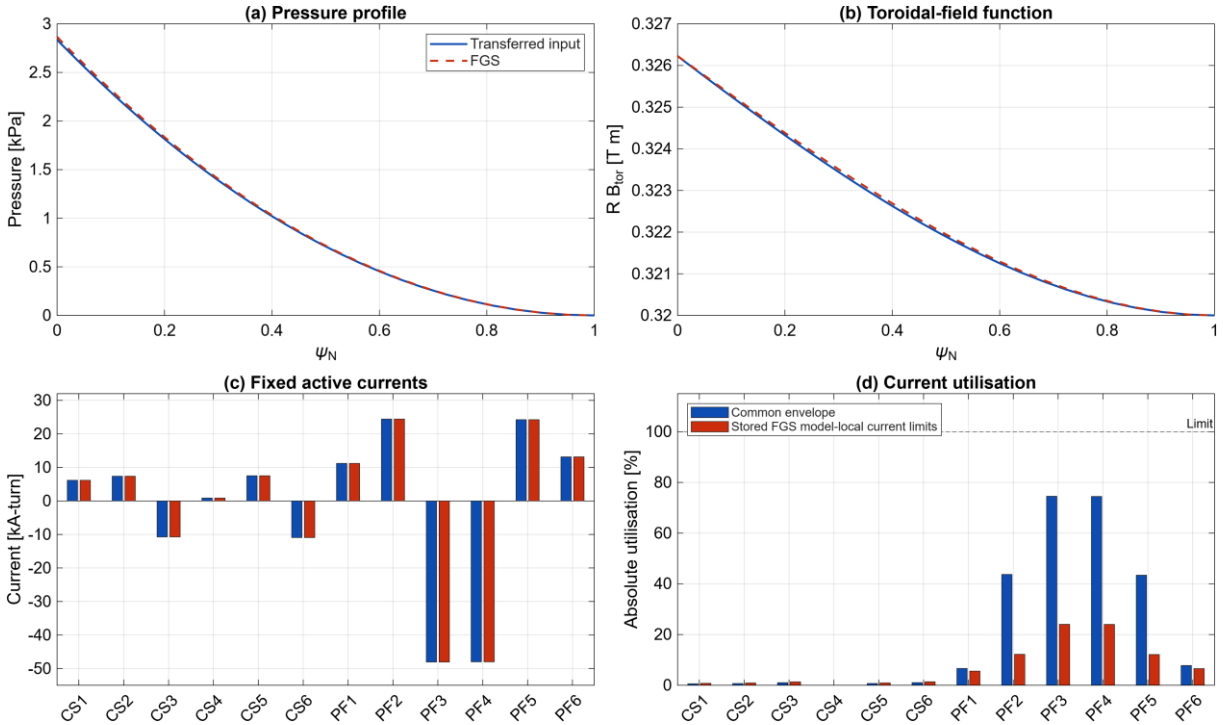



*Figure 4.37. Transferred input and FGS result: (a) pressure and (b) toroidal-field function versus normalised flux, (c) unchanged active-circuit currents, and (d) absolute current utilisation relative to the common scenario envelope and stored FGS model-local current limits. Profile samples use $\psi_N = pQ^2$ from the stored MEQ radial coordinate [12].*

### 4.4.2 FGS with MAXFEA Data

This case initialises FGS directly from the single-null MAXFEA equilibrium in trust_dsn_eof.mat. The adapter interpolates the flux map onto a 66 × 65 grid, reconstructs the plasma-current distribution and fits the MEQ profile basis. Currents assigned to the physical PF circuits include the MAXFEA virtual-control contributions and are expressed in ampere-turn coordinates. No intermediate FBT or LIUQE solve is used; the configuration has one plasma domain and no passive-current degrees of freedom [12].

Table 4.17 compares the MAXFEA reference, the transferred MEQ input and the FGS result. Conversion to MEQ already changes $I_p$ from 110.000 to 110.753 kA, while $q_A$ and $\beta_p$ become 1.31377 and 0.087433, respectively. These transferred quantities are used as the global constraints for FGS.

*Table 4.17. MAXFEA reference, transferred MEQ input and FGS result [12].*

| Quantity | MAXFEA | Input | FGS result |
|---|---|---|---|
| Plasma current, $I_p$ [kA] | 110.000 | 110.753 | 110.753 |
| Magnetic-axis radius, $R$ [m] | 0.319690 | 0.318979 | 0.318979 |
| Magnetic-axis height, $Z$ [m] | −0.001113 | 0.000027 | 0.000027 |
| Lower X-point radius, $R$ [m] | 0.273021 | 0.273227 | 0.273227 |
| Lower X-point height, $Z$ [m] | −0.224062 | −0.224187 | −0.224187 |
| On-axis safety factor, $q_A$ | 1.3811 | 1.3138 | 1.3138 |
| Poloidal beta, $\beta_p$ | 0.1151 | 0.0874 | 0.0874 |
| Normalised internal inductance, $l_i(3)$ | 0.8038 | 0.7628 | 0.7628 |
| Plasma kinetic energy, $W_k$ [J] | 163.385 | 161.726 | 161.726 |
| Plasma volume, $V_p$ [m³] | 0.127888 | 0.129451 | 0.129451 |
| Toroidal beta, $\beta_t$ [%] | 0.3916 | 0.3304 | 0.3304 |

The FGS result retains the transferred $I_p$, $q_A$, $\beta_p$ and plasma kinetic energy of 161.726 J. The flux maps differ by no more than $2.78 \times 10^{-17}$ Wb, and the active-current vectors are identical. The input and FGS columns of Table 4.17 therefore coincide at the precision shown.

The magnetic axis, boundary-defining lower X-point, plasma volume and internal inductance also retain their transferred values. Relative to MAXFEA, the magnetic axis lies 0.711 mm inward and 1.140 mm higher, while the lower X-point lies 0.207 mm outward and 0.125 mm lower. These differences arise during transfer to MEQ; the input and FGS boundary contours coincide in Figures 4.38 and 4.40.

The pressure profile is unchanged by FGS, reaching 3.039 kPa in both MEQ states. All twelve active currents remain fixed, with 563.885 kA-turn in absolute value in each CS circuit. PF5 reaches 98.58% of the common scenario envelope, while the CS circuits reach 70.49% of the stored FGS model-local current limits. Both comparisons remain below 100%.

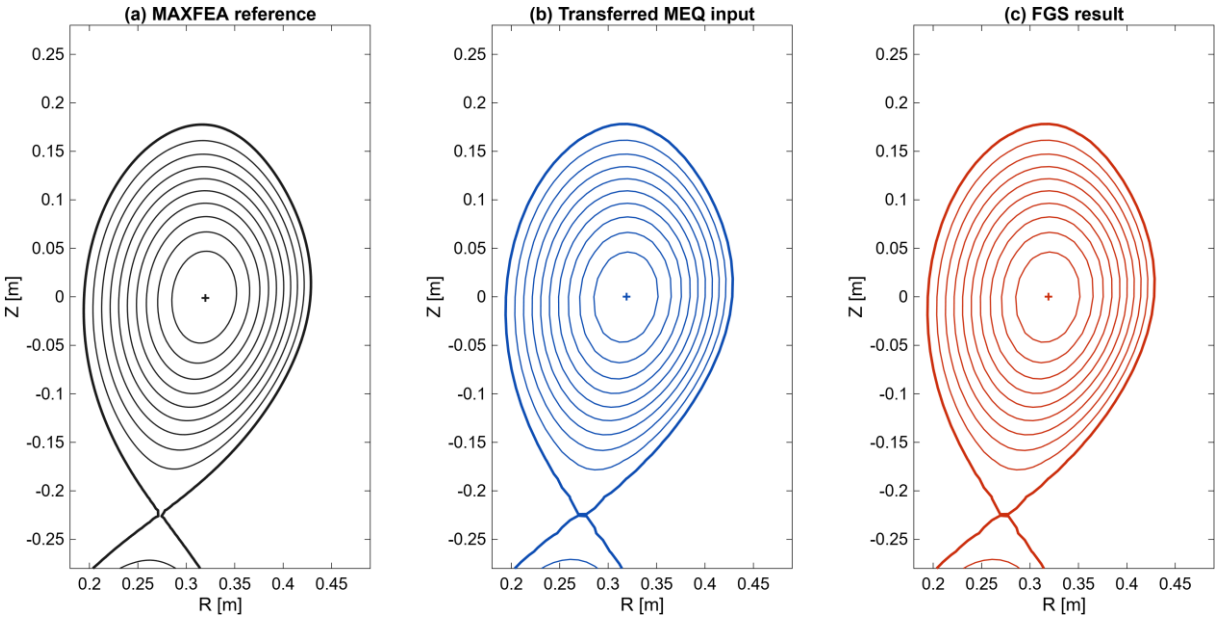



*Figure 4.38. MAXFEA reference, transferred MEQ input and FGS result on identical R/Z axes with unit spatial aspect ratio. Thin contours show normalised flux from 0.1 to 0.9; the thick contour marks $\psi_N = 1$, and plus signs (+) mark the magnetic axes. The input and FGS contours coincide [12].*

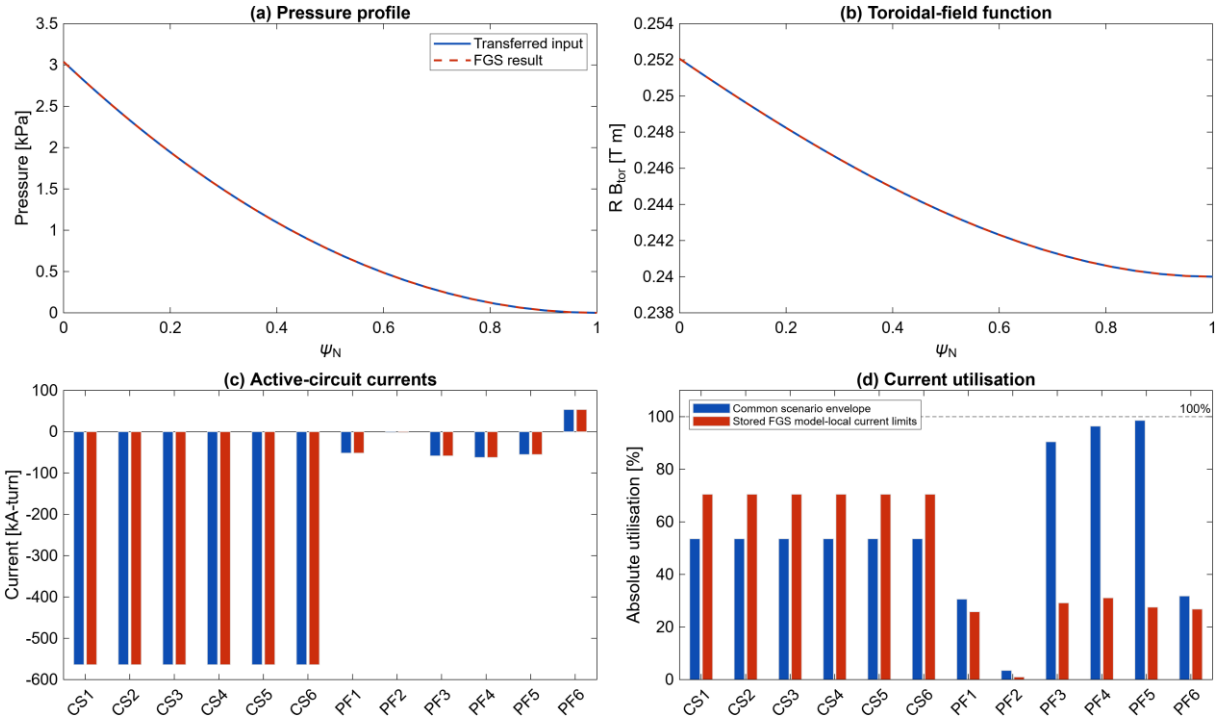



*Figure 4.39. Transferred-input and FGS pressure and toroidal-field-function profiles versus normalised flux, fixed active currents in kA-turn, and absolute current utilisation relative to the common scenario envelope and stored FGS model-local current limits. Blue denotes the input and red the FGS result [12].*

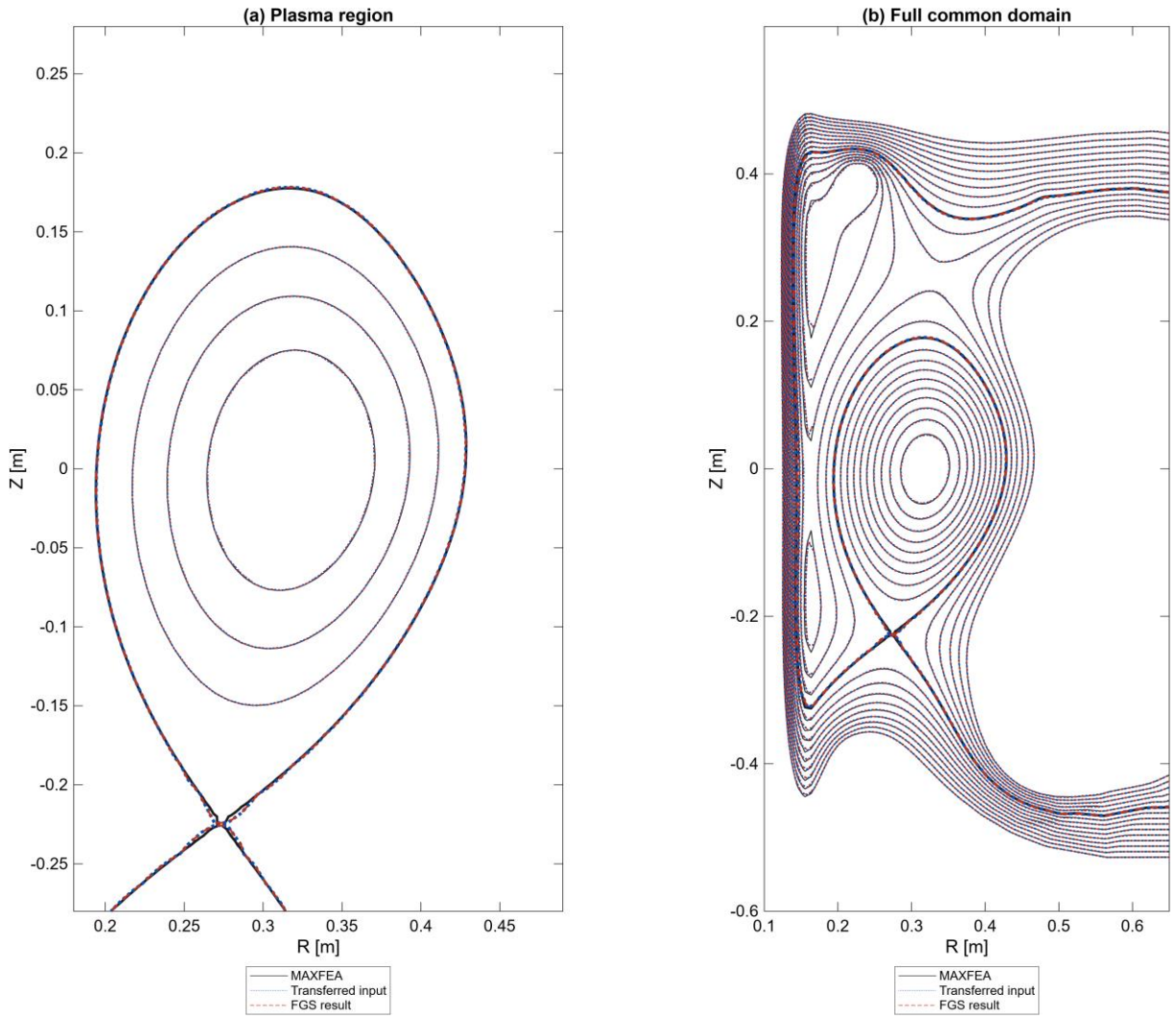



*Figure 4.40. Normalised-flux contours of MAXFEA (black), the transferred input (blue dotted) and FGS (red dashed) in the plasma region and over the full common domain. Each map is normalised using its own axis and boundary flux; thick contours mark $\psi_N = 1$. The input and FGS contours coincide [12].*

The FBT-derived case shows a 4.306 mm radial-axis adjustment after the FGS calculation, whereas the MAXFEA-derived result retains the transferred MEQ geometry. Active currents remain fixed in both cases.

## 4.5 Cross-Code Validation

This comparison evaluates the reliability and consistency of the MEQ reconstruction workflow by benchmarking its results against those obtained with EFIT and MAXFEA, two well-established and widely used equilibrium codes, for six TRUST reference equilibria in the presence of noise in the magnetic signals. MAXFEA supplies the reference geometry, and the comparison uses the distances between the reconstructed last closed flux surface and prescribed locations on the limiter. The study therefore tests boundary agreement under a specified synthetic measurement model; it does not constitute validation against experimental measurements [12].

### 4.5.1 Reference Cases and Diagnostic Geometry

The comparison covers double-null (DN), lower single-null (LSN) and upper single-null (USN) configurations at the start and end of flat top (SOF and EOF), giving six reference equilibria. Each equilibrium is reconstructed separately [12].

The magnetic diagnostic layout adopted for this numerical study is identical for all six cases and is shown in the left panel of Figure 4.41. It comprises 36 flux-loop locations and 36 magnetic-probe locations. Two orthogonal magnetic-field components are evaluated at each probe location, giving 72 magnetic-field channels and 108 magnetic observations per equilibrium [12].

The right panel marks the 16 fixed locations on the limiter used to evaluate the LCFS–limiter gaps. The same locations and numbering are used for MAXFEA, EFIT and MEQ in all six cases. These points define the geometric sampling positions for the boundary comparison [12].

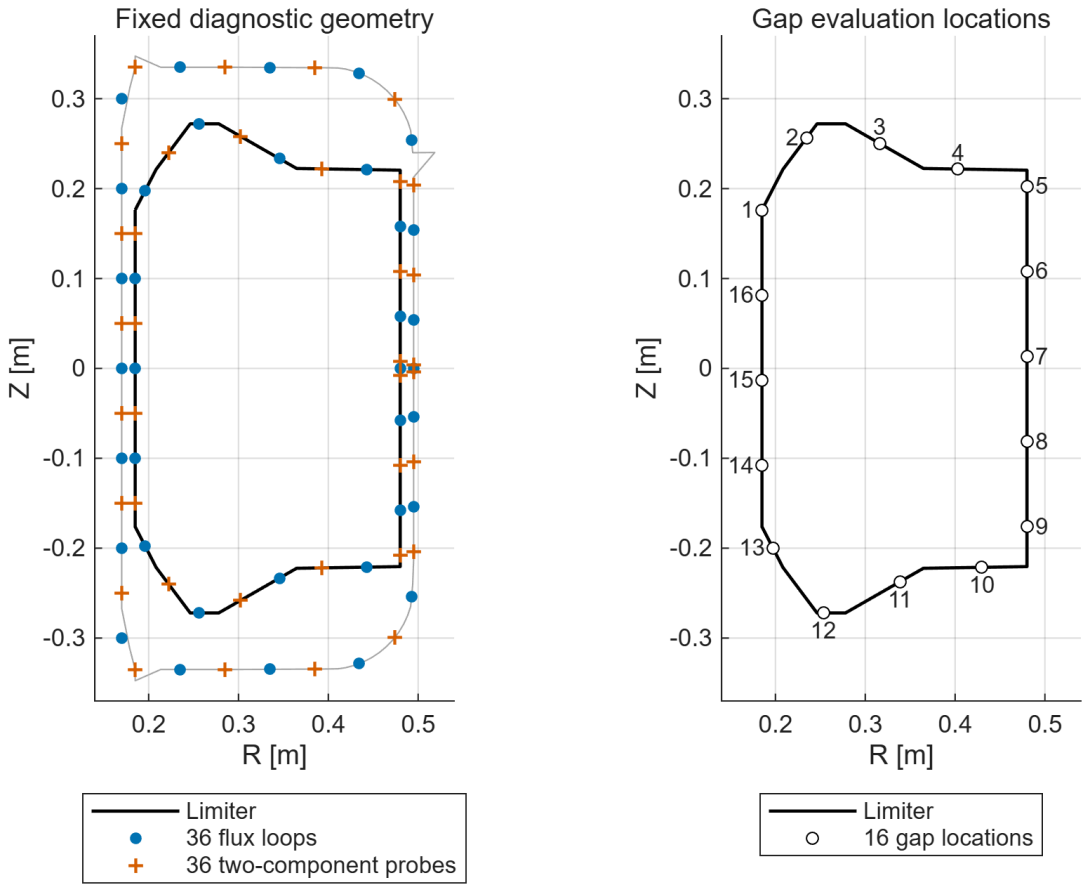



*Figure 4.41. Magnetic diagnostic layout (left) and the 16 limiter locations used to evaluate the boundary gaps (right). Both layouts are common to all six cases; gap numbering is retained in the case comparisons [12].*

### 4.5.2 Measurement Noise and Reconstruction Inputs

The measurement model adds an absolute Gaussian contribution and a contribution proportional to the magnitude of the unperturbed signal [12]. Writing the two standard-normal contributions as $\xi_a$ and $\xi_r$, the perturbed measurement is

$$\tilde{y} = y + \sigma_{abs}\xi_a + \sigma_{rel}|y|\xi_r$$

The absolute term represents a noise floor that remains finite when the signal approaches zero, while the relative term grows with signal amplitude. For independent draws, their variances add, giving an effective standard deviation of $\sqrt{{\sigma_{abs}}^2 + (\sigma_{rel}|y|)^2}$. Thus, the model does not impose a fixed 3% error on every measurement. Table 4.18 gives the parameters used for flux loops and magnetic probes; the random seed is 20 [12].

*Table 4.18. Absolute and relative Gaussian noise parameters. The 3% value is the coefficient of the signal-dependent term, not the total relative uncertainty near a signal zero [12].*

| Signal | Absolute coefficient | Relative coefficient |
|---|---|---|
| Flux loops | $1.0 \times 10^{-4}\ \frac{Wb}{rad}$ | 0.03 |
| Magnetic probes | $5.0 \times 10^{-4}$ T | 0.03 |

The two contributions have equal standard deviation at a flux-loop signal magnitude of $3.33 \times 10^{-3}\ \frac{Wb}{rad}$ and a magnetic-field magnitude of $1.67 \times 10^{-2}$ T. Below these values the absolute term dominates. The flux-loop unit refers to the poloidal flux per radian used in the EFIT input convention and must be retained when comparing the absolute error with the signal [12].

The EFIT calculation distinguishes boundary-based preparation from magnetic reconstruction. Its r-mode input contains 96 boundary targets from MAXFEA. The k-mode input supplies the 36 flux-loop and 72 magnetic-field observations, together with the coil-current and plasma-current constraints, without prescribing those boundary coordinates. The saved EFIT maps use a 65 × 65 grid over $R$ = 0.05–0.65 m and $Z$ = −0.60–0.60 m. The MEQ results belong to the campaign's FBT/LIUQE branch and are denoted MEQ below. The comparison assesses the final reconstructed boundaries, rather than equating the intermediate equilibria or the internal residual definitions of the two codes [12].

### 4.5.3 Boundary Agreement

Here, a gap is the LCFS–limiter distance evaluated at one of the 16 prescribed limiter locations. Let $g_{ref,i}$ and $g_{c,i}$ denote this distance for MAXFEA and reconstruction c (EFIT or MEQ), respectively, at location i. Their signed difference, $\Delta g_{c,i} = g_{c,i} - g_{ref,i}$, is the gap discrepancy relative to MAXFEA. A negative discrepancy means that the reconstructed LCFS–limiter gap is smaller than the MAXFEA reference gap, placing the reconstructed boundary closer to the limiter at that location; a positive discrepancy means that it is farther away. Agreement with MAXFEA is determined by the absolute magnitude of the discrepancy: the closer it is to zero, the better the agreement, regardless of sign.

The RMS and maximum absolute discrepancy summarise these differences over the 16 locations:

$$RMS_c = \sqrt{\frac{1}{16}\sum_{i=1}^{16}\left(\Delta g_{c,i}\right)^2} \qquad E_{max,c} = max_i|\Delta g_{c,i}|$$

These are statistics of the gap discrepancies, rather than of the gap distances themselves. They do not measure the maximum displacement of the complete LCFS or the X-point position error. Table 4.19 reports the RMS, maximum absolute discrepancy and signed mean, which retains the direction of the average offset.

Overall, MEQ shows closer agreement with the MAXFEA reference than EFIT according to the gap-based metrics. MEQ yields a lower RMS discrepancy in four of the six equilibria, while EFIT gives only marginally lower RMS values for the LSN EOF and USN EOF cases (3.60 versus 3.70 mm and 3.37 versus 3.42 mm, respectively). More notably, MEQ produces a lower maximum discrepancy in all six cases. The signed mean discrepancies are also consistently closer to zero for MEQ, whereas EFIT exhibits negative mean values in all cases, ranging from −2.69 to −5.49 mm. Across these six archived reconstructions, the EFIT results show a consistent tendency toward smaller LCFS–limiter gaps than the MAXFEA reference, while the MEQ results show substantially smaller overall offsets [12].

*Table 4.19. Discrepancies of the 16 LCFS–limiter gaps relative to MAXFEA, in millimetres. RMS and maximum are unsigned; the mean retains the sign of the discrepancy. These are spatial summaries of each saved reconstruction, not statistics over repeated noise realisations [12].*

| Case | EFIT RMS | EFIT Max. absolute | EFIT Mean | MEQ RMS | MEQ Max. absolute | MEQ Mean |
|---|---|---|---|---|---|---|
| DN SOF | 6.64 | 14.59 | −5.49 | 3.47 | 11.02 | +1.20 |
| DN EOF | 3.85 | 10.19 | −2.89 | 2.00 | 3.54 | +0.21 |
| LSN SOF | 5.03 | 12.47 | −4.18 | 1.49 | 3.84 | +0.83 |
| LSN EOF | 3.60 | 9.89 | −2.93 | 3.70 | 7.01 | +1.12 |
| USN SOF | 4.30 | 11.36 | −3.37 | 1.56 | 2.29 | +0.18 |
| USN EOF | 3.37 | 8.82 | −2.69 | 3.42 | 5.75 | −1.76 |

Figures 4.42–4.53 examine the six cases individually. Each boundary plot is followed by a graph of the gap values in millimetres for MAXFEA, EFIT and MEQ. At each index, the vertical separation from the MAXFEA curve is the magnitude of the discrepancy, irrespective of sign. A smaller physical gap is not, by itself, a better result. The connecting lines guide the eye between the discrete locations; the tables report the signed discrepancies relative to MAXFEA.

### 4.5.3.1 *Double Null at the Start of Flat Top*

MEQ follows the MAXFEA gaps more closely at 15 of the 16 locations, reducing the RMS discrepancy from 6.64 to 3.47 mm. The largest departure is at gap 12 in the lower region: the 69.09 mm reference gap becomes 54.50 mm in EFIT and 80.11 mm in MEQ. The reconstructions therefore lie on opposite sides of the reference at this sample. MEQ gives particularly close agreement at gaps 5–11, with discrepancies below 1 mm, but the lower-region mismatch remains appreciable. At gap 15, EFIT is slightly closer.

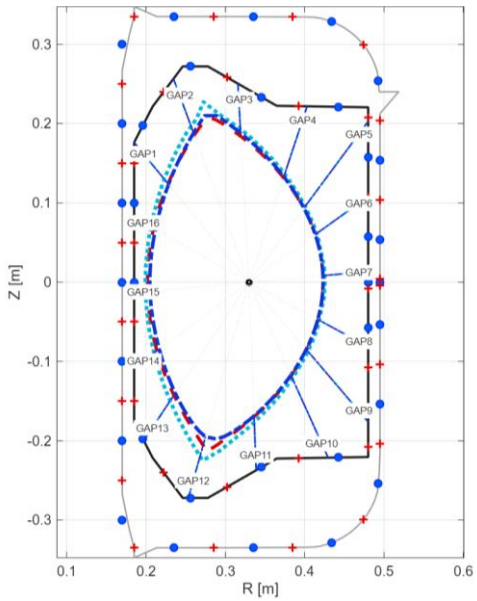



*Figure 4.42. DN SOF: plasma boundaries and diagnostic layout. MAXFEA is blue, EFIT cyan and MEQ red [12].*

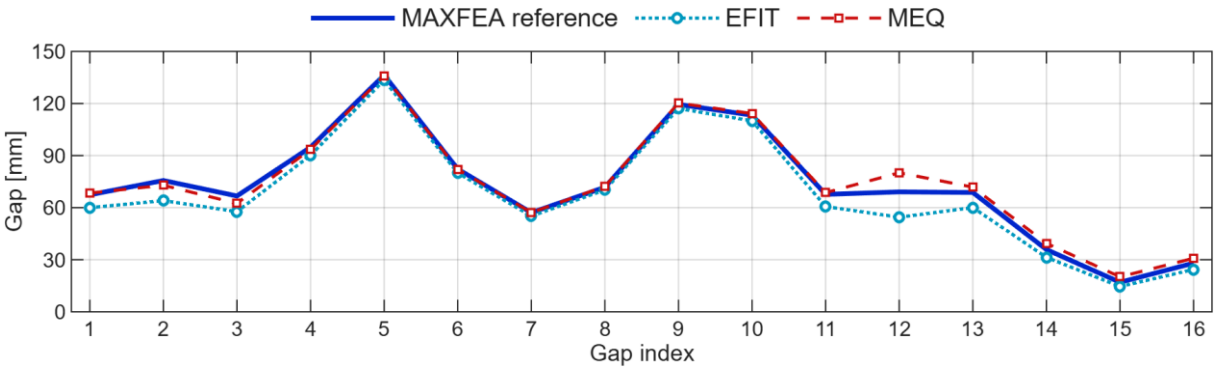



*Figure 4.43. DN SOF: gap values at the 16 limiter locations. The MAXFEA curve is the reference; all six case graphs use the same axes [12].*

*Table 4.20. DN SOF: signed gap discrepancies from MAXFEA, in mm [12].*

| Gap | EFIT − MAXFEA | MEQ − MAXFEA | Gap | EFIT − MAXFEA | MEQ − MAXFEA |
|---|---|---|---|---|---|
| 1 | −7.41 | +1.02 | 9 | −2.37 | +0.77 |
| 2 | −11.58 | −2.76 | 10 | −3.28 | +0.93 |
| 3 | −9.05 | −4.26 | 11 | −7.01 | +0.98 |
| 4 | −4.63 | −1.28 | 12 | −14.59 | +11.02 |
| 5 | −2.88 | −0.35 | 13 | −8.69 | +3.08 |
| 6 | −2.23 | −0.10 | 14 | −4.50 | +3.60 |
| 7 | −1.96 | +0.06 | 15 | −2.43 | +3.15 |
| 8 | −1.55 | +0.51 | 16 | −3.64 | +2.80 |

#### 4.5.3.2 *Double Null at the End of Flat Top*

Both reconstructions have smaller RMS discrepancies than in DN SOF: 3.85 mm for EFIT and 2.00 mm for MEQ. EFIT's main departure is now at gap 2 in the upper region, where it falls 10.19 mm below MAXFEA. MEQ is closer at 11 locations, although EFIT agrees better at gaps 7–8 and along the inboard side at gaps 14–16. The largest MEQ discrepancy is +3.54 mm at gap 15. The improved overall agreement at EOF thus coexists with a modest inboard departure in MEQ; it does not imply that MEQ is closer everywhere.

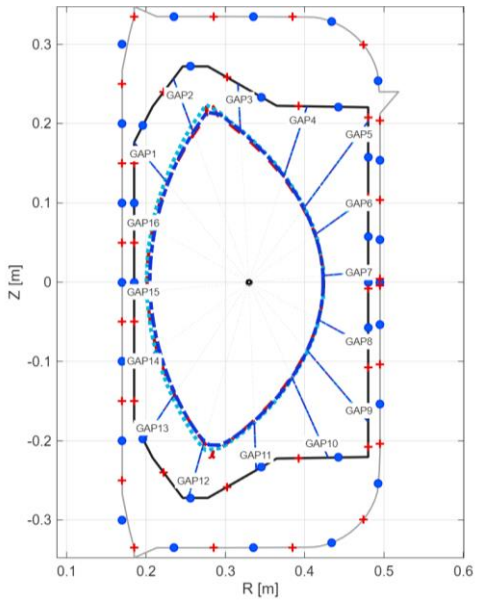



*Figure 4.44. DN EOF: plasma boundaries and diagnostic layout. MAXFEA is blue, EFIT cyan and MEQ red [12].*

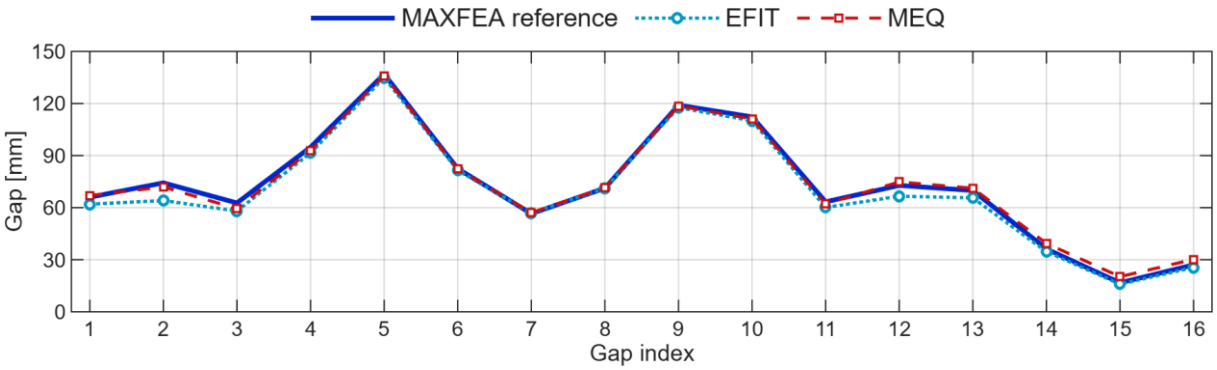



*Figure 4.45. DN EOF: gap values at the 16 limiter locations. The MAXFEA curve is the reference; all six case graphs use the same axes [12].*

*Table 4.21. DN EOF: signed gap discrepancies from MAXFEA, in mm [12].*

| Gap | EFIT − MAXFEA | MEQ − MAXFEA | Gap | EFIT − MAXFEA | MEQ − MAXFEA |
|---|---|---|---|---|---|
| 1 | −4.04 | +0.99 | 9 | −1.39 | −0.53 |
| 2 | −10.19 | −2.37 | 10 | −2.23 | −1.16 |
| 3 | −4.64 | −3.30 | 11 | −3.26 | −1.20 |
| 4 | −3.33 | −1.95 | 12 | −6.20 | +2.00 |
| 5 | −2.08 | −0.93 | 13 | −4.24 | +1.30 |
| 6 | −1.01 | −0.17 | 14 | −1.55 | +3.19 |
| 7 | +0.36 | +0.55 | 15 | −0.77 | +3.54 |
| 8 | −0.07 | +0.24 | 16 | −1.52 | +3.14 |

#### 4.5.3.3 *Lower Single Null at the Start of Flat Top*

This case gives the clearest improvement across the sampled boundary: MEQ is closer to MAXFEA at all 16 locations. Its RMS discrepancy is 1.49 mm, compared with 5.03 mm for EFIT. EFIT gives smaller gaps than the reference throughout, with its largest departure, −12.47 mm, at gap 12 near the lower divertor region. MEQ's largest departure occurs at the same location but has the opposite sign, +3.84 mm. Along gaps 6–11 and 13–16, the MEQ discrepancy remains below 1 mm. The improvement therefore extends well beyond the location responsible for the largest EFIT error.

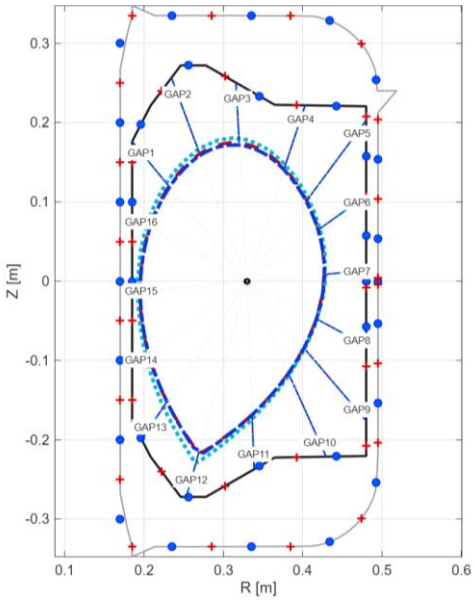



*Figure 4.46. LSN SOF: plasma boundaries and diagnostic layout. MAXFEA is blue, EFIT cyan and MEQ red [12].*

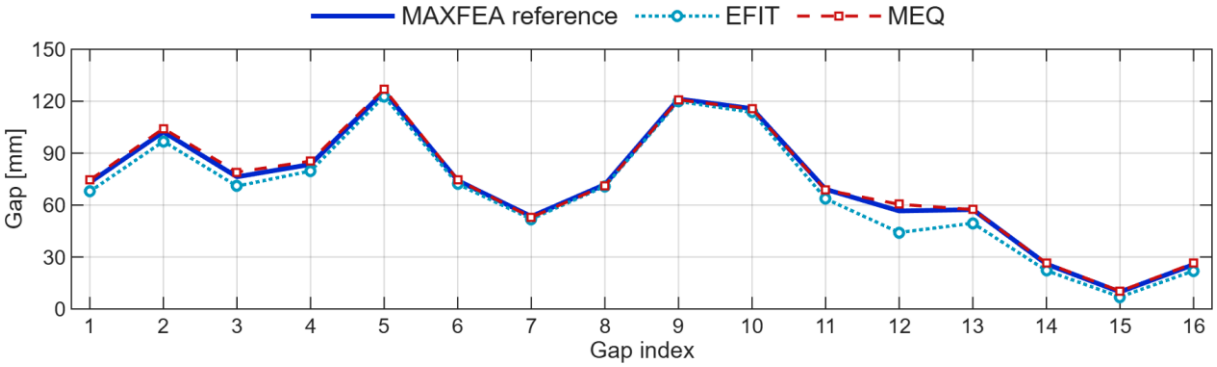



*Figure 4.47. LSN SOF: gap values at the 16 limiter locations. The MAXFEA curve is the reference; all six case graphs use the same axes [12].*

*Table 4.22. LSN SOF: signed gap discrepancies from MAXFEA, in mm [12].*

| Gap | EFIT − MAXFEA | MEQ − MAXFEA | Gap | EFIT − MAXFEA | MEQ − MAXFEA |
|---|---|---|---|---|---|
| 1 | −5.20 | +1.39 | 9 | −1.36 | −0.39 |
| 2 | −5.49 | +1.93 | 10 | −2.16 | −0.19 |
| 3 | −5.38 | +2.71 | 11 | −5.45 | −0.35 |
| 4 | −3.80 | +1.90 | 12 | −12.47 | +3.84 |
| 5 | −2.94 | +1.31 | 13 | −7.80 | +0.04 |
| 6 | −2.08 | +0.45 | 14 | −3.78 | +0.58 |
| 7 | −1.49 | −0.51 | 15 | −2.87 | +0.44 |
| 8 | −1.03 | −0.59 | 16 | −3.63 | +0.76 |

#### 4.5.3.4 *Lower Single Null at the End of Flat Top*

The two criteria favour different reconstructions here. MEQ is closer at ten locations and reduces the largest absolute discrepancy from 9.89 to 7.01 mm, yet its RMS is slightly higher: 3.70 mm against 3.60 mm for EFIT. MEQ reduces EFIT's pronounced departure at gap 12, but differs from MAXFEA by about 5–7 mm at gaps 1 and 14–16 on the inboard side. These departures contribute enough to the RMS to offset the improvement elsewhere. The result illustrates why a lower local maximum, or a larger number of closer samples, need not give a lower RMS.

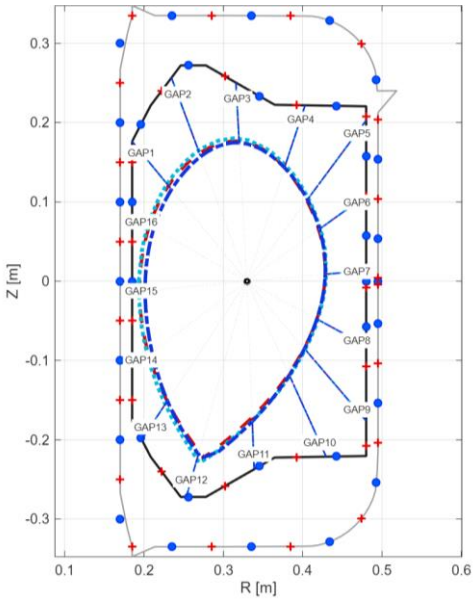



*Figure 4.48. LSN EOF: plasma boundaries and diagnostic layout. MAXFEA is blue, EFIT cyan and MEQ red [12].*

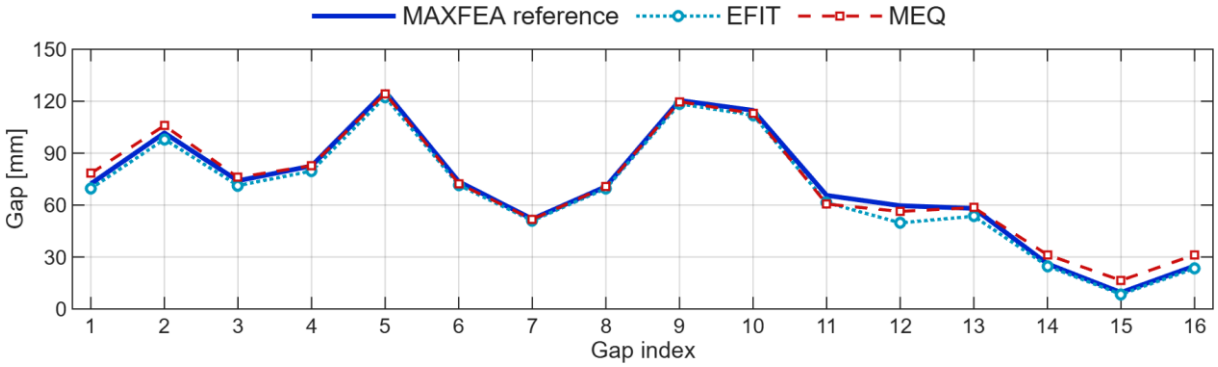



*Figure 4.49. LSN EOF: gap values at the 16 limiter locations. The MAXFEA curve is the reference; all six case graphs use the same axes [12].*

*Table 4.23. LSN EOF: signed gap discrepancies from MAXFEA, in mm [12].*

| Gap | EFIT − MAXFEA | MEQ − MAXFEA | Gap | EFIT − MAXFEA | MEQ − MAXFEA |
|---|---|---|---|---|---|
| 1 | −2.78 | +6.01 | 9 | −1.98 | −0.71 |
| 2 | −3.66 | +4.23 | 10 | −2.79 | −1.80 |
| 3 | −2.76 | +1.92 | 11 | −4.17 | −5.01 |
| 4 | −3.00 | +0.20 | 12 | −9.89 | −3.32 |
| 5 | −3.25 | −1.32 | 13 | −4.55 | +0.45 |
| 6 | −2.06 | −1.17 | 14 | −1.30 | +5.22 |
| 7 | −0.96 | −0.17 | 15 | −1.03 | +7.01 |
| 8 | −1.16 | −0.04 | 16 | −1.47 | +6.41 |

#### 4.5.3.5 *Upper Single Null at the Start of Flat Top*

MEQ shows a much smaller discrepancy than EFIT in the upper region. At gap 2, the EFIT difference is −11.36 mm, whereas MEQ differs from MAXFEA by only +0.72 mm. Across all samples, the RMS falls from 4.30 to 1.56 mm and the largest absolute discrepancy from 11.36 to 2.29 mm. MEQ is closer at 12 locations; EFIT is closer at outboard gaps 6–8 and at inboard gap 15. Thus, the advantage of MEQ in this case is chiefly the removal of the larger upper-region mismatch, with smaller differences between the codes over the rest of the sampled boundary.

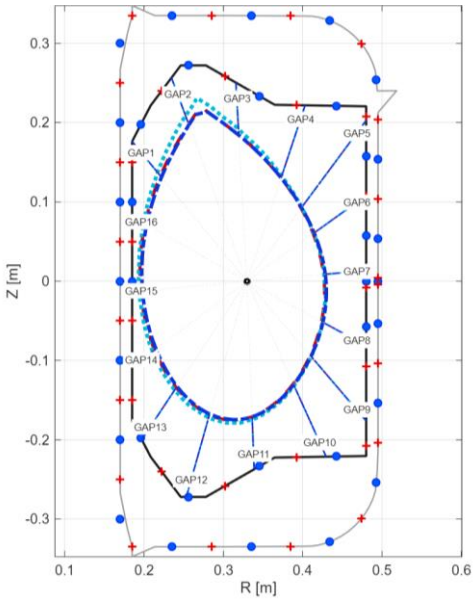



*Figure 4.50. USN SOF: plasma boundaries and diagnostic layout. MAXFEA is blue, EFIT cyan and MEQ red [12].*

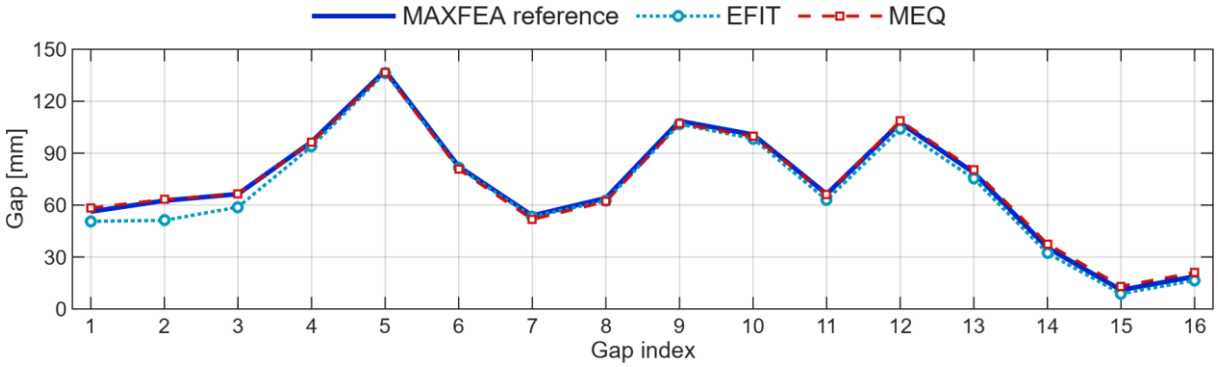



*Figure 4.51. USN SOF: gap values at the 16 limiter locations. The MAXFEA curve is the reference; all six case graphs use the same axes [12].*

*Table 4.24. USN SOF: signed gap discrepancies from MAXFEA, in mm [12].*

| Gap | EFIT − MAXFEA | MEQ − MAXFEA | Gap | EFIT − MAXFEA | MEQ − MAXFEA |
|---|---|---|---|---|---|
| 1 | −5.63 | +2.10 | 9 | −1.87 | −1.44 |
| 2 | −11.36 | +0.72 | 10 | −2.35 | −1.07 |
| 3 | −7.55 | +0.05 | 11 | −3.37 | −0.09 |
| 4 | −2.68 | −0.18 | 12 | −3.62 | +1.07 |
| 5 | −1.26 | −0.87 | 13 | −3.54 | +1.70 |
| 6 | −0.88 | −1.50 | 14 | −3.23 | +2.16 |
| 7 | −0.59 | −2.21 | 15 | −2.12 | +2.22 |
| 8 | −1.52 | −2.02 | 16 | −2.32 | +2.29 |

#### 4.5.3.6 *Upper Single Null at the End of Flat Top*

At EOF, MEQ again closely matches the upper reference gap: its discrepancy at gap 2 is −0.21 mm, compared with −8.82 mm for EFIT. Elsewhere, however, MEQ gives smaller gaps than MAXFEA over gaps 3–12, reaching −5.75 mm at gap 11 in the lower region. EFIT is closer at ten of the 16 samples and has a marginally smaller RMS, 3.37 mm against 3.42 mm. MEQ still has the smaller maximum discrepancy. The broader set of moderate MEQ departures explains why matching the upper gap so well does not produce the better RMS for the complete set.

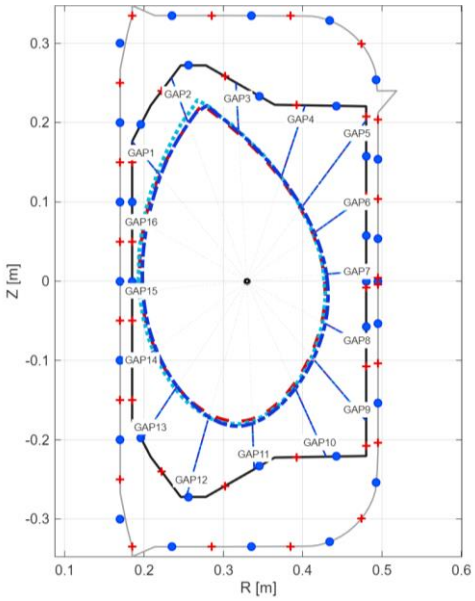



*Figure 4.52. USN EOF: plasma boundaries and diagnostic layout. MAXFEA is blue, EFIT cyan and MEQ red [12].*

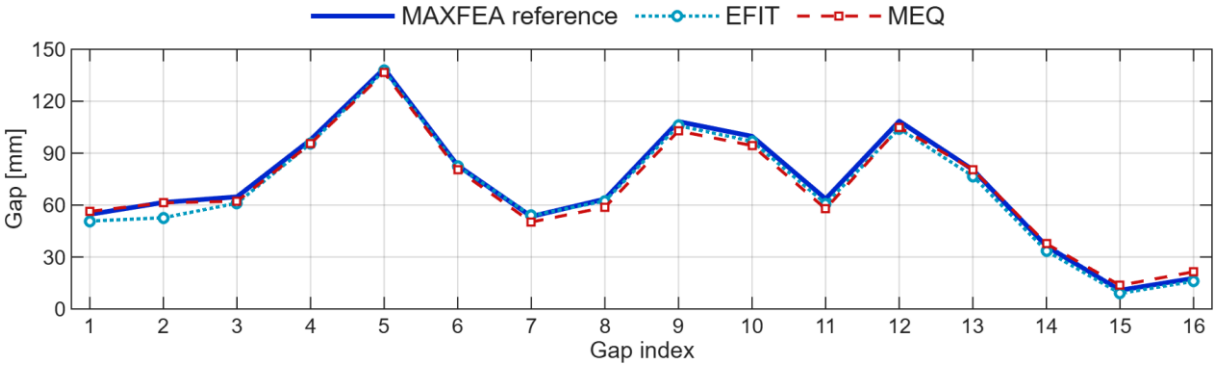



*Figure 4.53. USN EOF: gap values at the 16 limiter locations. The MAXFEA curve is the reference; all six case graphs use the same axes [12].*

*Table 4.25. USN EOF: signed gap discrepancies from MAXFEA, in mm [12].*

| Gap | EFIT − MAXFEA | MEQ − MAXFEA | Gap | EFIT − MAXFEA | MEQ − MAXFEA |
|---|---|---|---|---|---|
| 1 | −3.89 | +1.72 | 9 | −2.41 | −5.47 |
| 2 | −8.82 | −0.21 | 10 | −2.84 | −5.33 |
| 3 | −3.67 | −2.55 | 11 | −2.74 | −5.75 |
| 4 | −2.12 | −2.22 | 12 | −4.33 | −3.59 |
| 5 | −1.10 | −2.35 | 13 | −3.73 | +0.13 |
| 6 | −0.45 | −2.53 | 14 | −2.84 | +1.34 |
| 7 | +0.45 | −3.23 | 15 | −1.83 | +2.84 |
| 8 | −1.08 | −4.74 | 16 | −1.72 | +3.77 |

#### *4.5.3.7 Overall Comparison and Scope*

MEQ has the smaller RMS gap discrepancy in four of the six cases and the smaller maximum absolute gap discrepancy in all six. The comparison is less clear at EOF for the two single-null configurations: EFIT gives a slightly smaller RMS gap discrepancy, despite retaining a larger local maximum. This distinction matters because a reconstruction can distribute moderate errors around the boundary while another agrees more closely over much of the contour but departs more strongly in one region.

Across the six cases, EFIT predicts a smaller LCFS–limiter distance than MAXFEA in 94 of the 96 gap evaluations (16 per case), placing its reconstructed boundary closer to the limiter at those locations. The mean signed discrepancy relative to MAXFEA is negative in every case. This indicates a tendency to underestimate the reference gaps; it does not imply better agreement, which depends on how close the discrepancies are to zero [12].

MEQ has both signs, and its case means range from −1.76 to +1.20 mm. These patterns describe the six reconstructions considered here. Their origin cannot be attributed to measurement noise alone: profile assumptions, boundary preparation, discretisation and reconstruction constraints can also contribute.

Both reconstructions are reported as converged for all six cases. The resulting millimetre-scale RMS discrepancies provide a quantitative check of boundary agreement for this diagnostic array and these perturbed inputs. With one reconstruction per case, the present comparison does not provide a statistical uncertainty or a probability of successful reconstruction. Repeated noise realisations would be needed to quantify variability under the selected measurement model. Agreement of the sampled gaps also leaves pressure and current profiles, internal inductance and exact X-point positions as separate quantities to assess [12].

# 5 FULL-PULSE SIMULATION: COUPLING METIS WITH FBT

## 5.1 INTRODUCTION TO METIS

METIS is a tokamak system code designed for rapid integrated scenario modelling. It combines zero-dimensional descriptions of heat and particle transport, normalised through scaling laws, with one-dimensional current diffusion and two-dimensional equilibrium calculations. This reduced formulation supports time-dependent scenario studies while retaining the coupling between pressure, current distribution and magnetic geometry [14].

Input waveforms specify quantities such as plasma current, heating power, line-averaged density and plasma geometry. The LCFS can be described through geometric moments or a set of $R$ and $Z$ coordinates. METIS calculates the corresponding plasma evolution, providing global quantities and radial profiles that can be used as references for further equilibrium studies [14].

## 5.2 METIS–FBT COUPLING WORKFLOW

The coupling uses the numerical TRUST/METIS scenario to define a sequence of inverse free-boundary problems in the TRUST geometry. For each selected time, the reference plasma quantities and boundary are supplied to FBT, which calculates the corresponding equilibrium and active-circuit currents. The data flow is one-way: the FBT solution is not fed back into the METIS calculation [12].

The global plasma targets passed to FBT are $I_p$, poloidal beta $\beta_p$ and on-axis safety factor $q_A$. The toroidal-field parameter $RB_t$ is supplied separately. These quantities specify the requested plasma state, while the reference LCFS provides its geometric target. At each sample, that contour is represented by 64 pairs of $R$ and $Z$ coordinates from the geometry associated with the scenario. The figures label this prescribed contour “METIS LCFS”; the FBT boundary is calculated from the resulting equilibrium [12].

The FBT input also includes a reference vector for the 12 active circuits. These currents serve as targets in the inverse calculation and can differ from the final solution. FBT adjusts the active currents and its equilibrium variables to fit the requested plasma and boundary constraints within the configured optimisation bounds. The calculated current vectors are expressed in ampere-turn coordinates, consistently with the TRUST model used in Chapter 4. Table 5.1 summarises the reference inputs supplied to FBT [12].

*Table 5.1. Reference inputs supplied to FBT in the METIS-to-FBT coupling [12].*

| Quantity | Role in FBT | Status |
|---|---|---|
| $I_p$, $\beta_p$, $q_A$ | Global equilibrium targets | Prescribed targets |
| $RB_t$ | Toroidal-field parameter | Supplied input |
| Reference LCFS | 64 $R$–$Z$ contour samples per time | Shape target |
| 12-circuit current vector | Reference for inverse current calculation | Current reference |

The sequence is quasi-static. Each sample represents an equilibrium calculation; circuit dynamics are not integrated between successive times. In this configuration, passive currents are zero and voltage-limit checking is disabled. The coupling therefore provides the equilibria and active-current demand associated with the reference scenario [12].

## 5.3 Reference Scenario and Equilibrium Sequence

The calculation comprises 44 equilibria between 10 and 438 ms. Samples are separated by 10 ms through 430 ms, followed by a final sample at 438 ms. Six states cover the initial current rise (10–60 ms), 32 have an $I_p$ target of 110 kA (70–380 ms), and six cover the subsequent decrease (390–438 ms). The first target is 17.1875 kA: the sequence begins with an already formed plasma and ends during the reference current decrease. “Full pulse” denotes this sampled scenario interval; gas breakdown and final plasma extinction are outside its scope [12].

Figure 5.1 shows the three global plasma targets supplied to FBT at the selected times. The points are reference inputs to the inverse equilibrium calculation [12].

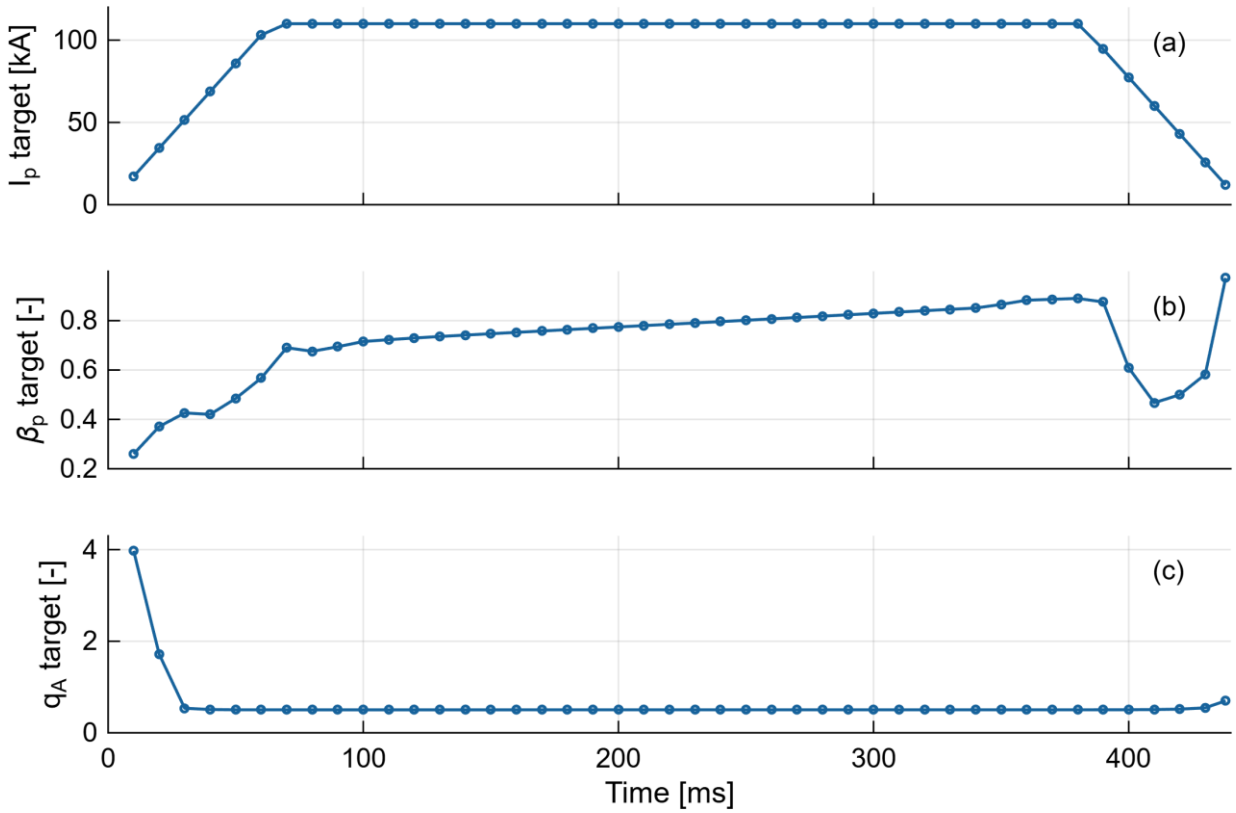



*Figure 5.1. Reference $I_p$, $\beta_p$ and $q_A$ supplied to FBT at the 44 sampled times. Markers identify the sampled inputs; connecting lines are visual guides [12].*

All 44 FBT states satisfy the stored convergence criterion. They require 6–33 iterations, and the largest residual is $5.8301 \times 10^{-4}$ for a tolerance of $6 \times 10^{-4}$. Across the scan, the maximum absolute differences from the global targets are 0.5522 A in $I_p$, $1.1818 \times 10^{-5}$ in $\beta_p$ and $3.5129 \times 10^{-4}$ in $q_A$. These values measure how closely FBT satisfies the imposed global constraints; boundary agreement is examined separately in the contour overlays [12].

Figure 5.2 shows the initial state at 10 ms. The central FBT boundary closely follows the approximately elliptical reference contour, and the calculated plasma current is 17.188 kA. The difference between the reference and calculated global quantities is small on the scale of the normalised bars.

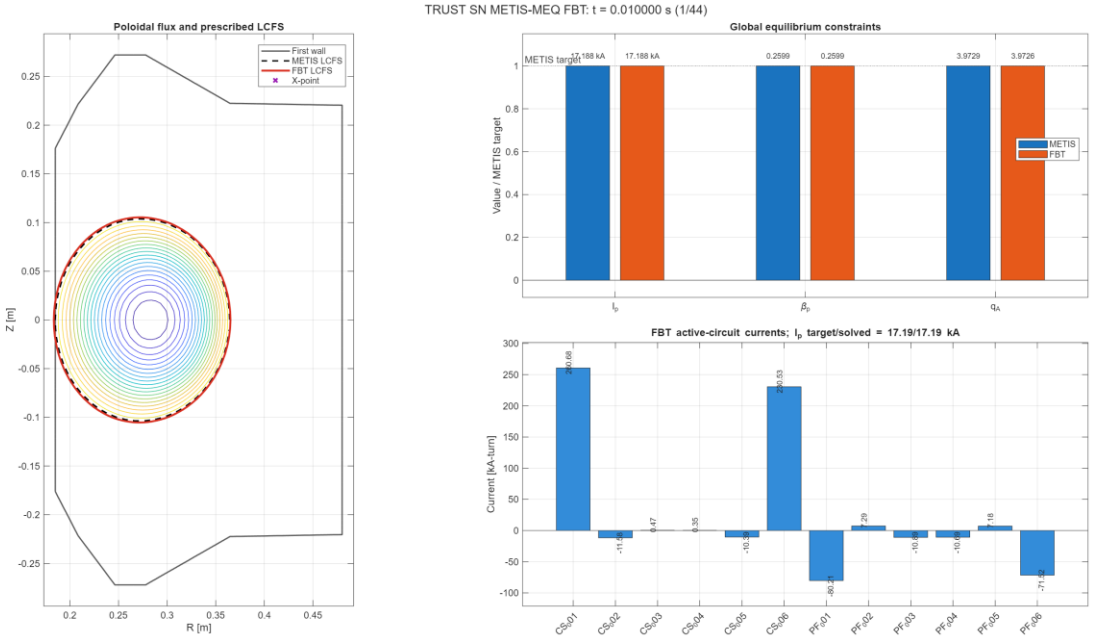



*Figure 5.2. METIS-to-FBT coupling at 10 ms. The dashed black contour is the reference LCFS and the red contour is the FBT boundary. The upper-right panel compares global targets with FBT values, normalised by the respective targets; the lower-right panel gives active-circuit currents in kA-turn [12].*

At 70 ms, the first sampled flat-top state has a lower X-point and an $I_p$ target of 110 kA (Figure 5.3). The FBT contour follows the reference shape, with visible local differences around the upper and lower parts of the boundary. During the sampled flat top, the targets continue to change even though $I_p$ remains fixed: $\beta_p$ rises from 0.69034 at 70 ms to 0.89013 at 380 ms [12].

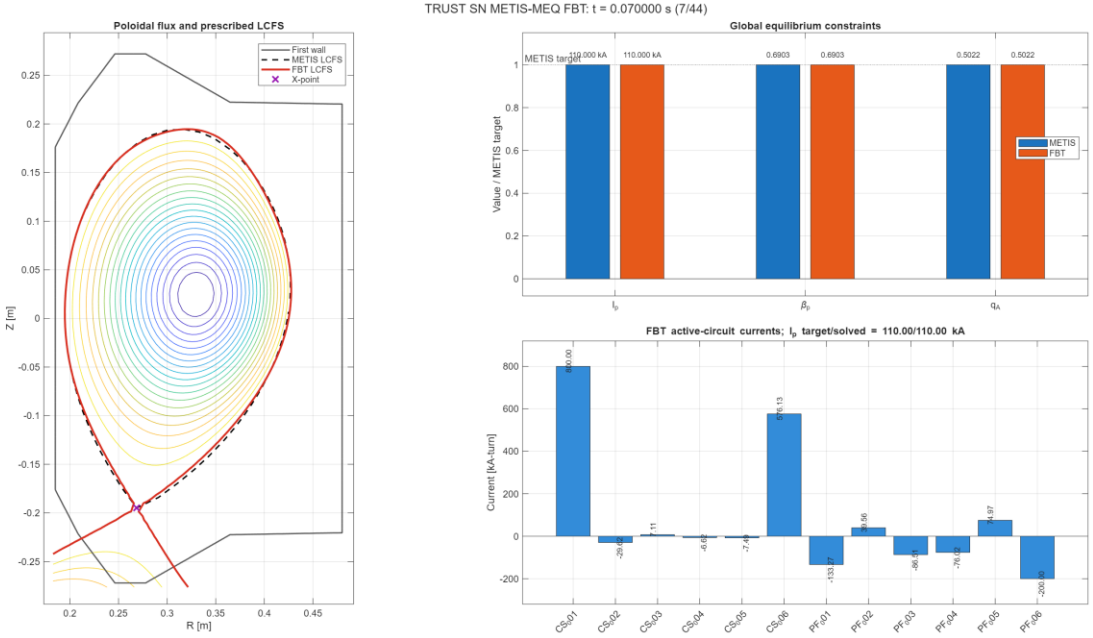



*Figure 5.3. First sampled flat-top equilibrium at 70 ms. Boundary, target-comparison and current conventions are the same as in Figure 5.2 [12].*

At the final sampled time, 438 ms, the target $I_p$ is 12.031 kA and the central plasma contour has returned to an approximately elliptical shape (Figure 5.4). The central closed FBT contour remains close to the reference in this overlay.

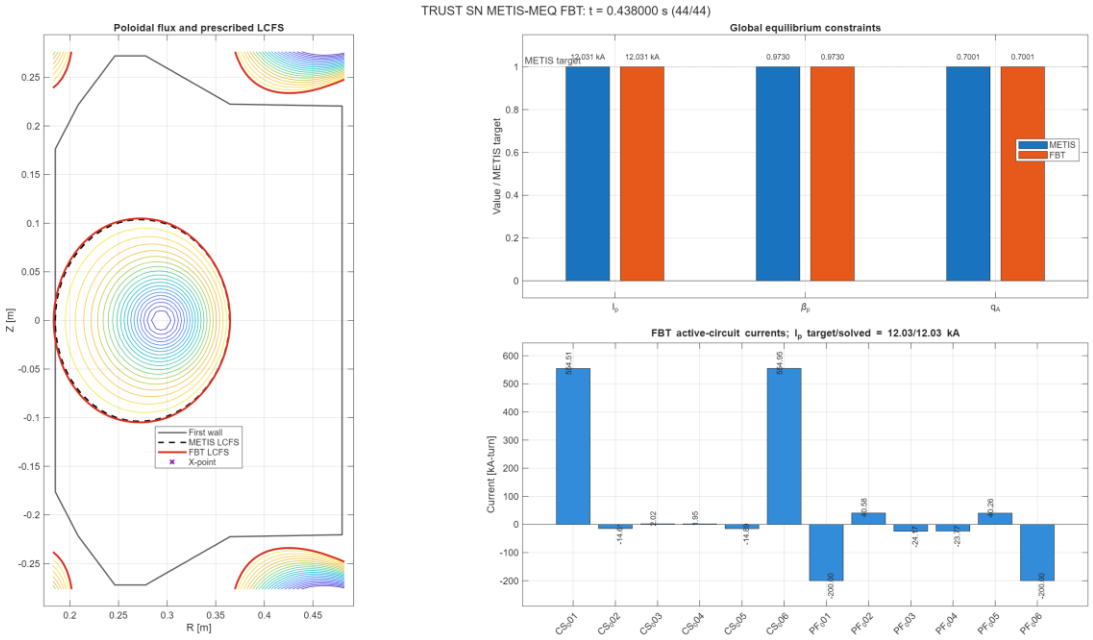



*Figure 5.4. Final sampled equilibrium at 438 ms. The central closed FBT contour is compared with the reference LCFS. The right-hand panels show the global-target comparison and the active currents, using the conventions of Figure 5.2 [12].*

Figures 5.2–5.4 give three detailed examples at 10, 70 and 438 ms. Figure 5.5 assembles nine selected FBT boundaries to show the formation and disappearance of the lower X-point. Figures 5.1 and 5.6 report the global targets and active currents, respectively, at all 44 sampled times.

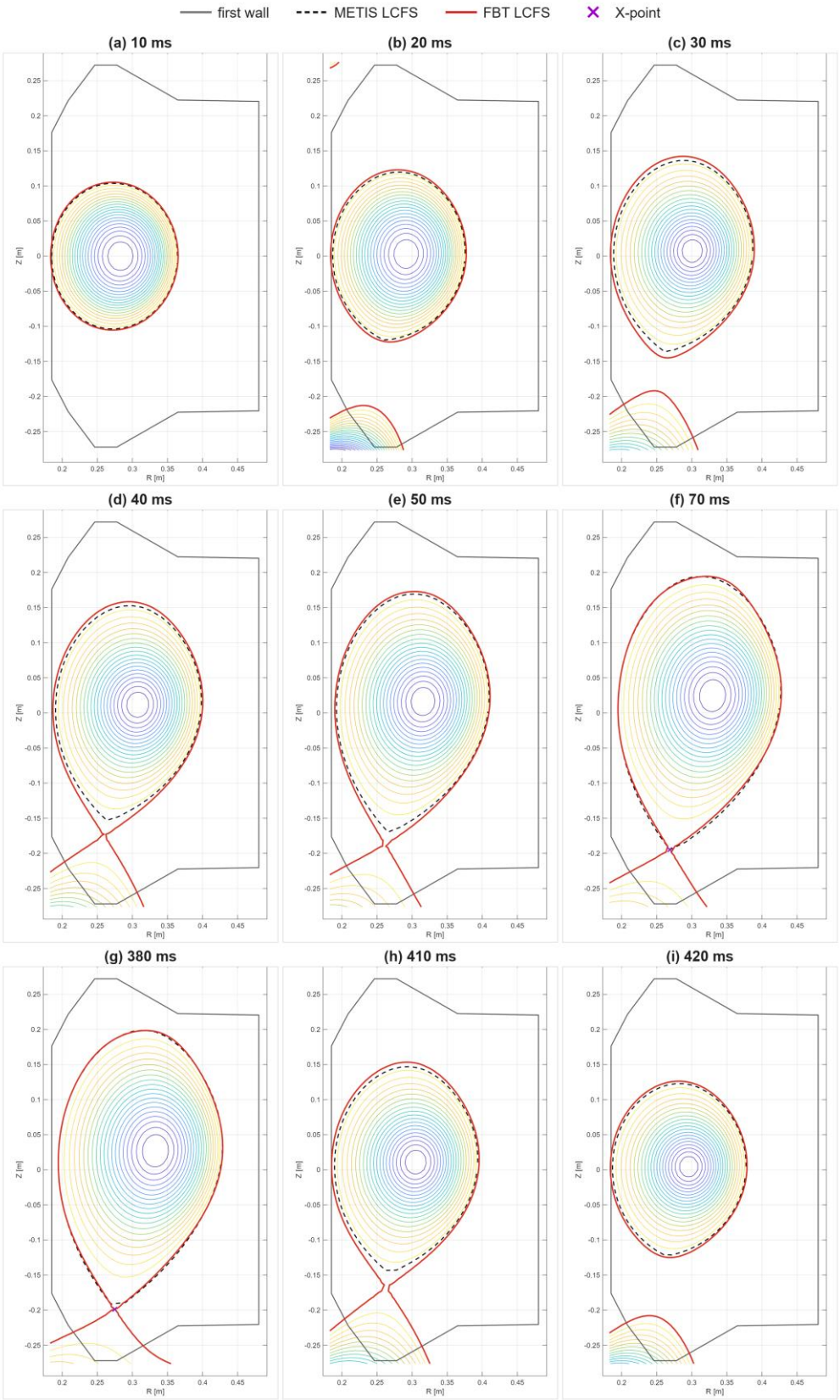



*Figure 5.5. METIS–FBT equilibrium evolution. (a–f) Formation of the lower single-null at 10, 20, 30, 40, 50 and 70 ms. (g–i) Late transition from lower single-null to limiter at 380, 410 and 420 ms. All panels use the same R–Z view: grey denotes the TRUST first wall, dashed black the reference METIS LCFS, and red the calculated FBT boundary. In the FBT solutions, the lower X-point appears by 40 ms and disappears between 410 and 420 ms [12].*

## 5.4 Active-Current Demand and Assessment Limits

Figure 5.6 shows the active currents calculated at all 44 sampled times. At 10 ms, the largest active current is 260.68 kA-turn in CS1, while CS6 carries 230.53 kA-turn. At 70 ms, CS1 reaches +800 kA-turn and PF6 reaches −200 kA-turn. These two currents have the same values in the 380 ms solution, while CS6 changes from 576.13 to 619.17 kA-turn between the two flat-top samples [12].

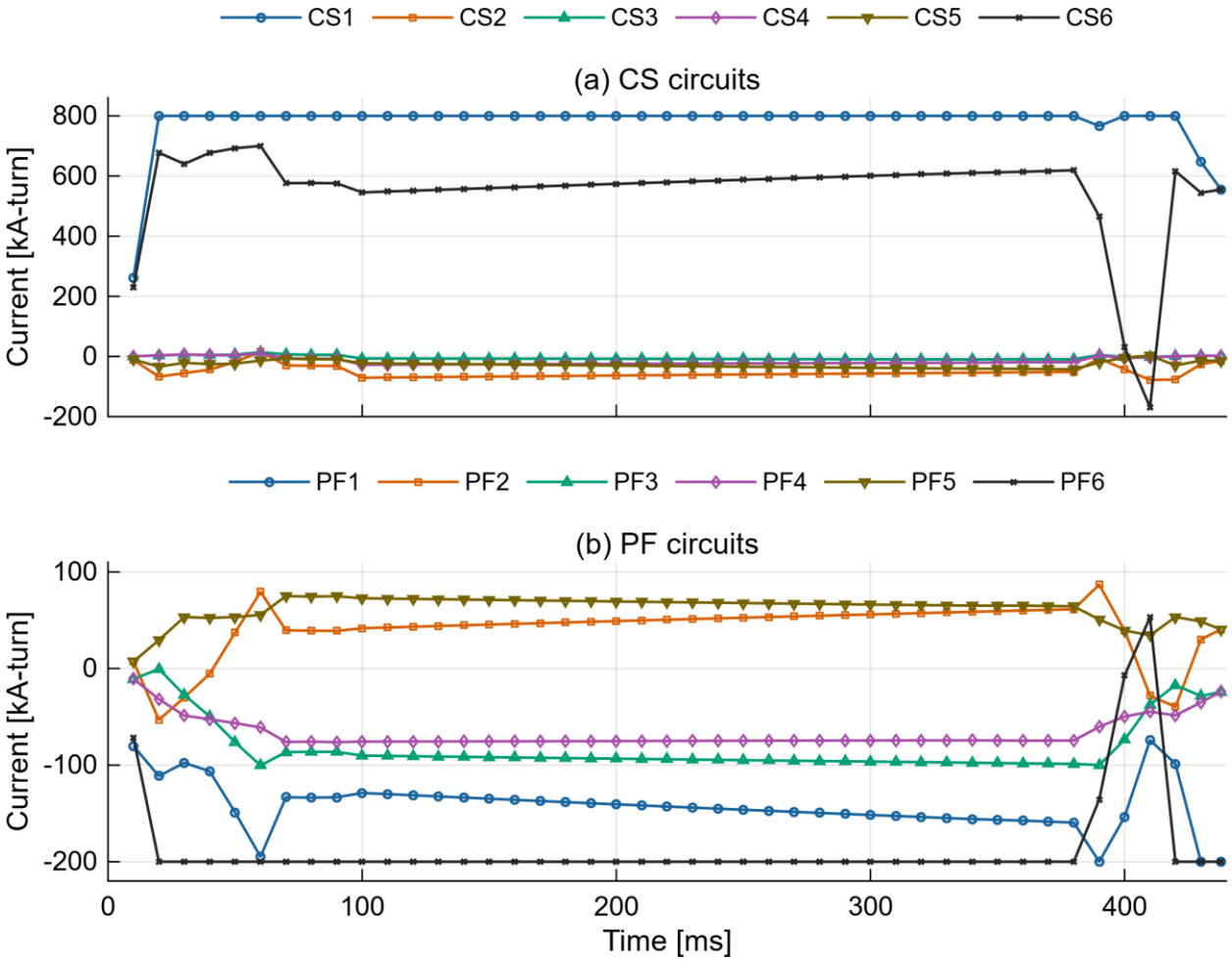



*Figure 5.6. FBT active currents at all 44 times, in ampere-turn coordinates: (a) CS circuits; (b) PF circuits. Markers denote the calculated states and lines connect successive samples without representing circuit time integration. The assessment envelope and optimisation bounds are listed separately in Table 5.2 [12].*

The complete scan also shows a change of sign in CS6 during the reference current decrease: it carries −169.27 kA-turn at 410 ms and +616.43 kA-turn at 420 ms. These values belong to successive quasi-static FBT solutions [12].

At 438 ms, the current vector remains different from that of the initial low-current equilibrium: CS1 and CS6 carry 554.51 and 554.95 kA-turn, respectively, while PF1 and PF6 are both approximately −200 kA-turn. These values describe the solution obtained for the final set of plasma and boundary targets [12].

The actual FBT optimisation bounds in this scan are ±800 kA-turn for the CS circuits and ±200 kA-turn for the PF circuits. All 44 current vectors satisfy these inequalities, with some currents reaching a bound. Table 5.2 compares the maximum absolute current in each circuit against both its optimisation bound and the separate common TRUST scenario envelope. Each utilisation is the maximum absolute current divided by the corresponding positive limit, multiplied by 100 [12].

*Table 5.2. Peak current demand over the 44 FBT states. Bounds are given as positive magnitudes; the actual inequalities are symmetric. Percentages use unrounded currents and limits [12].*

| Circuit | Peak \|I\| [kA-turn] | Common envelope [kA-turn] | Common-envelope utilisation [%] | FBT optimisation bound [kA-turn] | Optimisation-bound utilisation [%] |
|---|---|---|---|---|---|
| CS1 | 800.00 | 1053.00 | 75.97 | 800.00 | 100.00 |
| CS2 | 78.18 | 1053.00 | 7.42 | 800.00 | 9.77 |
| CS3 | 14.14 | 1053.00 | 1.34 | 800.00 | 1.77 |
| CS4 | 27.77 | 1053.00 | 2.64 | 800.00 | 3.47 |
| CS5 | 42.71 | 1053.00 | 4.06 | 800.00 | 5.34 |
| CS6 | 700.10 | 1053.00 | 66.49 | 800.00 | 87.51 |
| PF1 | 200.00 | 168.65 | 118.59 | 200.00 | 100.00 |
| PF2 | 86.89 | 55.85 | 155.59 | 200.00 | 43.44 |
| PF3 | 100.32 | 64.47 | 155.62 | 200.00 | 50.16 |
| PF4 | 76.22 | 64.47 | 118.23 | 200.00 | 38.11 |
| PF5 | 74.97 | 55.85 | 134.25 | 200.00 | 37.49 |
| PF6 | 200.00 | 168.65 | 118.59 | 200.00 | 100.00 |

At least one circuit exceeds the common envelope in 42 of the 44 states. The CS currents remain below that envelope throughout the scan, whereas each PF circuit exceeds its own assessment value at one or more times. The largest utilisation is 155.62% in PF3 at 60 ms (100.32 kA-turn against 64.47 kA-turn), closely followed by 155.59% in PF2 at 390 ms (86.89 kA-turn against 55.85 kA-turn). Thus, the converged sequence fits the global plasma targets within the FBT optimisation bounds, while its calculated currents exceed the common scenario envelope over much of the scan. This comparison concerns the numerical assessment limits defined for the TRUST scenario [12].

# 6 Conclusions and Future Developments

## 6.1 Conclusions

This thesis adapted the MEQ equilibrium framework to the TRUST tokamak and applied it to equilibrium generation, magnetic reconstruction and static forward calculations. The TRUST-specific interfaces bring the machine geometry, active and passive circuits, diagnostic layout and local scenario data into a common modelling environment, while retaining the generic MEQ solver core. This provides a basis for using numerical reference equilibria and scenario targets throughout the design and reconstruction workflow.

The FBT calculations cover low-current and flat-top equilibria, including limiter, single-null and double-null configurations. All six examples satisfy their convergence criteria. The resulting equilibria show how the requested plasma parameters and boundary targets are realised through the active-circuit currents. The imported-boundary examples also reveal changes in geometry and current distribution when an external equilibrium is represented in the TRUST-MEQ model. Agreement with an imported contour and the associated current demand must therefore be assessed together: some converged solutions exceed the common TRUST scenario envelope while satisfying the separate FBT optimisation bounds.

The LIUQE results show close recovery of plasma current and boundary geometry in several of the synthetic reconstruction cases, with appreciably larger differences in internal quantities. In the FBT-based single-null and double-null examples, the null displacements are below one millimetre, whereas the reconstructed on-axis safety factor is approximately 44% below its reference value. Pressure-related quantities also differ, particularly in the limiter and chained MAXFEA cases. The external-measurement interface reproduces the direct single-null reconstruction at the reported precision when supplied with the same synthetic inputs. For the MAXFEA–FBT–LIUQE chains, separating the two stages shows that the largest changes in magnetic signals and active currents occur during the transfer to FBT, while LIUQE introduces further changes in the reconstructed profile quantities.

The FGS examples complement the inverse and reconstruction calculations by examining static equilibria with assigned active currents. In the FBT-derived case, the magnetic axis moves radially by 4.306 mm after the forward calculation, with the selected global plasma quantities retained. In the MAXFEA-derived case, the FGS result retains the transferred MEQ geometry and profiles; the differences from MAXFEA arise during conversion. The active currents remain fixed in both examples, consistently with the forward formulation. These comparisons distinguish the effect of transferring a source equilibrium from the subsequent static calculation.

The cross-code study provides a quantitative comparison of reconstructed boundaries against MAXFEA for six TRUST reference equilibria. At the sixteen sampled limiter locations, MEQ gives a lower RMS gap discrepancy than EFIT in four cases and a lower maximum absolute discrepancy in all six. The RMS discrepancies are on the millimetre scale for both codes, with EFIT giving slightly smaller RMS values in the two single-null end-of-flat-top cases. These results support the use of MEQ for the boundary reconstruction problems examined here. Their scope is the sampled geometry under the specified synthetic measurement conditions; recovery of the internal plasma state remains a separate part of the assessment.

The METIS–FBT coupling extends this work from individual equilibria to a sequence associated with a reference scenario. Plasma-current, poloidal-beta and on-axis safety-factor targets, together with the associated reference boundary, define the inverse problems whose solutions provide the required active currents. All 44 equilibria between 10 and 438 ms converge and closely satisfy the imposed global targets. However, 42 states exceed the common scenario envelope in at least one circuit, even though all satisfy the

actual FBT optimisation bounds. The coupling thus identifies substantial PF current demand over the sampled scenario. Its quasi-static formulation determines the equilibrium sequence; electrical accessibility between successive states requires a separate assessment.

The work establishes a TRUST-specific framework connecting equilibrium design, magnetic reconstruction and scenario-based current assessment. Its principal results concern both the agreement achieved with numerical references and the differences that remain in internal profiles, transferred equilibria and actuator demand. These findings define concrete priorities for the next stage of development.

## 6.2 Future Developments

The main priority is to consolidate magnetic reconstruction, with particular attention to the recovery of internal plasma quantities. The LIUQE examples show that a close boundary fit can coexist with substantial differences in the on-axis safety factor, poloidal beta and plasma kinetic energy. Future work should aim to reduce these discrepancies while preserving the geometric agreement already obtained. The assessment should follow both aspects explicitly, using the reference equilibria to compare reconstructed profiles and integral quantities alongside the magnetic signals and boundary.

A focused sensitivity study could examine the profile parameterisation, regularisation and available diagnostic constraints, starting from the cases with the largest internal discrepancies. The different treatment of the diamagnetic signal in the existing workflows deserves particular attention, together with the role of initialisation. Varying these choices separately for a fixed reference would help establish which changes improve the reconstructed internal state. The negative sampled pressure and conditioning warnings found in some direct MAXFEA reconstructions also provide specific issues to resolve. Successful improvement would combine closer reference agreement with physically admissible profiles and more consistent numerical behaviour across the tested cases.

For equilibria imported from MAXFEA, this study should retain the distinction between conversion and reconstruction. Comparing the original reference, the transferred state and the final reconstruction makes it possible to locate where profile and geometry differences enter the workflow. This would guide improvements in the handling of external equilibria without attributing every discrepancy to LIUQE. When experimental TRUST measurements become available, the same framework could be assessed with calibrated diagnostic data, extending the present numerical comparisons to experimental reconstruction.

A complementary development concerns the current demand identified by the METIS–FBT coupling. The states that exceed the common scenario envelope provide starting points for studying how changes in the reference boundary and plasma targets affect the required PF currents. Subsequent assessment of voltage demand and circuit evolution would then address the accessibility of the equilibrium sequence. This would connect the scenario study more closely to the operating constraints of TRUST.

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